Open-FDD Documentation

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1 Home

2 Open-FDD

> **Single repository:** the `open-fdd` engine (`RuleRunner`, rule YAML, column maps) lives in `open_fdd/` and ships to **PyPI**. Application code for the **platform** (FastAPI, React, SQL schema, VOLTTRON bridge helpers) lives in `afdd_stack/`. **Docker Compose** here starts **Postgres/TimescaleDB** (and optional Grafana or Mosquitto) only — **not** field buses. **BACnet, Modbus, and similar OT protocols run only inside each building's VOLTTRON** deployment; Open-FDD consumes **SQL** (and optional HTTP APIs), not raw device wires.

Default path (on-prem or cloud for the app tier) — from the repo root run `./afdd_stack/scripts/bootstrap.sh` to prepare **VOLTTRON Central / edge** via **volttron-docker** and optionally `--compose-db` for SQL. **Per-site VOLTTRON** on the building LAN uses **ZMQ** VIP and pub/sub for the platform message bus (not

RabbitMQ in this reference design). Run **FastAPI + React from source** when you need Brick CRUD, SPARQL, and REST. See **Site VOLTTRON and the data plane (ZMQ)**, **Getting started**, and **`afdd_stack/legacy/README.md`**.

Open-FDD is an open-source knowledge graph fault-detection platform for HVAC systems that helps facilities optimize their energy usage and cost-savings. Because it runs on-prem, facilities never have to worry about a vendor hiking prices, going dark, or walking away with their data. The platform is an AFDD stack designed to run inside the building, behind the firewall, under the owner's control. It transforms operational data into actionable, cost-saving insights and provides a secure integration layer that any cloud platform can use without vendor lock-in. U.S. Department of Energy research reports median energy savings of roughly 8–9% from FDD programs—meaningful annual savings depending on facility size and energy spend.

The building is modeled in a **unified graph**: Brick (sites, equipment, points), optional RDF from imports or legacy tooling, platform config, and—as the project evolves—other ontologies such as ASHRAE 223P, in one semantic model queried via SPARQL and serialized to `afdd_stack/config/data_model.ttl` when you run **FastAPI** from the repo. **Field discovery RDF** is not produced by Open-FDD services in the default architecture—VOLTTRON and your ETL own what lands in SQL and what you merge into the graph.

2.1 What it does

Open-FDD is an **edge analytics and rules engine** for building automation. It:

- **Receives** time-series and metadata written by **site VOLTTRON** historians and agents (**ZMQ** pub/sub on the VOLTTRON bus, **SQL** into Postgres/Timescale); optional **Open-Meteo** or other writers can still feed the same SQL schema depending on how you deploy jobs
- **Stores** telemetry in **Postgres/TimescaleDB** and models the building in a **unified RDF graph** (Brick + platform config + optional imported RDF)
- **Runs** YAML-defined FDD rules (bounds, flatline, hunting, expression) — in the platform this is moving toward **VOLTTRON agents** over historian/SQL data; the **`open_fdd`** engine remains usable standalone on pandas
- **Exposes** (when FastAPI is running) REST APIs for sites, points, equipment, data-model export/import, bulk timeseries and fault download, TTL generation, SPARQL validation — for agents and Open-Claw-style automation

- **Visualizes** via **Grafana** (optional compose profile), **VOLTTRON Central**, or the **React** dashboard when you run the frontend from source

Operators and integrators get full control, lower cost, and no vendor lock-in. Already powering HVAC optimization and commissioning workflows.

2.2 Quick start

```
git clone https://github.com/bbartling/open-fdd.git
cd open-fdd
./afdd_stack/scripts/bootstrap.sh --help
./afdd_stack/scripts/bootstrap.sh --doctor
./afdd_stack/scripts/bootstrap.sh --central-lab
# Optional local SQL (Open-FDD schema + historian-friendly Postgres):
./afdd_stack/scripts/bootstrap.sh --compose-db
# Pytest (+ optional frontend when OFDD_BOOTSTRAP_FRONTEND_TEST=1):
./afdd_stack/scripts/bootstrap.sh --test
```

Typical URLs (when you run components yourself):

What	URL	Notes
Postgres/ TimescaleDB	`127.0.0.1:5432` (default compose)	Database and port come from `afdd_stack/stack/docker-compose.yml`; see Security — database.
API (REST)	http://localhost:8000/docs	Run FastAPI with **`uvicorn`** from a dev venv — not started by default bootstrap. Reference: Appendix: API Reference.
Frontend (React)	http://localhost:5173	Dev server or static build when you run the frontend from **`afdd_stack/frontend/`** — not a Compose service here.
Grafana	http://localhost:3000	**Optional:** `docker compose --profile grafana …` from `afdd_stack/stack/` (see compose file).
VOLTTRON Central / edge	(your deployment)	Use upstream VOLTTRON docs for ports and TLS.

WebSockets: When the API is running, `/ws/events` uses the same auth model as REST (**JWT** or `OFDD_API_KEY` in the query string). See Security.

Legacy reverse proxy: `afdd_stack/stack/caddy/Caddyfile` remains in the tree as a **reference** if you put Caddy in front of API + React yourself. It is **not** started by the default compose file. Patterns for basic auth and TLS are in Security and Caddy.

2.3 Documentation

Section	Description
Engine (PyPI)	Library-only install, `RuleRunner`, column-map resolvers, rule YAML — **bbartling.github.io/open-fdd**
System Overview	Architecture, services, data flow
Modular Architecture	Collector/Model/Engine/Interface boundaries and mode-based bootstrap matrix.
Getting Started	Install, bootstrap, first run
Using the React dashboard	Overview, Config, Points, Data model, Faults, Plots, Web weather, System resources
Edge field buses (VOLTTRON)	**Open-FDD does not run BACnet/Modbus**; site VOLTTRON does. ZMQ message bus; historian → SQL.
Data modeling	Sites, equipment, points (CRUD), Brick TTL, 223P-aligned engineering metadata, SPARQL cookbook, AI-assisted tagging
Fault rules for HVAC	Rule types, expression cookbook, test bench rule catalog
Concepts	Site VOLTTRON + ZMQ data plane, Cloud export, …
Operations	Integrity sweep, Overnight review, Testing plan
How-to Guides	VOLTTRON Central + parity, Open Claw integration (HTTP + SSH operator for remote edge → Central), openfdd-engine vs `open_fdd.engine`, Engine-only / IoT, Data model engineering, Grafana dashboards (optional), Grafana SQL cookbook
Security & Caddy	API auth, optional Caddy/TLS patterns (custom deployments)

Section	Description
Configuration	Platform config, rule YAML
Appendix	API Reference — REST at a glance; Technical reference, Developer guide
Contributing	How to contribute; alpha/beta focus; bugs, rules, docs, drivers

Use the React frontend for config, sites, points, data model, faults, and plots. **API details** (CRUD, config, data-model, download, analytics; legacy BACnet proxy routes if you enable them) are summarized in Appendix: API Reference; full OpenAPI at `/docs` and `/openapi.json`. **Grafana** is optional; the React UI provides equivalent timeseries and fault views.

2.4 Stack (what ships in this repo)

Piece	Port / location	Purpose
Postgres/ TimescaleDB	5432 (default compose)	Open-FDD metadata + time series schema; compatible with VOLTTRON SQL historian when you share one server
Grafana	3000	**Optional** compose profile
Mosquitto	1883	**Optional** compose profile — generic MQTT broker **only if you add it**; not the VOLTTRON bus (MQTT how-to)
VOLTTRON	(per building)	**Site instances** on the LAN: drivers, **ZMQ** VIP/pub-sub, historian → SQL; Central/

Piece	Port / location	Purpose
FastAPI + React	8000 / 5173 when run locally	fleet UI — upstream docs **Source** under `afdd_stack/`; not started by default compose

Removed from Compose (see `afdd_stack/legacy/README.md`): Caddy, API container, frontend container, diy-bacnet-server, BACnet scraper, weather scraper, FDD loop container. **Do not** plan new BACnet/Modbus ingest through those services; use **site VOLTTRON** and SQL. The **Edge field buses** page states this explicitly and points to legacy material in git only.

2.5 Behind the firewall; cloud export is vendor-led

Open-FDD runs **inside the building, behind the firewall**. Analytics and rules run locally; telemetry stays on-premises unless you export it. Cloud/MSI vendors run their **own** export process (e.g. edge gateway or script) that **pulls** from the API over the LAN; Open-FDD does not push to the cloud. See the Cloud export example and `examples/cloud_export.py`. Data stays with the client; the open data model and APIs let you switch vendors without losing data or the model.

3 System Overview

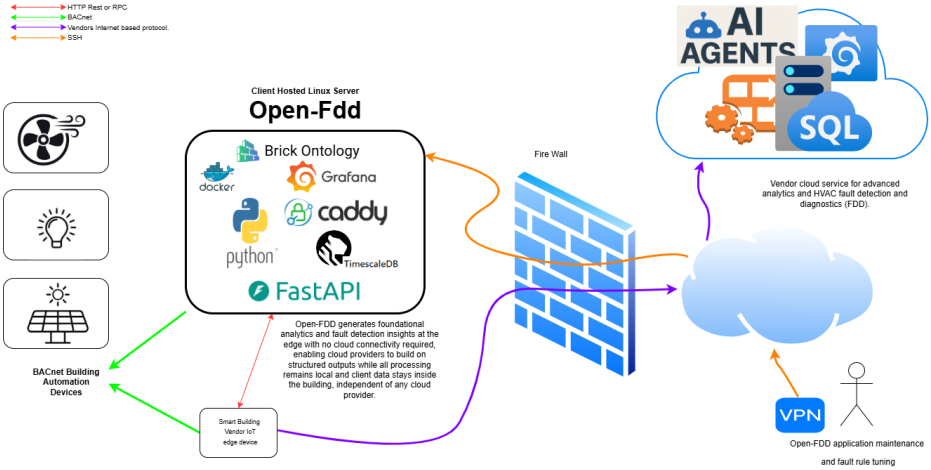
4 System Overview

This repository ships the **`open-fdd`** Python package on **PyPI**: YAML-defined **FDD rules on pandas** (`open_fdd.engine`), plus schema and reporting helpers. That engine can run **inside your own application**, inside **VOLTTRON agents**, or beside optional **FastAPI + React** when you run them from this monorepo. **Docker Compose** here is intentionally slim: **Postgres/TimescaleDB** (+ optional Grafana/Mosquitto), not field-protocol services.

The text below describes the **platform** in `afdd_stack/` within the **open-fdd monorepo**—knowledge graph, services, and data flow. **Open-FDD does not host BACnet, Modbus, or other OT buses.** Those live in **per-building VOLTTRON** deployments; telemetry reaches this stack through **SQL** (historian, ETL) and **identity mapping** (`external_id`, points). For engine-only usage, see **Engine-only / IoT** and **Getting started**.

Read first: Site VOLTTRON and the data plane (ZMQ) — ZMQ VIP / pub-sub on the VOLTTRON bus (not RabbitMQ in the reference design), optional cloud or on-prem for the app tier, same integration contract.

4.1 Architecture



Open-FDD Edge Platform Architecture

Integrators may run Open-FDD **beside** client-hosted services or pull from the local API over the LAN. **Field collection** is always **site-local VOLTTRON** (drivers, proxies, historians) on the OT network; the **semantic and FDD tier** reads **Postgres** and optional **FastAPI** routes.

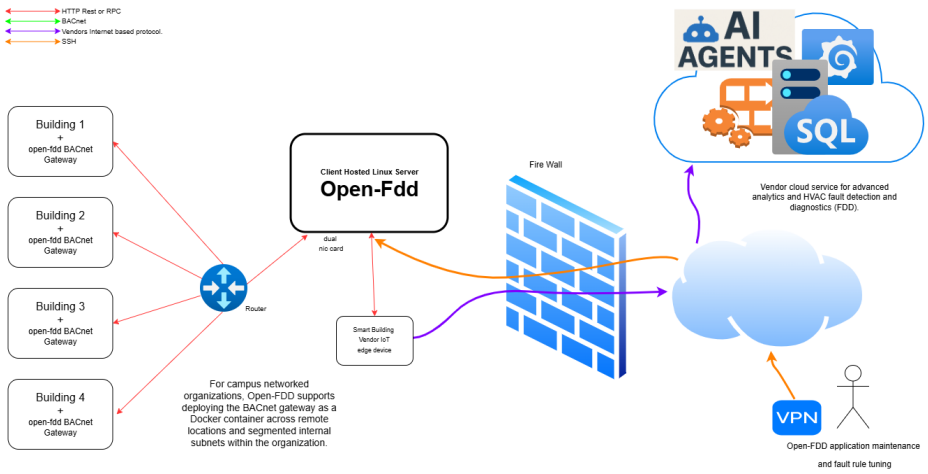
4.2 Services and components

Component	Description
VOLTTRON (per site)	**Required for field data:** platform driver, BACnet/Modbus **inside VOLTTRON only**, pub/sub over **ZMQ**, SQL historian. **VOLTTRON Central** for fleet UI — upstream docs.
TimescaleDB	

Component	Description
	PostgreSQL with TimescaleDB extension in compose. Open-FDD metadata + time-series tables; historian tables use VOLTTRON's **`tables_def`** when you colocate on one server.
FastAPI (optional, from source)	CRUD, data-model export/import, TTL, SPARQL, jobs. **Legacy** `/bacnet/*` proxy routes exist only if you run a separate lab gateway; they are **not** the default ingest path.
React (optional, from source)	Dashboard under `afdd_stack/frontend/` for modeling and plots.
Grafana (optional profile)	Pre-provisioned TimescaleDB datasource when you enable the **`grafana`** compose profile. Grafana SQL cookbook.

Removed from default Compose: see **`afdd_stack/legacy/README.md`** (Caddy, API/frontend containers, diy-bacnet-server, BACnet scraper, weather scraper, FDD loop container). Do not resurrect them for **new** BACnet/Modbus ingest.

4.3 Multi-building and "central" deployments



Open-FDD campus schematic

Each building runs **VOLTTRON** on the local LAN. Historians (or agents) write **SQL** toward a database you designate—on the edge appliance, a central VM, or a managed cloud Postgres **with**

appropriate network security. Open-FDD keeps **one semantic model** and fault pipeline keyed by **site** and **points**; it does **not** require raw BACnet to reach the app tier.

4.4 Data flow

1. **Ingestion: VOLTTRON** at the site writes time series into **Postgres/Timescale** (historian + `tables_def`). Align topic/point identity with Open-FDD **`points`** / **`external_id`** (VOLTTRON gateway and sync, **`openfdd_stack.volttron_bridge`**).
2. **Data model:** Brick (sites, equipment, points), platform config, optional RDF from **import** or legacy tooling — CRUD and SPARQL when FastAPI runs; TTL at `config/data_model.ttl`.
3. **FDD:** Rules run on pandas — **VOLTTRON agents**, scheduled jobs, or **`open_fdd.engine`** on DataFrames. **`fault_results`** remain the canonical output table.
4. **Visualization:** Grafana (optional), **VOLTTRON Central**, or React when you run it.

4.5 Ways to deploy

- **VOLTTRON + SQL (default):** **`./afdd_stack/scripts/bootstrap.sh`** — **volttron-docker**, optional **`--compose-db`** for Timescale + Open-FDD SQL init only.
- **SQL only:** `docker compose -f afdd_stack/stack/docker-compose.yml up -d` — **TimescaleDB** + init SQL. **`afdd_stack/legacy/README.md`** lists removed services.
- **Engine only:** `pip install open-fdd` and `RuleRunner` on pandas — **Engine-only / IoT**.
- **Manual / custom:** your processes; same rule YAML and `open_fdd.engine` APIs.

4.6 Key concepts

- **Sites** — Buildings or facilities.
- **Equipment** — Devices (AHUs, VAVs, heat pumps). Belong to a site.
- **Points** — Time-series references. **`external_id`** aligns with VOLTTRON historian / topic naming; **`rule_input`** for FDD columns; optional **`brick_type`**.
- **Fault rules** — YAML (bounds, flatline, hunting, expression). See Fault rules for HVAC.
- **Unified graph** — Brick + platform config (+ optional RDF). SPARQL and TTL when FastAPI runs.

5 Column map & resolvers

6 Column map & resolvers

YAML rules name their **`inputs`** with **logical keys**—often **Brick** class names (e.g. `Supply_Air_Temperature_Sensor`) on the stack, but the same mechanism supports **Haystack-style slugs**, **DBO** type names, or **223P-scoped** identifiers if those strings match your map. Your **`pandas`** frame uses **your** column names (`sat`, `AHU1_SupplyTemp`, …). See Expression Rule Cookbook — ontology labels.

`column_map` is the bridge: `dict[str, str]` where each **key** is a **logical label** (Brick class, or a string from the rule's optional **`haystack`** / **`dbo`** / **`s223`** / **`223p`** fields on that input), and each **value** is the **actual `DataFrame` column name**. `RuleRunner` tries those fields in order until one matches a map key; see Expression Rule Cookbook.

Pass it to **`RuleRunner.run(..., column_map=...)`**. You can build that dict by hand, load it from a manifest, or **merge** several sources with a composite resolver. Deriving the map from a Brick **`.ttl`** file is done on the **AFDD stack** (rdflib), not in the **`open-fdd`** package.

6.1 1. Plain dict (most IoT / warehouse flows)

```
runner.run(
    df,
    column_map={
        "Supply_Air_Temperature_Sensor": "sat",
        "Outside_Air_Temperature_Sensor": "oat",
    },
)
```

No TTL file required if you already know the mapping. Keys must match the **input names** used in your rule YAML (see Fault rules overview).

6.2 2. Brick TTL (AFDD stack only)

If you run the **Docker AFDD platform**, **`fdd-loop`** builds **`column_map`** from **`config/data_model.ttl`** using **`BrickTtlColumnMapResolver`** in **`openfdd_stack.platform.brick_ttl_resolver`** (depends on **rdflib** / **pyparsing**, declared in the stack's **`pyproject.toml`** — not in **`open-fdd`**).

For **engine-only** notebooks and pipelines, use a **manifest** (below) or a **plain dict** that lists the same logical keys your rules use.

6.3 3. Manifest file (`ManifestColumnMapResolver`)

JSON or **YAML**: either a flat `logical_name → column_name` object, or a top-level **`column_map:`** nested mapping.

```
from pathlib import Path
from open_fdd.engine.column_map_resolver import
ManifestColumnMapResolver

r = ManifestColumnMapResolver(Path("my_mapping.yaml"))
runner.run(df, column_map=r.build_column_map(ttl_path=Path(".")))
```

For a raw dict without the resolver class, use **`load_column_map_manifest(path)`** from the same module.

6.4 4. Composite priority (`FirstWinsCompositeResolver`)

Run multiple resolvers **in order**; for each logical key, the **first resolver that defines it wins** (for example: a base manifest, then an override file that only adds missing keys). On the stack you can compose the stack's **`BrickTtlColumnMapResolver`** with a **`ManifestColumnMapResolver`** in your own startup code.

```
from pathlib import Path
from open_fdd.engine.column_map_resolver import (
    FirstWinsCompositeResolver,
    ManifestColumnMapResolver,
)

composite = FirstWinsCompositeResolver(
    ManifestColumnMapResolver("base.yaml"),
    ManifestColumnMapResolver("extras.yaml"),
)
```

```
        runner.run(df,
column_map=composite.build_column_map(ttl_path=Path(".")))
```

Security: There is **no** env-driven “import a resolver class by string” — compose resolvers in code or a small startup script.

6.5 Examples in the repo

Minimal workshop — **`examples/column_map_resolver_workshop/`** (folder on GitHub):

Resource	Link
Run	`simple_ontology_demo.py`
Rule YAML (cookbook-style `inputs`)	`simple_ontology_rule.yaml`
Walkthrough	`README.md`
Top-level index	`examples/README.md`

AHU / RTU notebooks and sample rules — **`examples/AHU/`** (folder on GitHub):

Resource	Link
Sample YAML rules	`rules/sensor_bounds.yaml`, `rules/sensor_flatline.yaml`, `rules/sat_operating_band.yaml`
Notebooks	`RTU11_standardized_refactored.ipynb`, `RTU7_standardized_refactored.ipynb`, `RTU7_machine_learning.ipynb`
Sample CSV	`RTU11.csv`, `AHU7.csv`
Helpers	`openfdd_notebook_helpers_v2.py`

More narrative (Docker **`--mode engine`**, IoT pipelines, **`column_map` policy**): Engine-only & IoT pipelines. Curated doc links: Examples (repository).

6.6 See also

- Getting started — **`RuleRunner`** + **`column_map`** in one example
- Engine-only & IoT pipelines — integration patterns and resolver priority
- Fault rules overview — YAML **`inputs`** and Brick-oriented keys
- Expression rule cookbook — expression rules (same keying model)

- Examples (repository) — notebooks and CSVs under **`examples/AHU/`**

7 Getting Started

8 Getting Started

This page covers **prerequisites** and the **bootstrap script**: how to get the Open-FDD platform running. For configuration, data modeling, and rules, see the Documentation index.

8.1 Do this to bootstrap

1. **Prerequisites: Git** and **Docker** (including Compose v2). **Python 3** on the host for monorepo dev, **`--smoke-fdd-loop`**, and tooling — the bootstrap script no longer installs VOLTTRON in a host venv; use **volttron-docker** for the platform.

2. **Clone and run bootstrap** (from repo root):

```
git clone https://github.com/bbartling/open-fdd.git
cd open-fdd
./afdd_stack/scripts/bootstrap.sh --help
./afdd_stack/scripts/bootstrap.sh --doctor
```

Clone directory: matches the GitHub repo name **`open-fdd`**. If `cd` fails, list your home directory — a common mistake is cloning correctly but typing an **extra** hyphen in the folder name (see the exact spelling in the clone URL above).

VOLTTRON Central via Docker (when you have Docker):

```
./afdd_stack/scripts/bootstrap.sh --volttron-docker
cd "${OFDD_VOLTTRON_DOCKER_DIR:-$HOME/volttron-docker}"
# Build the image and run docker compose per that repository's
README.
```

One-shot lab (Timescale + stubs + clone volttron-docker):

```
./afdd_stack/scripts/bootstrap.sh --central-lab
```

Run the test suite (like the legacy stack's `--test`):

```
./afdd_stack/scripts/bootstrap.sh --test
# If pytest is missing: OFDD_BOOTSTRAP_INSTALL_DEV=1 ./
```

```
afdd_stack/scripts/bootstrap.sh --test
        # Optional: eslint + tsc + vitest in afdd_stack/frontend
        # OFDD_BOOTSTRAP_FRONTEND_TEST=1 ./afdd_stack/scripts/
bootstrap.sh --test
```

3. **Optional local database** (faults / sites / points schema + historian-friendly Postgres):

```
./afdd_stack/scripts/bootstrap.sh --compose-db
```

BACnet, Modbus, and other OT protocols are not implemented in this application — they run **only inside each building's VOLTTRON** instance (drivers, proxies, **ZMQ** VIP/pub-sub). Historians write **SQL** that Open-FDD consumes. FDD execution and operator UI beyond Central are typically **VOLTTRON** or optional **FastAPI/React from source**; this compose file does not start those services.

8.2 External Agentic AI (OpenAI-compatible)

Open-FDD does not embed an LLM. Instead, external AI agents (for example an OpenAI-compatible tool like Open-Claw) can take advantage of Open-FDD by calling its APIs:

1. Export the current data model JSON: `GET /data-model/export`
2. Fetch documentation as model context: `GET /model-context/docs` (optionally with `query=...` / keyword retrieval)
3. Import tagged JSON back into the platform: `PUT /data-model/import`

Manual Data Model export/import (JSON) always works without any AI.

See Open-Claw integration and API Reference for endpoint details.

The optional **MCP RAG** sidecar and **`--with-mcp-rag`** bootstrap flag were **removed**; use **`GET /openapi.json`** / **`/docs`** and **`GET /model-context/docs`** for agents instead.

8.3 Prerequisites

- **OS:** Linux for VOLTTRON edge (e.g. Raspberry Pi). Maintainers primarily test on **x86_64** and **ARM64**. Bootstrap shell is not aimed at Windows. Keep the system updated:

```
sudo apt update && sudo apt upgrade -y
```

- **Docker and Docker Compose: Optional** for **`--compose-db`** (Timescale + init SQL) and **recommended** for **volttron-docker** (VOLTTRON Central in a container). See Docker install.

- **Python:** Use a venv for monorepo dev tests: `pip install -e ".[dev]"` (see root README). For **`--smoke-fdd-loop`**, install **`pip install -e ".[stack]"`** in that venv or set **`PYTHONPATH`** from **`--print-paths`**.

- **Troubleshooting:** `./afdd_stack/scripts/bootstrap.sh --doctor` checks **git**, **python3**, **Docker**, and **Compose**.

- **Git:** To clone the project:

```
git clone https://github.com/bbartling/open-fdd.git
cd open-fdd
```

- **Field data:** Configure **VOLTTRON** at the site (Platform Driver, historian, **ZMQ** message bus per upstream docs — **not** RabbitMQ in this project's reference design). See **Site VOLTTRON and the data plane (ZMQ)**. Legacy FastAPI `/bacnet/*` proxy routes are **not** the default ingest path.

8.4 What the bootstrap script does

`afdd_stack/scripts/bootstrap.sh` (run from the **repo root**):

1. **`--central-lab`** — runs **`--compose-db`**, waits for Postgres, writes **`$VOLTTRON_HOME`** stubs (**`--volttron-config-stub`**, **`--write-env-defaults`**, **`--write-logrotate`**), verifies the FDD schema, then **`--volttron-docker`** (clone/update **volttron-docker**). Afterward, build the image and **`docker compose up`** from **`$HOME/volttron-docker`** (see that repo's README).
2. **`--volttron-docker`** (alias **`--clone-volttron-docker`**) — clone or update **volttron-docker** into **`OFDD_VOLTTRON_DOCKER_DIR`** (default **`$HOME/volttron-docker`**).
3. **`--compose-db`** — optional `docker compose … up -d db` for Timescale + `stack/sql` init.
4. **`--print-paths`** — prints `PYTHONPATH` so host tools can `import openfdd_stack`.
5. **`--doctor`** — checks **git**, **python3**, **docker**, **docker compose**, and the volttron-docker checkout path.
6. **`--test`** — runs **`pytest`** on `open_fdd/tests` and `afdd_stack/openfdd_stack/tests` from the repo root. Set **`OFDD_BOOTSTRAP_INSTALL_DEV=1`** for a one-shot **`pip install -e ".[dev]"`** before pytest. Set **`OFDD_BOOTSTRAP_FRONTEND_TEST=1`** to also run **`npm ci`**, **`npm run lint`**, **`npm run build`**, and **`npm run test`** under

`afdd_stack/frontend`. Optional **`OFDD_PYTEST_ARGS`** adds extra pytest flags (shell word-splitting applies).

Bootstrap options: Run `./afdd_stack/scripts/bootstrap.sh --help`. The old “one command brings up API + Caddy + field-bus containers in Docker” flow is **removed**; see **`afdd_stack/legacy/README.md`** for what changed.

Open-FDD API from source: If you run **`uvicorn`** for local development, configure secrets and auth in **`afdd_stack/stack/.env`** as described under **Security — authentication** (API keys, JWT, dashboard user). That file is **not** populated by the slim bootstrap; compose no longer starts the API container.

Optional Grafana: With Docker available, from `afdd_stack/stack/` you can start the **`grafana`** profile (see compose file comments) for dashboards against the same Postgres/Timescale host; use the Grafana SQL cookbook.

VOLTTRON Central / edge UI: Use upstream VOLTTRON and Central docs for remote access, TLS, and operator UI — not Caddy from this repo.

8.5 After bootstrap

- **Database:** With **`--compose-db`**, Postgres listens on the host port defined in compose (see `afdd_stack/stack/docker-compose.yml`). Apply VOLTTRON SQL historian **`tables_def`** against the same server if you colocate historian tables with Open-FDD schema; keep backups and migrations explicit.
- **Agents:** Point **`PYTHONPATH`** at this repo (**`--print-paths`**) when packaging Open-FDD logic as VOLTTRON agents.

8.6 Next steps

- **How-to Guides** — Grafana (optional) and SQL cookbook.
- **Configuration** — Platform config, rule YAML.
- **Security** — API keys and JWT when running the FastAPI app from source.
- **Appendix: API Reference** — REST endpoints for agents and tooling.
- **VOLTTRON gateway and sync** — how field data and SQL fit together.
- **VOLTTRON Central and AFDD parity (monorepo)** — phased plan (one DB, mapping, FDD runner, Brick UI, multi-site) aligned with Central + **volttron-docker**.

For data modeling and fault rules: Data modeling, Fault rules for HVAC. **Data model export/import (JSON)** works without any AI—you can always export, tag manually or with an external LLM, and import.

9 Using the React dashboard

10 Using the React dashboard

The Open-FDD **React frontend** is available at **http://localhost:5173** (direct) or **http://localhost/** when **Caddy** is the entry point (see Security). Use it after Getting started. When auth is enabled, sign in at **`/login`** (JWT in the browser session + HttpOnly refresh cookie; see Security — authentication). Docker Compose defaults to **`VITE_API_BASE=/api`** so the UI talks to the API through Caddy on the same host/port.

10.1 Navigation

Page	Purpose
Overview	High-level summary: site selector, FDD/weather status, links into the app. The UI does not embed or run an LLM. **External agents** should follow the **help links** on Overview (published docs, PDF, LLM workflow, Open-Claw integration) rather than duplicating endpoint prose in the UI.
OpenFDD Config	View and edit platform config (GET/PUT `/config`): FDD rule interval, lookback, BACnet, Open-Meteo, **`graph_sync_interval_min`** (how often the API writes the merged graph—and Brick `ref`: from the DB—to `data_model.ttl`). **Overview** does not edit this; use **OpenFDD Config** → form → Save. Current settings are shown in a read-only summary first.
BACnet tools	**Step 1 — Sites:** create or delete sites (same API as before; moved here from the Data model page). **BACnet** tab: **Step 2 — BACnet discovery** (gateway selector, Who-Is,

Page	Purpose
	point discovery, **Add to data model**) and optional read/write RPC. **Modbus client** tab: Modbus TCP test bench (proxied through the same gateway) and **Add rows to data model** with per-point `modbus_config`. Modbus reads follow the point's **polling** flag and the platform's shared scrape cycle (see `openfdd_stack/platform/drivers/modbus_tcp.py`); they do **not** use BACnet's separate interval-based discovery loop.
Points	List points (optionally by site). Shows external ID, equipment, Brick type, FDD input, unit, polling, last value and time. Optional BACnet panel for discovery from this page; primary discovery flow is **BACnet tools**.
Data model	Equipment tree, points, export/import, TTL view, SPARQL. Edit equipment and points; run data-model export/import and SPARQL. **Sites** are created on **BACnet tools** (Step 1). Includes a **Danger zone** at the bottom (see below).
Energy Engineering	Per-**site** energy / savings calculation specs: **tree by equipment** (site-level bucket + each device), **row actions (⋮)** or **right-click** a calc to enable / disable / delete, **Seed default penalty catalog (18)** (disabled rows from energy penalty equations), **Download export JSON** and **Apply import** for the same LLM loop as Data Model BRICK (AI-assisted energy calculations). Form + preview for predefined calc types; **Equipment metadata** tab (energy-first sizing fields; legacy controls/docs under Advanced). TTL: `ofdd:EnergyCalculation`, optional `ofdd:penaltyCatalogSeq`, `brick:isPartOf` the site.
Analytics	Placeholder for future trend dashboards and fault-duration analytics; points users to Energy Engineering for site-specific savings assumptions today.

Page	Purpose
Faults	Active fault states and definitions. Filter by site/equipment.
Plots	CSV Plotter workbench (Plotly-style). Load Open-FDD export CSV by site/range/points or drag-drop any CSV; choose X and multiple Y columns; toggle lines/points/both; optionally overlay faults and export CSV joined with `fault_*` 0/1 signals.
Web weather	Open-Meteo weather charts (temp, RH, wind, radiation, etc.) when weather is enabled.
System resources	Host and container metrics (when host-stats is running): memory, load, disk, **per-container CPU/memory** (table + time-series charts). Status badges (green/yellow/red) indicate resource pressure.
Stack status	Overview shows API, BACnet gateway, and MQTT bridge status with **green / yellow / red** indicators (e.g. MQTT bridge connected = green, enabled but disconnected = yellow).

10.2 Common workflows

- **Change platform config:** OpenFDD Config → edit fields → Save. Changes take effect on the next FDD run or scraper cycle.
- **Discover BACnet points: BACnet tools** → Step 1 (site if needed) → Step 2 (Who-Is, point discovery, Add to data model). You can also use the BACnet panel on **Points**.
- **Inspect or edit the data model:** Data model → browse equipment and points, export/import, view TTL, run SPARQL.
- **Plot and explore CSV:** Plots → choose source (**Open-FDD** or **Upload CSV**) → load/drop file → pick X and Y columns → chart. Optional: select faults and export CSV joined with fault activity columns (`fault_<fault_id>`).
- **Check faults:** Faults → see active faults and definitions; combine with Plots to correlate with sensor data.

For API integration (curl, scripts), see Appendix: API Reference and Swagger at http://localhost:8000/docs.

10.3 Energy Engineering (LLM export / import)

Select a **site** in the header, then open **Energy Engineering → Energy calculations**. The **tree** lists **Site-level** calcs and one folder per equipment asset; **right-click** a calculation to enable, disable, or delete it. Use **Seed default penalty catalog** to insert 18 disabled engineering narratives (energy penalty equations). Use **Download export JSON** to produce the same bundle as `GET /energy-calculations/export` (includes embedded **`calc_types`** and **`penalty_catalog`**). Paste the LLM's **`energy_calculations`** array (or full export object) into **Apply import** — equivalent to `PUT /energy-calculations/import`. Step-by-step and copy-paste prompt: AI-assisted energy calculations.

10.4 Data model Brick: danger zone

The **Data model** page ends with a **Danger zone** card. There are two separate actions (lower risk first in the UI, maximum risk last):

Tier	UI label	What it does
Lower risk	Reset graph to DB-only	Calls `POST /data-model/reset` only. Clears the in-memory RDF graph, removes BACnet discovery triples and orphan blank nodes, rebuilds Brick triples from the **current** Postgres state, and rewrites `config/data_model.ttl`. **Does not** delete sites, equipment, or points. Use on a test bench when discovery left stale BACnet RDF but you want to keep DB rows. Use **Check integrity** on the same page if you need orphan warnings (reset does not run that check).
Maximum risk	Remove all sites and reset graph	Deletes **every site** via `DELETE /sites/{id}` (cascade removes equipment, points, and related data), then calls `POST /data-model/reset`. **Irreversible** data loss for that relational model (including time-series tied to

Tier	UI label	What it does
		those points), not “just” clearing the TTL file.

How this fits: Postgres holds the authoritative sites/equipment/points (including `external_id` for time-series columns). The in-memory graph merges Brick (from DB) with BACnet discovery RDF. The TTL file on disk is a snapshot of that graph; the API also persists periodically and on import/reset.

For the same **help links** as the Overview page (GitHub Pages, PDF, LLM prompt, Open-Claw integration), see README — Online Documentation in the repo.

11 Modular Architecture

12 Modular Architecture

Open-FDD separates concerns into **collector**, **model**, **engine**, and **interface** modules. The **default bootstrap** in this repo is **VOLTTRON-first** (`./afdd_stack/scripts/bootstrap.sh`) and no longer maps those modules to **`docker compose` services** the way the old full stack did.

12.1 Module contracts (conceptual)

Module	Concern	Typical runtime today
Collector	Field protocols + historian (**ZMQ** bus in VOLTTRON, not RabbitMQ here)	**Per-site VOLTTRON** platform driver + SQL historian; optional ETL into `timeseries_readings`
Model	Brick / data-model CRUD, SPARQL, TTL	**FastAPI** (+ optional React) **from source** when you need the semantic layer
Engine		**VOLTTRON agents** or **`open_fdd.engine`** on DataFrames;

Module	Concern	Typical runtime today
	Pandas / YAML FDD execution	historical **`fdd-loop`** container removed from default compose
Interface	REST, OpenClaw-style HTTP, export/ import	**FastAPI** when running **`uvicorn`**; **VOLTTRON Central** for fleet/edge UI

12.2 Legacy Docker mode matrix

The one-command **`./scripts/bootstrap.sh --mode …`** matrix applied when Compose shipped **`api`**, **`bacnet-scraper`**, **`fdd-loop`**, **Caddy**, etc. That stack is **removed** from the default `afdd_stack/stack/docker-compose.yml`. For a concise list of what disappeared and how to run **SQL-only** compose, see **`afdd_stack/legacy/README.md`**.

If you maintain a **private fork** or custom compose that restores those services, you can still think in terms of:

- **Collector slice:** DB + **site VOLTTRON** historians (SQL) + topic mapping
- **Model slice:** DB + API + frontend + reverse proxy
- **Engine slice:** DB + scheduled FDD + optional weather

… but upstream Open-FDD docs assume **VOLTTRON** owns the collector slice for new deployments.

12.3 Feature coverage (conceptual)

Feature	Collector (VOLTTRON)	Model (FastAPI+React)	Engine (agents / library)
Time-series ingest	yes (historian / ETL)	no	reads DB or frames
CRUD + SPARQL	no	yes (when API up)	no
React / Central UI	Central (edge)	dev server / build	—
FDD rules	via agents	via API/jobs if enabled	`RuleRunner`

12.4 Migration stance

The repository stays a **single codebase**: `open_fdd` on PyPI, `openfdd_stack` for platform and VOLTTRON helpers. Module boundaries help you choose **which processes to run** without implying every process still has a matching container in this repo.

12.5 External IoT / “we already have collection + modeling”

If another system owns **ingestion** and **semantic modeling**, use `open_fdd.engine` on **pandas** with the **same YAML** as the platform, or wire a thin agent to SQL. See Engine-only deployment and external IoT pipelines. For `openfdd_engine` vs `open_fdd.engine`, see The optional openfdd-engine package.

13 Edge field buses (VOLTTRON)

14 Edge field buses (VOLTTRON)

Open-FDD does not implement BACnet, Modbus, or other field buses. Those protocols run **only inside each building’s VOLTTRON** deployment (platform driver, BACnet proxy, Modbus devices, etc., per **your** VOLTTRON configuration). Telemetry reaches Open-FDD through **SQL** (historian, ETL) and **topic identity** aligned with `points` / `external_id`—not through this repository hosting OT traffic.

Message bus: VOLTTRON’s default inter-platform path is **ZeroMQ (ZMQ)** VIP and pub/sub as described in upstream docs. **RabbitMQ is not** part of Open-FDD’s reference architecture; if you add AMQP for a custom integration, treat it as **out of scope** for these docs.

Where to read next

- **Site VOLTTRON and the data plane (ZMQ)** — canonical on-prem vs cloud + ZMQ framing.
- **VOLTTRON gateway, FastAPI, and data-model sync** — SQL, optional API, Central.

- **VOLTTRON Central and AFDD parity** — bootstrap and historian colocation.

Legacy material (removed from navigation)

Older releases documented **diy-bacnet-server**, **bacnet-scraper**, and FastAPI **JSON-RPC proxy** flows. Those containers are **not** started by default Compose anymore. Historical detail lives only in **`afdd_stack/legacy/README.md`** and in **git history** for deleted pages (`overview`, `fault_verification`, `graph_context`, `gateway_rpc_contract`). Do not plan new deployments against that path unless you maintain a **private fork**.

Upstream field stack: VOLTTRON readthedocs.

15 Overview

16 Data model flow

Open-FDD uses a **unified graph**: one semantic model that combines Brick (sites, equipment, points), **optional** BACnet-shaped or other RDF from **imports / legacy tooling**, platform config, and **engineering / 223-style topology** (`ofdd:*` schedule fields and `s223:*` connection patterns from **Data Model Engineering**). **Telemetry** arrives from **VOLTTRON → SQL**; CRUD and import align **`external_id`** with those readings. **Backend SPARQL** runs over the in-memory graph (rdflib). Rules resolve inputs via `ofdd:mapsToRuleInput` in the TTL. See Data model engineering (Brick + 223P MVP) and **Site VOLTTRON and the data plane (ZMQ)**.

16.1 Flow

```
Sites + Equipment + Points (DB)  ← single source of truth
  └─ equipment.metadata.engineering (optional JSON → ofdd/s223 in
TTL)
         │
         ▼
  Data-model export / CRUD
         │
         ▼
  Brick TTL (config/data_model.ttl)  ← Brick section reserialized
on every create/update/delete; same file may include optional
imported RDF + engineering (one file for SPARQL)
         │
```

```
        ▼
  FDD column_map (external_id → rule_input)
        │
        ▼
  RuleRunner
```

CRUD and Brick TTL sync: The database is the single source of truth. Every **create**, **update**, or **delete** on **sites**, **equipment**, or **points** (via API or data-model import) triggers a reserialize: the Brick TTL file (`config/data_model.ttl`, or `OFDD_BRICK_TTL_PATH`) is regenerated from the current DB and written to disk. So the Brick model is always in sync with CRUD. Deleting a site, device (equipment), or point also cascades to dependent data (timeseries, fault results, etc.) as in a typical CRUD app; see Danger zone — CRUD deletes.

16.2 TTL structure

```
:oat_sensor a brick:Outside_Air_Temperature_Sensor ;
    rdfs:label "OAT (°F)" ;
    ref:hasExternalReference [
        a ref:TimeseriesReference ;
        ref:hasTimeseriesId "oat_sensor" ;
        ref:storedAt "postgresql://localhost:5432/openfdd/
timeseries_readings"
    ] ;
    brick:isPointOf :ahu_7 ;
    ofdd:mapsToRuleInput "oat" .
```

- Brick classes define sensor types
- `ofdd:mapsToRuleInput` maps to FDD DataFrame columns
- `ref:hasExternalReference` links points to BACnet/timeseries external systems
- `rdfs:label` for display

16.3 Data modeling process (discover → tag → import → validate)

1. **Discover** — BACnet discovery and/or manual entry populate **sites**, **equipment**, and **points** in the DB.
2. **Export** — Use **GET /data-model/export** (or the Export card on the Data Model Protocols page) to get JSON for tagging.
3. **Tag** — Either **manual**: copy JSON → use an external LLM or human → paste back (or automate with an external agent like Open-Claw). If your agent needs platform documentation as context, fetch it from `GET /model-context/docs`.

4. **Import — PUT /data-model/import** (or the Import card / auto-import from the agent) writes tagged points and optional equipment relationships into the DB and reserializes the Brick TTL.
5. **Validate** — Use the **Data Model Testing** page (SPARQL, “Summarize your HVAC”) to confirm the model; treat the result as pass or fail.

16.4 Data-model API

Endpoint	Description
`GET /data-model/export`	Single export route: BACnet discovery + DB points (optional `?bacnet_only=true`, `?site_id=...`). Use for AI-assisted tagging.
`PUT /data-model/import`	Import JSON: **points** (required) and optional **equipment** (feeds/fed_by). Creates/updates points; does not accept sites or equipments.
`GET /data-model/ttl`	Generate Brick TTL from DB (and in-memory BACnet graph). Optional `?save=true` to write to file.
`POST /data-model/sparql`	Run SPARQL query against the current data model.

16.5 Validation

Use SPARQL to validate:

- All rule inputs have `ofdd:mapsToRuleInput`
- Equipment types and points are consistent
- Brick schema compliance (optional)

17 Sites

18 Sites

Sites represent buildings or facilities. Equipment and points are scoped to a site.

18.1 Structure

Field	Description
`id`	UUID primary key
`name`	Display name

18.2 Scoping

All time-series and equipment queries can be filtered by `site_id`. Grafana dashboards typically scope by site.

18.3 API

- `GET /sites` — List sites
- `GET /sites/{id}` — Get one
- `POST /sites` — Create
- `PATCH /sites/{id}` — Update
- `DELETE /sites/{id}` — Delete (cascades to equipment, points, timeseries, fault results; see Danger zone)

19 Equipment

20 Equipment

Equipment represents physical devices or systems (AHUs, VAVs, heat pumps, weather stations). Points may belong to equipment.

20.1 Structure

Field	Description
`id`	UUID primary key
`site_id`	FK to sites
`name`	Display name (e.g. “AHU-7”)
`equipment_type`	Optional type string
`external_id`	Optional raw identifier from source

20.2 Hierarchy

```
Site
 └── Equipment (AHU, VAV, etc.)
      └── Points (sensors, setpoints)
```

20.3 API

- `GET /equipment` — List equipment (filter by site)
- `GET /equipment/{id}` — Get one
- `POST /equipment` — Create
- `PATCH /equipment/{id}` — Update
- `DELETE /equipment/{id}` — Delete (cascades to points and their timeseries; see Danger zone)

21 AI-assisted data modeling

22 AI-assisted Brick tagging

Related (LLM + engineering): LLM workflow — copy/paste prompt, AI-assisted energy calculations, Data model engineering (Brick + 223P), and Open Claw integration (HTTP agents plus **section 1f**: SSH operator for remote VOLTTRON edge → Central, with links to official VOLTTRON docs).

Open-FDD supports **AI-assisted data modeling**: use **GET /data-model/export** to dump **existing DB points** (and optional graph-backed fields) as JSON, then have an **LLM or human** add **Brick point classes** (`brick_type`), **Brick equipment classes** (`equipment_type` — best guess the operator confirms), rule inputs,

and equipment relationships — and send the result to **PUT /data-model/import**. **Live BACnet/Modbus ingest is not performed by Open-FDD**; site **VOLTTRON** historians populate `timeseries_readings` when `external_id` lines up. Tagging is how you make the **semantic model** rule-ready.

This workflow is intended for **mechanical engineers and building operators** who need the data model tagged so that FDD rules in `stack/rules/` have the correct inputs and only the right points are logged long-term.

22.1 Workflow (export → tag → import)

1. **Discover (optional / legacy lab only)** — If you still run a **separate** JSON-RPC gateway, you may use **POST /bacnet/*** routes to seed BACnet-shaped RDF. **Default path:** define **`external_id`** to match **VOLTTRON** historian output and use CRUD/import without wire discovery from this app.

2. **Sites and equipment** — Create the building/site and equipment (AHUs, VAVs, zones) via **POST /sites** and **POST /equipment**. Keep `site_id` UUIDs from export when present. For equipment assignment on import, you can use `equipment_id` UUIDs or `equipment_name` (resolved/created under `site_id`).

3. **Export — GET /data-model/export** (or `?bacnet_only=true` to filter rows that still carry BACnet-style metadata). Returns a single JSON array. **`site_id` / `site_name` are pre-filled** when you pass **`?site_id=`** (UUID or site name), when the **Data model** UI has a site selected in the top bar, or when the database has **exactly one site** — so LLMs can assign `equipment_name` safely. With **multiple sites** and no `site_id` filter, those fields stay null until you scope the export.

4. **Tag** — Send the export JSON to an LLM or edit manually. For each point set:

   - **site_id**, **external_id** (time-series key)
   - **brick_type** (e.g. `Supply_Air_Temperature_Sensor`, `Zone_Air_Temperature_Sensor`) — Brick **point** class
   - **equipment_type** (e.g. `Air_Handling_Unit`, `Variable_Air_Volume_Box`) — Brick **1.4 equipment** class local name when you set **equipment_name**, so **Data Model Testing** preset buttons (AHUs, VAVs, chillers, …) return meaningful counts after import. Use the **best defensible guess**; the operator corrects rare mistakes with one-click SPARQL checks. See LLM workflow canonical prompt and the stack repo file `frontend/src/data/brick-1.4-query-class-allowlist.ts`.
   - **rule_input** (name FDD rules use)

    - **unit** when known (e.g. `degF`, `%`, `cfm`, `0/1` for binary). Units are stored in the data model and TTL; the frontend uses them for Plots axis labels and grouping (e.g. temperatures on one axis, humidity on another). Use standard abbreviations so Plots and exports stay consistent.
    - **equipment_id** (optional) or **equipment_name** (optional, with site_id)
    - **polling: Legacy** flag from removed BACnet scraper flows; **VOLTTRON** logging is driven by historian config, not this field. You may still set it for forks that read export JSON.
    - For **equipment relationships** (Brick feeds/isFedBy), optional **engineering**, and **equipment_type** on named equipment, use the import **equipment** array. UUID-based and name-based forms are both supported; name-based relationship rows require `site_id`.

5. **Import — PUT /data-model/import** with body: **points** (array) and optional **equipment** (array for feeds/fed_by, **equipment_type**, **engineering**, etc.). The API accepts **only** these two keys — no `sites`, `equipments`, or `relationships`. The backend creates/updates points and equipment relationships, then rebuilds the RDF and TTL.

6. **Time series** — Ensure **VOLTTRON** (or your ETL) writes readings for each **`external_id`** you care about. Grafana and FDD both read the same SQL-backed model.

22.2 API contract (import)

- **points** (required): Each row is a point to **create** (omit `point_id`; set `site_id`, `external_id`, `bacnet_device_id`, `object_identifier`) or **update** (set `point_id` and fields to change). `site_id` should remain the export UUID when present. `equipment_id` is optional if `equipment_name` is provided with `site_id`.
- **equipment** (optional): Updates equipment with Brick feeds/isFedBy, **`equipment_type`** (Brick class for `rdf:type`), **`engineering`**, **`metadata`**. Rows may use UUID fields (`equipment_id`, `feeds_equipment_id`, `fed_by_equipment_id`) or name fields (`equipment_name`, `feeds`, `fed_by`) with `site_id` for resolution.
- **No other top-level keys:** The API model uses **`extra="forbid"`** — unknown keys such as `sites` or `relationships` produce **422 Unprocessable Entity** (they are not silently ignored).

Historian alignment: Prefer matching **GET /data-model/export** rows to the **topic / point names** your **site VOLTTRON** historian uses. See **Site VOLTTRON and the data plane (ZMQ)**.

22.3 LLM prompt and agent guidelines

Where this is documented: the tagging rules and agent flow are described **on this page** (you are in the right place) and summarized under **LLM tagging workflow** in the Technical reference. The README points here under **AI and data modeling**.

Canonical prompt text: The **full** copy-paste system prompt (points + equipment rules, strict JSON output, **fault-first pre-flight** for polling) is on the docs site under **LLM workflow — Copy/paste prompt template**. Published hub: **Data modeling** (same content as the repo after Pages build). You can save the same text to a local file (e.g. `pdf/canonical_llm_prompt.txt`) for agents; keep it in sync when you change instructions. The running API does **not** auto-inject this into chat UIs; load it yourself or point agents at the published section. For **HTTP model context**, use **`GET /model-context/docs`**, which serves **`pdf/open-fdd-docs.txt`** (or **`OFDD_DOCS_PATH`**)—regenerate that bundle after doc changes so agents see updates. Agents that can **fetch the web** may also load the modeling hub or the template anchor directly.

What the prompt must cover: It should be generic for **any site** (single building, campus, or tenant). The only input that changes per run is the export JSON from **GET /data-model/export** (optionally `?site_id=YourSiteName`). The LLM must preserve all fields, add **brick_type**, **rule_input**, **unit** (when known), and **polling**, and use equipment by name and **site_id** from the export. **Best practice:** the operator should state **which faults/rules** will run and supply **YAML (or snippets) where possible** before treating **polling** as final; the canonical prompt enforces that as **pre-flight / job context** or a **conservative draft**.

> **Ontology / column_map resolvers:** On the full stack, **`fdd-loop`** still resolves rule inputs from the **Brick** model in **`data_model.ttl`** by default (**`BrickTtlColumnMapResolver`** in **`openfdd_stack.platform.brick_ttl_resolver`**). Tagging with **`brick_type`** / **`rule_input`** is what keeps rules aligned with timeseries columns—you do not add a separate manifest to the LLM import JSON for that path. Optional **`ManifestColumnMapResolver`** / multi-ontology manifests are for library or custom pipelines; see the callout on LLM workflow.

For exact schema details and import body (points + equipment only), see the Technical reference. It defines the primary task (Brick tagging), the export → tag → import flow, polling semantics, equipment feeds, and the **unit** field (e.g. degrees-fahrenheit, percent) used by the frontend and stored in the RDF/TTL.

22.4 Rules for this project (what to send the LLM)

Start here for real jobs: which Open-FDD faults you will run, plus **actual YAML** for those rules when you have it — that yields the best **polling** and **rule_input** choices.

To have the LLM align **rule_input**, Brick types, **units**, and **polling** with your FDD rules, include:

- **Fault rules overview** — FDD rule types and YAML format.
- **Expression Rule Cookbook** — AHU, chiller, weather, and advanced recipes; **rule_input** examples (e.g. sat, rat, zone_temp, sf_status).
- Your project's **YAML rule files** (e.g. from `stack/rules/`) — Paste snippets so the LLM uses the same input names.

See LLM workflow (export + rules + validate → import) for the full one-shot flow and validating LLM output so it parses on the backend.

22.5 See also

- LLM workflow — One prompt + export JSON + optional rules; validate with schema/Pydantic; then import and run FDD/tests.
- AI-assisted energy calculations — **Next phase:** export/import JSON for site-scoped savings specs (`calc_types` + `energy_calculations`) after points and equipment exist. For **human-edited** nameplate / sizing fields (parallel to `equipment[].engineering` on import), use **Energy Engineering → Equipment metadata** in the UI or keep engineering in the same import JSON per Data model engineering.
- Data model overview — Flow (DB → TTL → FDD)

22.6 External agent integration (Open-Claw or any OpenAI-compatible LLM)

Where it runs: Outside Open-FDD (in your external LLM/Open-Claw environment). Open-FDD stays the single source of truth; your agent orchestrates `GET /data-model/export` → LLM tagging → `PUT /data-model/import`.

How it's invoked: Your external agent calls `GET /data-model/export` (optionally with a site filter), tags the export via Open-Claw (OpenAI-compatible) using the canonical prompt, then imports validated JSON with `PUT /data-model/import`. If validation fails, include the error text and retry (prompt chaining).

What the agent does: It uses the same API endpoints as the manual process: `GET /data-model/export` (optionally filtered by site) → LLM tagging using the canonical prompt → validate that the returned JSON matches Open-FDD's import schema → `PUT /data-model/import` (or return JSON for the engineer to import).

22.6.1 External retry loop (prompt chaining)

The agent is typically **not** a long-lived process; your external loop exports the model, tags it, validates the output, and retries when needed:

1. **Export** — Call `GET /data-model/export` (optionally filtered by site) and send the export JSON to your agent.
2. **System prompt** — Use the full tagging instructions (this page + Technical reference, or your local `pdf/canonical_llm_prompt.txt` if you maintain one) as the Open-Claw/OpenAI-compatible **system** prompt.
3. **User message** — Send the export JSON as the user message. Optionally prepend any engineer description of feeds/fed_by topology and assumptions so the LLM can choose correct Brick types and relationships.
4. **LLM call** — Call your Open-Claw/OpenAI-compatible LLM once (for typical payloads). For very large exports, your agent may split into chunks and merge results. Request JSON output (for example with `response_format={"type": "json_object"}`).
5. **Validation** — The response is parsed as JSON and validated with **DataModelImportBody** (same schema as import). If validation fails:
   - **Prompt chaining:** When validation fails, append the error text to the next attempt so the model can correct the JSON.
   - **Retry** — Retry up to a configurable max attempts in your agent.
6. **Agent log** — Optionally log attempts/outcomes so the engineer can inspect what changed between retries.
7. **Optional import** — Either return JSON for review, or immediately call `PUT /data-model/import` after validation.

So the agent uses **retry** with **prompt chaining**: keep the same export/rules, and after a validation failure include the error so the LLM can correct its output.

22.7 Automating tagging (external agents)

1. Call `GET /data-model/export` to get the export JSON (optionally with `?site_id=...`).
2. Fetch Open-FDD documentation context for the LLM: `GET /model-context/docs` (use `query=...` to retrieve relevant sections).
3. Prompt/tag with Open-Claw (OpenAI-compatible) to produce import JSON that matches the Open-FDD schema.
4. Validate and import:
   - either return JSON for the engineer to review, or
   - call `PUT /data-model/import` with the validated JSON.

22.8 Troubleshooting import failures (fixing your tagged JSON)

Sometimes the LLM returns valid JSON that passes schema validation, but `PUT /data-model/import` still rejects it during referential checks.

22.8.1 `site_id` must be a UUID

If the UI shows an error like:
`site_id must be a valid UUID ... Got: 'BensOffice'` it means a **UUID field** holds a human-readable value (often the site name was pasted into `points[].site_id`).

Human fix (pick the path that matches your row shape):

1. **Preferred:** Replace the bad value with the **real site UUID** from `GET /data-model/export` or `GET /sites` (same site the row belongs to). This is always safe for **updates** (`point_id` set) and for **creates**, and it keeps **name-based** `equipment[]` rows valid — those rows still **require** a proper `site_id` UUID alongside `equipment_name` / `feeds` / `fed_by`.
2. **Creates only (`point_id` omitted):** If the backend should resolve the site by name, you may set `site_id` to `null` (or remove the key) **and** set `site_name` to an existing site name (see import logic in the API). Do **not** use this as a blanket fix for every row: any row that uses **equipment by name** must keep a valid `site_id` UUID.
3. **Bad equipment or point UUIDs:** If the error is about `point_id` or `equipment_id`, remove or replace only those fields with IDs from your DB/export — don't strip `site_id` from unrelated rows.

If `PUT /data-model/import` rejects otherwise-valid JSON, re-run tagging with prompt chaining: include the import error text in your next LLM attempt so it can correct UUIDs/references, or return the JSON for a human to edit and import manually.

22.9 Possible extension: AI assist on Data Model Testing

A future **Data Model Testing** page could offer a second AI assist (same chat style as on Data Model Protocols): the engineer describes what they see (e.g. SPARQL summary, missing relationships, or test failure), and the model suggests **changes as import JSON** (points/equipment) so the engineer can apply and re-test. Under the hood this would use a **separate prompt** from the tagging prompt:

- **Input:** Current data model context (e.g. TTL snippet or SPARQL "Summarize your HVAC" result) plus the engineer's message (e.g. "Add feeds/fed-by between AHU-1 and VAV-1", "Fix the brick_type for SA-T").
- **Output:** Same schema as the tagging flow — valid **points** and **equipment** import JSON only — so the same **PUT /data-model/import** and validation path apply. The engineer reviews, applies, then re-runs SPARQL or predefined tests and passes/fails.

That way: **Setup** = tag from export (same instructions as above, optionally stored in `pdf/canonical_llm_prompt.txt`); **Testing** = revise from current model + chat (optional second file e.g. `pdf/canonical_llm_prompt_testing.txt`). Both use the same API and retry/prompt-chaining behavior; only the system prompt and user message differ.

- Appendix: API Reference — Data model export/import, CRUD
- Site VOLTTRON and the data plane (ZMQ) — ingest and `external_id` alignment
- Fault rules — Brick-driven rules in `stack/rules/`

23 Points

24 Points

Points are time-series references in the Open-FDD data model. They link **telemetry that already arrived in SQL** (historian / ETL keyed by `external_id`) to equipment and Brick semantics. **Open-FDD does not poll BACnet or Modbus**; site **VOLTTRON** does.

24.1 Structure

Each point has:

Field	Description
`id`	UUID primary key
`site_id`	FK to sites
`equipment_id`	FK to equipment (nullable)
`external_id`	Raw identifier from the **writer** (e.g. VOLTTRON historian topic / device path); unique per site
`fdd_input`	Column name used by FDD rules (e.g. `oat`, `sat`); data-model API uses `rule_input`
`brick_type`	Optional Brick class (e.g. `Outside_Air_Temperature_Sensor`)
`unit`	Optional unit of measure
`description`	Optional
Optional legacy metadata	
`bacnet_device_id`	Optional. May be filled from **imports** or historical tooling; **not** used by Open-FDD to poll BACnet.
`object_identifier`	Optional. Same — metadata only in the default architecture.
`object_name`	Optional. Human-readable name (often aligns with `external_id`).
`modbus_config`	Optional. **Legacy / lab** column; **ingest Modbus in VOLTTRON**, then map SQL rows to `external_id`.

24.2 Time-series data

Readings are stored in `timeseries_readings`:

Column	Type	Description
`point_id`	UUID	FK to points
`ts`	timestamp	Timestamp (UTC)
`value`	float	Numeric value

TimescaleDB hypertable, optimized for range scans and downsampling.

24.3 Layers and mapping

- **VOLTTRON / SQL layer (default):** Writers (historian, agents) insert rows into `timeseries_readings` keyed by `point_id` / `external_id`. Define points in CRUD or import so **`external_id`** matches what VOLTTRON publishes. Use **`openfdd_stack.volttron_bridge`**. See **Site VOLTTRON and the data plane (ZMQ)**.
- **Weather layer:** Points with `external_id` = `temp_f`, `rh_pct`, `dewpoint_f`, etc. come from the Open-Meteo weather fetch when enabled. They are linked to synthetic equipment **Open-Meteo** (type **Weather_Service**) per site. RDF marks `ofdd:dataSource "open_meteo"`.
- **Rule layer:** `fdd_input` / `rule_input` maps to DataFrame column names used by YAML rules.

The data-model API and Brick TTL coordinate `external_id` ↔ `rule_input` ↔ `brick_type`. Open-FDD may emit Brick external references for interoperability; **BACnet/Modbus on the wire remain in VOLTTRON**. See External representations.

24.3.1 Optional BACnet-style metadata

`bacnet_device_id` / `object_identifier` may still be populated for **imports, migration, or SPARQL examples**. They do **not** cause Open-FDD to poll BACnet in the default deployment. See **Edge field buses (VOLTTRON)**.

24.4 API

- `GET /points` — List points (filter by site or equipment)
- `GET /points/{id}` — Get one
- `POST /points` — Create

- `PATCH /points/{id}` — Update
- `DELETE /points/{id}` — Delete (cascades to timeseries_readings; see Danger zone)

24.5 Queries

```sql
SELECT ts, p.external_id, tr.value
FROM timeseries_readings tr
JOIN points p ON p.id = tr.point_id
WHERE p.external_id = 'temp_f'
  AND ts > NOW() - INTERVAL '1 day'
ORDER BY ts DESC;
```

25 LLM workflow (export + rules + validate → import)

26 LLM workflow: one prompt, export JSON, rules, and validated import

This page describes a **single upload** workflow for mechanical engineers: send the **canonical prompt**, the **data-model export JSON**, and **fault/rule context** (strongly recommended) to an LLM; get back import-ready JSON; **validate** it so you know it will parse on the Open-FDD backend; then **PUT /data-model/import** and run FDD (or Sparkl/tests) as needed.

After the data model is stable: use **AI-assisted energy calculations** for phase two — `GET /energy-calculations/export?site_id=…` (bundle includes **`calc_types`**, **`penalty_catalog`** (18 default narratives), and `energy_calculations`), LLM-authored or operator-edited rows, then `PUT /energy-calculations/import`. Optionally **seed default penalty rows** first. Same site-scoping rules as points and equipment.

Best practice: Decide **which Open-FDD faults and rules** you will run *before* finalizing `polling`. The canonical prompt is **fault-first**: it tells the model to gather job context (faults, YAML, units, production vs bench, weather scope) or to stay conservative on polling until that context exists.

> **Automated path available:** External OpenAI-compatible agents (for example Open Claw) can automate the same flow by calling `GET /data-model/export`, fetching platform documentation context from `GET /model-context/docs`, and then calling `PUT /data-model/import` with validated import JSON. For HTTP auth, split workspaces, and **SSH-based operator** steps (remote VOLTTRON edge → VOLTTRON Central), see **Open Claw integration** — especially **section 1f** there, which points at official VOLTTRON docs. The manual copy-paste workflow below always works too.

> **Web-connected agents:** If your LLM can fetch HTTPS documentation, point it at the published **Data modeling** hub and this page's template anchor **Copy/paste prompt template (recommended)** in addition to (or alongside) `GET /model-context/docs`, so instructions stay aligned with the live docs site.

> **Full AFDD stack vs column-map resolvers:** This workflow still targets **Brick** point classes (`brick_type`), **`rule_input`**, and related import fields because **rule YAML** on the platform uses **Brick-class logical names**. In **`fdd-loop`**, the default **`BrickTtlColumnMapResolver`** (**`openfdd_stack.platform.brick_ttl_resolver`**) builds **`column_map`** from **`config/data_model.ttl`** (same semantic model you enrich with import JSON). The LLM does **not** emit a separate "ontology manifest" for PyPI-style **`ManifestColumnMapResolver`** —that path is for **library / custom** pipelines (Engine-only deployment and external IoT pipelines, column map resolver workshop). If you ever run the loop with a **manifest-only** resolver, keep **rule YAML** and **`brick_type`** aligned with whatever logical keys your rules use.

26.1 What you upload to the LLM

1. **The canonical prompt** — Use **AI-assisted data modeling** (section *LLM prompt and agent guidelines*), the inline template **below** on this page ("Copy/paste prompt template"), or the Technical reference — LLM tagging workflow. You can copy from this page or keep an optional local mirror (e.g. `pdf/canonical_llm_prompt.txt`) for agents. The prompt must tell the LLM to return only `{"points": [...], "equipment": [...]}` with Brick types, rule_input slugs, equipment_name, feeds/fed_by, polling, and units (subject to the **pre-flight / job context** rules in the template).

2. **The export JSON** — From **GET /data-model/export** (optionally `?site_id=YourSiteName`). Each row includes point fields (`point_id`, `bacnet_device_id`, `object_identifier`, `object_name`, `external_id`, `site_id`, `site_name`, `equipment_id`, `equipment_name`,

`brick_type`, `rule_input`, `unit`, `polling`, …). Rows may also include **`engineering`** (and **`equipment_metadata`**) — a **per-equipment** mirror from the DB so the LLM sees rated/submittal context next to points. On **import**, write engineering updates under **`equipment[].engineering`**, not as ad-hoc point fields (see Data model engineering and `examples/223P_engineering/`). Unimported BACnet rows have `point_id: null` and null tagging fields until you import.

3. **Faults and rules for this job (strongly recommended)** — So the LLM can align **rule_input**, **Brick types**, **units**, and especially **polling** with what you will actually run in Open-FDD:
   - Paste **YAML** from your project's rules (e.g. from `stack/rules/` or your own rule files). **This is the best input for correct polling decisions** — the model can see exactly which point inputs each rule uses.
   - Or point the LLM at the **Fault rules overview** and **Expression Rule Cookbook** (AHU, chiller, weather, advanced recipes) and list which fault IDs or recipes apply.

> **Polling and YAML:** Without fault/rule context, many BACnet points may be tagged correctly but should still stay `polling: false`. The canonical prompt defaults to **asking for job context first** or emitting a **conservative draft** (mostly `polling: false`) until you provide which faults run and ideally the **actual rule YAML** for the job.

26.2 Copy/paste prompt template (recommended)

Use this as your LLM **system** or **developer** prompt when transforming `GET /data-model/export` into import JSON. This is the **canonical** copy-paste text for the published docs (save to a local file such as `pdf/canonical_llm_prompt.txt` if you want a path for agents or runbooks).

```
You are transforming Open-FDD export JSON into Open-FDD import
JSON.

I will paste JSON from:
GET /data-model/export?site_id=<site_id>

Your job is to return ONLY valid JSON with EXACTLY these two top-
level keys:

{
  "points": [...],
```

```
  "equipment": [...]
}
```

Do not return markdown.
Do not return explanations.
Do not return comments.
Do not return any extra top-level keys.

Exception for multi-turn chat: if fault/rule job context is still missing and you must ask pre-flight questions first, you may reply with plain-language questions only in that turn (see PRE-FLIGHT below). When you emit import JSON, the STRICT OUTPUT REQUIREMENTS at the end apply.

PRE-FLIGHT / JOB CONTEXT (required before final polling decisions)

Polling must be driven by the **actual faults and rules** the operator plans to run in Open-FDD — not by “this point looks generally useful.”

Before deciding final polling values, establish job context.

Required pre-flight question: What faults/rules are you going to run in Open-FDD?

Ask or confirm the following (plain language) if not already answered:

2. Can you provide the actual YAML rule files or rule snippets for this job?
3. Are units imperial or metric for this job?
4. Is this a production/live HVAC job or a bench/demo/test bench?
5. Do you want weather-related rules or weather polling included?
6. Should polling be limited to points required by the selected faults, or also include extra plotting/trending points the operator explicitly wants?

HARD RULE — if faults/rules context is missing:

- Either (a) ask the pre-flight questions above and wait (no import JSON in that turn), OR (b) return import JSON that is an explicitly **conservative draft**: set polling=false for every point unless it is clearly essential; do **not** enable broad polling coverage by guesswork.
- Do not treat aggressive polling=true choices as “final” until the operator has described which faults run and (ideally) supplied YAML or clear rule snippets.

If YAML rules or snippets **are** provided:

- Align brick_type, rule_input, unit, and polling with those rules' inputs.
- Set polling=true primarily for points **required** by the selected faults, plus any points the operator **explicitly** approved for plotting/trending.
- Do not turn on polling for unrelated "nice to have" points unless the operator asked for broader trending.

After faults/rules are known, you must:

- Infer which **Brick classes** and **rule_input** slugs those rules require (from YAML expressions and the Expression Rule Cookbook when YAML is thin). On the full Open-FDD stack, the engine maps those Brick names to trend/BACnet columns from the published data model (TTL), so keep **brick_type** and **rule_input** consistent with the YAML—not a separate hand-authored column_map in this JSON.
- Set **polling=true** for points that clearly supply those required inputs once tagged.
- Leave **polling=false** for unrelated points unless the operator explicitly asked for broader trending/plotting.

Example: If the operator runs only **sensor-bounds** and **sensor-flatline**-style temperature rules, BACnet **commands**, **setpoints**, **occupancy/schedules**, and **unrelated weather/Open-Meteo** points usually stay **polling=false** unless the operator asked to trend them or a named rule truly needs them.

POINT RULES

For each point in the input:

1. KEEP every existing field exactly as provided, including:
- point_id
- bacnet_device_id
- object_identifier
- object_name
- external_id
- site_id
- site_name
- equipment_id
- any other existing point fields present in the export

Note: the export may include **engineering** / **equipment_metadata** on each row (a mirror of **equipment**-level metadata). For **PUT /data-model/import**, rated/submittal engineering belongs under **equipment[].engineering** (see EQUIPMENT RULES). Do not add spurious extra keys on point objects if your validator rejects them; fold engineering into the **equipment** array once per **equipment_name** + **site_id**.

2. ADD or FILL these fields:
- brick_type (Brick **point** class — sensor, command, setpoint, …)
- rule_input
- polling
- unit
- equipment_name
- equipment_type (Brick **equipment** class — see §4 — powers Data Model Testing summary buttons)

3. brick_type:
Choose the best matching Brick class for the point.
Use a standard Brick class name such as:
- Supply_Air_Temperature_Sensor
- Return_Air_Temperature_Sensor
- Mixed_Air_Temperature_Sensor
- Zone_Air_Temperature_Sensor
- Damper_Position_Command
- Supply_Air_Flow_Sensor
- Static_Pressure_Sensor
- Occupancy_Command
Prefix with "brick:" only if the input already uses that style consistently. Otherwise omit the prefix.

4. equipment assignment:
Assign points to equipment by NAME ONLY.
Example:
- "equipment_name": "AHU-1"
- "equipment_name": "VAV-1"

Do NOT use equipment_id or any UUID for equipment relationships.

Conservative equipment_name inference:
- Set or change **equipment_name** only when **strongly supported** by BACnet **device** grouping, consistent **object_name** / **external_id** patterns, and any operator brief — not from a single ambiguous point name.
- If grouping is unclear, keep the export’s **equipment_name** / **equipment_id** relationship as-is rather than inventing AHUs/VAVs.

equipment_type (Brick 1.4 equipment class — best LLM guess, human verifies with one click):
- When you set **equipment_name** on points (or list equipment in **equipment[]**), also set **equipment_type** to the **most specific defensible Brick equipment class** as a **bare local name** (no `brick:` prefix), matching what Open-FDD writes to RDF for `rdf:type`.
- Goal: the operator can open **Data Model Testing → Summarize your HVAC** and use preset buttons (**AHUs**, **VAV boxes**, **Chillers**, **Central plant**, **Meters**, …) without re-

tagging. Those queries filter on Brick **1.4** classes such as:
  - **Air_Handling_Unit** — central air handler (supply/return/mixed air, fan, coils context)
  - **Variable_Air_Volume_Box** or **Variable_Air_Volume_Box_With_Reheat** — VAV / fan-powered terminal
  - **Chiller**, **Boiler**, **Cooling_Tower**, **Water_Pump**, **Heat_Exchanger**
  - **Chilled_Water_System**, **Condenser_Water_System**, **Hot_Water_System** — distribution systems when the row truly represents that system
  - **HVAC_Zone** — thermal / zoning context when points are zone-level only
  - **Building_Electrical_Meter** — building-level electrical meter equipment (not every kW sensor)
  - Generic fallback: **Equipment** (Open-FDD default) when class is unclear — better than mis-typing a VAV as a Chiller.
- **Same names as the UI presets** are maintained in this monorepo under [`afdd_stack/frontend/src/data/brick-1.4-query-class-allowlist.ts`](https://github.com/bbartling/open-fdd/blob/main/afdd_stack/frontend/src/data/brick-1.4-query-class-allowlist.ts) (regression test: [`data-model-testing-queries.brick.test.ts`](https://github.com/bbartling/open-fdd/blob/main/afdd_stack/frontend/src/data/data-model-testing-queries.brick.test.ts)).
- On **equipment[]** rows, include **equipment_type** whenever you include **equipment_name** + **site_id** so created/updated equipment gets the correct type before points attach.

5. unit:
Fill unit when known.
Use standard abbreviations consistent with the job units mode from pre-flight (imperial vs metric):
- temperature -> "degF" or "°F" (imperial) / "degC" or "°C" (metric) as appropriate
- percentage -> "percent" or "%"
- airflow -> "cfm" (imperial) or align with metric job conventions when stated
- binary / boolean -> "0/1"
- power -> "W"
- irradiance -> "W/m²"
Preserve unit strings already present in the export when they are consistent with the stated job mode; do not rewrite known-good values unless the operator asked for conversion.
If unknown, use null.

Conservative units (power, flow, energy):
- Do **not** force **unit** for ambiguous power, flow, or energy points (e.g. kW vs W, CFM vs percent of max, BTU vs MBH) without **operator confirmation** or an unambiguous BACnet engineering unit in context.
- Prefer **null** over a confident-looking guess that could mislead FDD and plots.

6. polling:
Set polling=true **primarily** for points required by the **specific faults/rules** the operator intends to run, plus any **operator-approved** plotting/trending points (see pre-flight question 6).
Set polling=false for points not needed by those rules or approved use cases.
Do **not** enable broad polling just because a point looks generally useful.
If fault/rule context has not been provided, follow the HARD RULE in PRE-FLIGHT (conservative draft or ask first).
If YAML or rule snippets were provided, align polling (and brick_type, rule_input, unit) with those rules.

7. rule_input:
Only populate rule_input when it is actually needed for disambiguation or explicit aliasing.

Use rule_input in these cases:
- two or more points in the same equipment or rule scope share the same Brick class
- a rule needs a stable explicit alias that cannot be reliably inferred from Brick class alone
- the point name clearly indicates a meaningful distinction such as pre/post, entering/leaving, heating/cooling, min/max, supply/return when same-class ambiguity exists

Example:
If one AHU has two Supply_Air_Temperature_Sensor points:
- SAT before coil -> rule_input: "sat_pre"
- SAT after coil -> rule_input: "sat_post"

If only one point of that Brick class exists in scope and it is unambiguous, set:
- "rule_input": null

Do NOT invent unnecessary rule_input values for every point.

8. weather / Open-Meteo / network weather points:
Do not invent weather equipment or weather points that are not in the export.
If the export already contains Open-Meteo or other weather/network points, keep those rows in the output. Set polling=true only when weather-related faults/rules are in scope, or the operator explicitly wants weather polling/plotting; otherwise prefer polling=false for weather rows when fault context does not require them.
Do not add new weather points unless the operator or attached rules explicitly require them.

9. fallback behavior:

If uncertain:
- brick_type = null
- rule_input = null
- unit = null
- polling = false
- equipment_type = omit or null (backend defaults to generic Equipment — use this instead of a wrong class)

REAL-JOB / CONSERVATIVE MODE (not optional for production buildings)

Bench and demo setups can be forgiving; **on a real live HVAC job** the model must not drift from discoverable truth.

Do not invent or guess:

- Extra BACnet devices, synthetic weather stations, or integration “placeholder” equipment
- Equipment rows or feeds/fed_by topology that are not in the export, the user’s brief, or an attached as-built
- Point rows that were not in the export (no hallucinated objects)
- **Engineering** numerics (HP, tons, MBH, design CFM, kW) on **equipment[].engineering** without operator-provided submittal/spec text

Preserve identity exactly (character-for-character when the export provides them):

- `bacnet_device_id`, `object_identifier`, `object_name`, `external_id`, `point_id`, `site_id`, `site_name`, `equipment_id`

When unsure, prefer the safer default:

- `null` for unknown Brick type or unit — not a best guess
- `polling: false` unless the point is clearly needed for the faults/rules or plotting/trending the operator asked for
- Omit `feeds` / `fed_by` rather than inferring ductwork relationships
- Saying (in a side channel) “cannot determine X from export” is better than fabricating X in JSON

Demo vs live: Use pre-flight answers: test-bench convenience must not override the rules above on a production import.

Operator review before import (short checklist):

1. Compare row count and key BACnet fields to the latest `GET /data-model/export` — no mystery devices.
2. Confirm every non-null `site_id` is still a UUID from `GET /sites`.

3. `PUT dry-run` is not currently supported by the Open-FDD API for `PUT /data-model/import`; use schema/Pydantic validation first, and if you need a no-risk rehearsal, run the same `PUT` against a staging instance while comparing inputs from `GET /data-model/export` and site UUIDs from `GET /sites`.
4. After import, verify a handful of BACnet reads match the gateway for the same object ids.
5. On **Data Model Testing**, run **AHUs**, **VAV boxes**, **Class summary**, etc., to confirm **equipment_type** choices match reality (quick human QA).

EQUIPMENT RULES

Create an "equipment" array with one entry per equipment.

Each equipment item must use this shape:

```
{
  "equipment_name": "AHU-1",
  "equipment_type": "Air_Handling_Unit",
  "site_id": "<same site_id as the points>",
  "feeds": ["VAV-1"]
}
```

or

```
{
  "equipment_name": "VAV-1",
  "equipment_type": "Variable_Air_Volume_Box",
  "site_id": "<same site_id as the points>",
  "fed_by": ["AHU-1"]
}
```

Rules:
- Use equipment names only
- Do not use equipment UUIDs
- Preserve the exact site_id from the export
- Set **equipment_type** on each row (see §4) so Brick `rdf:type` matches **Data Model Testing** preset queries (Brick **1.4** vocabulary).
- Include feeds/fed_by only when supported by the provided data or clearly specified by the user
- Do not invent mechanical relationships unless they are explicitly given or obvious from the provided context

Optional — engineering metadata (`equipment[].engineering`):
- Open-FDD **GET /data-model/export** duplicates **equipment.metadata.engineering** on each point row as **engineering** for LLM context. On import, put updates under **equipment[]** using the same **equipment_name** + **site_id**,

```
with an **engineering** object (nested sections such as
**mechanical**, **electrical**, **controls**, **topology**,
**documents** — see project examples).
- You **may** help capture **rated** data: design CFM, cooling/
heating capacity, fan or pump motor nameplate (HP/kW/FLA), coil or
heat-exchanger ratings, feeder/panel references, **when the
operator provides submittal, schedule, or as-built values** (or
asks you to transcribe pasted specs).
- Do **not** invent numeric ratings (HP, tons, MBH, CFM, kW) from
BACnet object names alone.
- Preserve existing **engineering** from the export when present
unless the operator asks to correct it.

---------------------------------------------------
STRICT OUTPUT REQUIREMENTS
---------------------------------------------------

When emitting import JSON:
- Return ONLY valid JSON
- EXACTLY two top-level keys: "points" and "equipment"
- No "sites"
- No "relationships"
- No "equipments"
- No prose
- No markdown
- No comments

If duplicate external_id appears for the same site, keep the last
occurrence.

Now process the following input JSON:
```

After the final line, paste the **export JSON** (or send it as the next user message). This template works for manual copy-paste and external agents (including Open-Claw workflows).

Again, for agents with web access: Data modeling (GitHub Pages) and this template section.

26.3 Where the rules live

What	Where
Fault rules overview	docs/rules/overview — FDD rule types, YAML format, Brick-driven inputs.
Expression Rule Cookbook	docs/expression_rule_cookbook — AHU, chiller, weather, and advanced

What	Where
	recipes; **rule_input** examples and expression patterns.
Actual YAML rule files	`stack/rules/` in the repo (or your `rules_dir`). Paste snippets into the LLM session so it uses the same input names **and** can set polling only where rules need data.

The cookbook is **not** a fault rule file itself; it's documentation. The **rules you want to use** are the YAML files in `stack/rules/` (or your project's rules). For the LLM, prefer pasting that YAML; you can also say "use rule_input slugs from the Expression Rule Cookbook (sat, rat, zone_temp, …)" when YAML is not at hand — but polling decisions will be less certain until you tie them to concrete faults.

26.4 Engineering metadata (fan HP, pumps, coils, design CFM, …)

Yes — the **export** exposes equipment-level engineering on each row as **`engineering`** (and raw **`equipment_metadata`**). The **import** accepts the same data on **`equipment[]`** via **`engineering`**, merged into `equipment.metadata.engineering` in PostgreSQL and emitted into the knowledge graph (see **`open_fdd/platform/data_model_ttl.py`**).

Use this for **nameplate / submittal** style fields (design CFM, cooling/heating capacity, motor HP or FLA, feeder panel, topology sketches, source drawing references). **FDD rules still run on time-series columns** from polled points; engineering scalars are for **context**, SPARQL, dashboards, and downstream analytics unless you extend the runner.

- **Docs:** Data model engineering (Brick + 223P MVP)
- **Example import:** `examples/223P_engineering/engineering_import_example.json`
- **UI: Energy Engineering → Equipment metadata** edits the same `equipment.metadata.engineering` store (and optional topology JSON) without a separate import round-trip when you prefer forms over LLM JSON.

The canonical prompt tells the LLM to **only** fill or change engineering when the operator supplies evidence — not to invent tons, HP, or CFM from BACnet names alone.

26.5 Feeds / fed_by (HVAC topology)

- **feeds** — This equipment supplies another (e.g. AHU feeds VAV).
- **fed_by** — This equipment is supplied by another (e.g. VAV fed_by AHU).

If topology is not known confidently, omit feeds/fed_by rather than guessing. You can add or refine relationships later.

26.6 Validate before import (so backend CRUD accepts it)

The Open-FDD **PUT /data-model/import** endpoint expects a body that matches the **DataModelImportBody** Pydantic model: exactly `points` (array) and optional `equipment` (array). If the LLM returns extra keys, wrong types, or invalid UUIDs, the API returns **422 Unprocessable Entity**.

To avoid that:

1. **Instruct the LLM** — In your prompt, add: "Return only valid JSON that conforms to the Open-FDD import schema: top-level keys `points` and `equipment` only; each point has the fields listed in the prompt; equipment items use `equipment_name`, `site_id`, optional `feeds` / `fed_by`, optional `engineering` / `metadata`."

2. **Use the API's JSON Schema** — The OpenAPI spec at **GET /openapi.json** (or **GET /docs** and "openapi.json") includes a **schema** for the import body. You can:

   - Export that schema (e.g. the `DataModelImportBody` and nested `PointImportRow` / `EquipmentImportRow` from the spec) and give it to the LLM: "Return JSON that validates against this schema."
   - Or run a local **validation step** before pasting into the UI: validate the LLM's JSON against the same schema (e.g. with a small script or tool that loads the schema and runs `jsonschema.validate`). If it passes, PUT /data-model/import will accept it (aside from referential issues like missing site_id).

3. **Common failure: `site_id` is not a UUID** — `PUT /data-model/import` expects `points[].site_id` to be either:

   - a UUID string (as returned by `GET /data-model/export`), or
   - `null`/omitted (so the backend can resolve it from `points[].site_name`).

If you see an error like: `site_id must be a valid UUID from GET /sites ... Got: 'BensOffice'` then your LLM replaced the UUID with the human-readable site name. Reinforce: **never** replace `site_id` with `site_name`; if the export has no UUID, keep `site_id` null and keep `site_name`.

4. **Pydantic in the repo** — The backend defines the import shape in **open_fdd/platform/api/data_model.py**: `DataModelImportBody`, `PointImportRow`, `EquipmentImportRow`. A script or pipeline can import those models and validate the LLM output (e.g. `DataModelImportBody.model_validate(json.loads(llm_output))`) before returning it to the human. That way the human only sees JSON that is known to parse on the backend.

26.7 Mechanical engineer flow (short)

1. **Create site** (and optionally equipment) via API or UI; note **site_id**.
2. **Export** — GET /data-model/export?site_id=YourSiteName (or no filter for full dump).
3. **Upload to LLM** — Paste (a) the canonical template above (or your saved copy of the same text), (b) **fault/rule context** (which faults you run + **YAML snippets when possible**), then (c) export JSON. Optionally include the import JSON Schema so the LLM returns a valid payload.
   - **Fault-first:** Answer the pre-flight questions (or let the model ask them) before treating polling as final.
   - **UUID reminder:** Never replace `points[].site_id` with a human-readable site name; keep the UUID from the export (see **Validate before import** below).
4. **Validate** — Run schema validation or Pydantic validation on the LLM reply so you know it will parse on the backend.
5. **Import** — PUT /data-model/import with the validated JSON.
6. **Run FDD / tests** — Trigger an FDD run or Sparkl (or other) tests as needed for the project.

26.8 See also

- AI-assisted data modeling — Export → tag → import and API contract.
- Data model engineering — `equipment.metadata.engineering`, import/export, TTL / SPARQL.
- Technical reference — PyPI vs repo, LLM tagging workflow; full prompt is above on this page.

- Fault rules overview and Expression Rule Cookbook — Rules and rule_input reference.

27 SPARQL cookbook

28 SPARQL cookbook

Open-FDD keeps one **knowledge graph** (Brick + BACnet + platform config) in `config/data_model.ttl`. All queries in this cookbook run **via the REST API only** — use **POST /data-model/sparql** with a JSON body `{"query": "..."}`. Do not query the TTL file directly; the API is the single entry point so the same flow works for UIs, scripts, and tests.

28.1 Data Model Protocols vs Data Model Testing

The frontend has two tabs:

Tab	Route	Purpose
Data Model Protocols	`/data-model`	Sites, equipment, import/export JSON, view TTL. Build or edit the data model (sites, equipment, points) and paste AI-tagged JSON to import.
Data Model Testing	`/data-model-testing`	**Summarize your HVAC** (one-click SPARQL buttons: Sites, AHUs, Zones, Building, VAV boxes, etc.) and **Custom SPARQL** (textarea, Run, upload .sparql file). Same queries as this cookbook.

Use **Data Model Testing** to run SPARQL in the browser. Use **Data Model Protocols** to manage sites, equipment, and import.

28.2 How to run SPARQL

Method	Use
Data Model Testing (UI)	Open **Data Model Testing** → click a predefined button (Sites, AHUs, Zones, …) or paste SPARQL in Custom SPARQL → Run SPARQL. Optional: check “Include BACnet device and point IDs” for `bacnet_device_id` / `object_identifier` in results.
Swagger	http://localhost:8000/docs → **POST /data-model/sparql** → body `{"query": "SELECT ..."}` → Execute.
curl	`curl -X POST http://localhost:8000/data-model/sparql -H "Content-Type: application/json" -d '{"query":"SELECT ?s ?p ?o WHERE { ?s ?p ?o } LIMIT 10"}'`
Upload .sparql file	**Data Model Testing** → Custom SPARQL → “Upload .sparql file”, then Run; or **POST /data-model/sparql/upload** with a `.sparql` file.

Response: JSON with a `bindings` array; each element is a map of variable names to values (e.g. `{"site_label": {"value": "DemoSite"}}`). Use the `value` field for literals and the full object for URIs.

28.3 Prefixes and graph contents

The graph uses these namespaces. Include them in your queries:

Prefix	Namespace
`brick:`	https://brickschema.org/schema/Brick#
`rdfs:`	http://www.w3.org/2000/01/rdf-schema#
`ref:`	https://brickschema.org/schema/Brick/ref#
`ofdd:`	http://openfdd.local/ontology#
`bacnet:`	http://data.ashrae.org/bacnet/2020#
`:` (default)	http://openfdd.local/site#

Graph contents: Brick triples (sites, equipment, points from the DB), Brick external references (`ref:BACnetReference`, `ref:TimeseriesReference`), BACnet triples (devices and objects from discovery), and **ofdd:PlatformConfig** (platform config from GET/PUT /config). All are queryable together.

28.4 Recipe 1: Platform config

Platform config is stored as RDF (same graph). Use it to verify or audit settings after PUT /config or bootstrap.

All config triples:

```
PREFIX ofdd: <http://openfdd.local/ontology#>
SELECT ?p ?v WHERE {
  ?c a ofdd:PlatformConfig .
  ?c ?p ?v .
}
```

Example bindings: `ofdd:ruleIntervalHours`, `ofdd:bacnetServerUrl`, `ofdd:bacnetEnabled`, etc. Predicates are camelCase (e.g. `ruleIntervalHours`). Use this to confirm config is in the graph after seeding.

28.5 Recipe 2: Sites and counts

All sites with labels:

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?site ?site_label WHERE {
  ?site a brick:Site .
  ?site rdfs:label ?site_label .
}
```

Site and aggregate counts (equipment count, point count per site):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?site_label (COUNT(DISTINCT ?equipment) AS ?
equipment_count) (COUNT(?point) AS ?point_count) WHERE {
  ?site a brick:Site .
  ?site rdfs:label ?site_label .
  ?equipment brick:isPartOf ?site .
  ?point brick:isPointOf ?equipment .
}
GROUP BY ?site ?site_label
```

Use this to validate the data model after CRUD or import (see Getting started).

28.6 Recipe 3: Equipment and points by site

Equipment labels for a given site (replace `"YourSiteLabel"` with your site name):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?eq_label WHERE {
  ?eq brick:isPartOf ?site .
  ?site rdfs:label "YourSiteLabel" .
  ?eq rdfs:label ?eq_label .
}
```

Point labels for a site (points under equipment under site):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?pt_label WHERE {
  ?pt brick:isPointOf ?eq .
  ?eq brick:isPartOf ?site .
  ?site rdfs:label "YourSiteLabel" .
  ?pt rdfs:label ?pt_label .
}
```

Equipment feeds (Brick `feeds` relationships for a site):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?eq_label ?feeds_label WHERE {
  ?eq brick:isPartOf ?site .
  ?site rdfs:label "YourSiteLabel" .
  ?eq rdfs:label ?eq_label .
  ?eq brick:feeds ?other .
```

```
  ?other rdfs:label ?feeds_label .
}
```

In the **Data Model Testing** UI, the **Feed topology** button runs a graph-wide query that lists both `brick:feeds` and `brick:isFedBy` edges (with optional site / equipment labels). See `afdd_stack/frontend/src/data/data-model-testing-queries.ts` (`equipment_feeds_topology`).

28.7 Recipe 3b: External references (Brick v1.4)

List point-to-reference relationships:

```
PREFIX ref: <https://brickschema.org/schema/Brick/ref#>
SELECT ?point ?rep ?type WHERE {
  ?point ref:hasExternalReference ?rep .
  ?rep a ?type .
}
LIMIT 200
```

BACnet references from point metadata:

```
PREFIX ref: <https://brickschema.org/schema/Brick/ref#>
PREFIX bacnet: <http://data.ashrae.org/bacnet/2020#>
SELECT ?point ?oid ?device WHERE {
  ?point ref:hasExternalReference ?rep .
  ?rep a ref:BACnetReference ;
       bacnet:object-identifier ?oid ;
       bacnet:objectOf ?device .
}
```

Timeseries references:

```
PREFIX ref: <https://brickschema.org/schema/Brick/ref#>
SELECT ?point ?tsid ?store WHERE {
  ?point ref:hasExternalReference ?rep .
  ?rep a ref:TimeseriesReference ;
       ref:hasTimeseriesId ?tsid ;
       ref:storedAt ?store .
}
```

28.8 Recipe 4: BACnet devices and objects

After **POST /bacnet/point_discovery_to_graph**, the graph contains BACnet devices and their objects.

All BACnet devices:

```
PREFIX bacnet: <http://data.ashrae.org/bacnet/2020#>
SELECT ?dev WHERE { ?dev a bacnet:Device }
```

Object count per device:

```
PREFIX bacnet: <http://data.ashrae.org/bacnet/2020#>
SELECT ?dev (COUNT(?obj) AS ?n) WHERE {
  ?dev a bacnet:Device .
  ?dev bacnet:contains ?obj .
}
GROUP BY ?dev
```

Object names for a device (e.g. device instance 3456789):

```
PREFIX bacnet: <http://data.ashrae.org/bacnet/2020#>
SELECT ?name WHERE {
  ?dev a bacnet:Device ;
       bacnet:device-instance 3456789 ;
       bacnet:contains ?obj .
  ?obj bacnet:object-name ?name .
}
```

Use these to confirm discovery results are in the graph and to cross-check with GET /data-model/export.

BACnet device and object addresses for points with polling = true (join Brick points to BACnet via matching label and object-name):

```
PREFIX bacnet: <http://data.ashrae.org/bacnet/2020#>
PREFIX ofdd: <http://openfdd.local/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?device_instance ?device_address ?object_identifier ?
point_label
WHERE {
  ?point ofdd:polling true ;
         rdfs:label ?point_label .
  ?bacnet_obj bacnet:object-name ?point_label ;
              bacnet:object-identifier ?object_identifier .
  ?device bacnet:contains ?bacnet_obj ;
          bacnet:device-instance ?device_instance ;
          bacnet:device-address ?device_address .
}
ORDER BY ?device_instance ?object_identifier
```

Returns one row per polling point that has a matching BACnet object: `device_instance`, `device_address`, `object_identifier` (e.g. `analog-input,1`), and `point_label`. Use this for scraper config or to list "which BACnet addresses are in the data model and polled."

Why "No bindings (empty result)"? The query above joins **Brick** points to **BACnet** objects. If your graph has no BACnet triples (no discovery has been run, or you only have Brick from a TTL file), there are no `bacnet:Device`, `bacnet:contains`, or `bacnet:object-name` triples, so the join returns nothing. Use the query below when you only have Brick data.

Polling points with a Brick type (no BACnet) — use this when the graph has no BACnet data; it returns points that have `ofdd:polling true` and a Brick class (e.g. `Return_Air_Temperature_Sensor`):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX ofdd: <http://openfdd.local/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?brick_class ?point_label ?unit
WHERE {
  ?point ofdd:polling true ;
         rdfs:label ?point_label ;
         a ?brick_type .
  FILTER(STRSTARTS(STR(?brick_type), STR(brick:)))
  BIND(REPLACE(STR(?brick_type), "https://brickschema.org/schema/
Brick#", "") AS ?brick_class)
  OPTIONAL { ?point ofdd:unit ?unit . }
}
ORDER BY ?point_label
```

Data Model Protocols: Equipment table vs "View full data model (TTL)" — The **Equipment** table on **Data Model Protocols** comes from the **database** (REST API: sites, equipment, points). **Data Model Testing** (SPARQL) and **"View full data model (TTL)"** on Protocols use the **in-memory graph**, which is synced from the DB (Brick) plus BACnet discovery. If you see a full TTL with equipment but "No equipment configured" in the table, the DB may be empty or out of sync: the graph can be loaded from `config/data_model.ttl` on startup, but the UI list is always from the API/DB. Use the site selector in the top bar ("All sites" vs a specific site); ensure sites and equipment exist in the DB (e.g. create via the UI or import).

28.9 Recipe 5: FDD — points and rule mapping

FDD rules resolve Brick classes to DataFrame columns via the TTL. These queries help you validate that points have the right Brick types and rule inputs for your rules (e.g. sensor_bounds or expression rules).

Brick class → label → rule input (timeseries reference and FDD input name):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX ofdd: <http://openfdd.local/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?brick_class ?label ?rule_input WHERE {
  ?point a ?brick_type .
  FILTER(STRSTARTS(STR(?brick_type), STR(brick:)))
  BIND(REPLACE(STR(?brick_type), "https://brickschema.org/schema/
Brick#", "") AS ?brick_class)
  ?point rdfs:label ?label .
  OPTIONAL { ?point ofdd:mapsToRuleInput ?rule_input . }
}
ORDER BY ?brick_class ?rule_input
LIMIT 50
```

- **`rdfs:label`** = `external_id` in the DB = timeseries key (column name in the FDD DataFrame).
- **`ofdd:mapsToRuleInput`** = `fdd_input` in the DB = name the rule uses when multiple points share a Brick class.

Brick classes used in the model (count per class):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
SELECT ?brick_class (COUNT(?point) AS ?count) WHERE {
  ?point a ?brick_type .
  FILTER(STRSTARTS(STR(?brick_type), STR(brick:)))
  BIND(REPLACE(STR(?brick_type), "https://brickschema.org/schema/
Brick#", "") AS ?brick_class)
}
GROUP BY ?brick_class
ORDER BY DESC(?count)
```

Use this to ensure every Brick class your rules reference (e.g. `Supply_Air_Temperature_Sensor`, `Zone_Temperature_Sensor`) has at least one point in the graph.

28.10 Recipe 6: FDD — time-series references for rules

Rules in `stack/rules/` (e.g. `sensor_bounds.yaml`, expression rules) declare **inputs** by Brick class only. The runner uses the Brick TTL (this graph) to resolve:

- **Brick class** → points of that type → **rdfs:label** (external_id) → column in the timeseries DataFrame.

So the **timeseries reference** for FDD is the point's **rdfs:label** (which equals `external_id` in the DB and is the key in `timeseries_readings` when joined via `point_id`). To list "which columns will this rule see for site X?":

Points that map to rule inputs for a site (equipment hierarchy + label = timeseries key):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX ofdd: <http://openfdd.local/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?equipment_label ?point_class ?label ?rule_input WHERE {
  ?site a brick:Site .
  ?site rdfs:label "YourSiteLabel" .
  ?equipment brick:isPartOf ?site .
  ?equipment rdfs:label ?equipment_label .
  ?point brick:isPointOf ?equipment .
  ?point a ?brick_type .
  FILTER(STRSTARTS(STR(?brick_type), STR(brick:)))
  BIND(REPLACE(STR(?brick_type), "https://brickschema.org/schema/
Brick#", "") AS ?point_class)
  ?point rdfs:label ?label .
  OPTIONAL { ?point ofdd:mapsToRuleInput ?rule_input . }
}
ORDER BY ?equipment_label ?point_class
```

Interpretation: For each row, **`label`** is the timeseries column name (external_id). The rule runner loads timeseries by (site, external_id) and builds the DataFrame; then Brick class (and optional rule_input) select which column is used for each rule input. So this query answers "what timeseries refs (columns) does the FDD pipeline have for this site?"

Polling points only (points that the BACnet scraper writes to; exclude non-polling):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX ofdd: <http://openfdd.local/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?label ?brick_class WHERE {
  ?pt a ?brick_type .
```

```
  FILTER(STRSTARTS(STR(?brick_type), STR(brick:)))
  BIND(REPLACE(STR(?brick_type), "https://brickschema.org/schema/
Brick#", "") AS ?brick_class)
  ?pt rdfs:label ?label .
  ?pt ofdd:polling true .
}
ORDER BY ?brick_class ?label
```

These queries mirror the logic in `scripts/automated_testing/sparql/` (e.g. 05_brick_rule_mapping.sparql); run them via the API for validation or for building a small “FDD readiness” report.

28.11 Recipe 7: Integrity and completeness

Graph check — Use **GET /data-model/check** for triple counts, sites count, BACnet device count, and orphan warnings. For custom checks, use SPARQL:

Count triples:

```
SELECT (COUNT(*) AS ?n) WHERE { ?s ?p ?o }
```

Equipment hierarchy snapshot (site → equipment type → point class → rule input):

```
PREFIX brick: <https://brickschema.org/schema/Brick#>
PREFIX ofdd: <http://openfdd.local/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?site_label ?equipment_label ?equipment_type ?point_class ?
rule_input WHERE {
  ?site a brick:Site .
  ?site rdfs:label ?site_label .
  ?equipment brick:isPartOf ?site .
  ?equipment rdfs:label ?equipment_label .
  ?equipment ofdd:equipmentType ?equipment_type .
  ?point brick:isPointOf ?equipment .
  ?point a ?brick_type .
  FILTER(STRSTARTS(STR(?brick_type), STR(brick:)))
  BIND(REPLACE(STR(?brick_type), "https://brickschema.org/schema/
Brick#", "") AS ?point_class)
  OPTIONAL { ?point ofdd:mapsToRuleInput ?rule_input . }
}
ORDER BY ?site_label ?equipment_label ?point_class
LIMIT 100
```

28.12 Predefined buttons (Data Model Testing)

The **Data Model Testing** tab has one-click buttons under **Summarize your HVAC** that run the same SPARQL as this cookbook.

Button	Recipe
Sites	Recipe 2 (sites with labels)
AHUs, Zones, Building, VAV boxes, VAVs per AHU	Count or list by Brick type
Chillers, Cooling towers, Boilers, Central plant	Count or list equipment types
HVAC equipment, Meters, Points, Class summary	Counts and Brick class summary

Checking **Include BACnet device and point IDs** on Data Model Testing runs a variant of each query that adds `bacnet_device_id` and `object_identifier` (for telemetry and algorithms).

28.13 SPARQL smoke suite (repo files)

The same queries used by **Data Model Testing** predefined buttons and by optional bench automation are checked in under **`openclaw/bench/sparql/`** in the open-fdd repository. Run them via **POST /data-model/sparql** (body `{"query": "..."}`), **POST /data-model/sparql/upload**, or **Data Model Testing** → Custom SPARQL (paste or upload). See also the directory README on GitHub for the mapping to UI buttons and cookbook recipes.

File	Role
`01_platform_config.sparql`	Recipe 1: platform config
`02_sites_labels.sparql`	Recipe 2 / **Sites** button
`02_site_counts.sparql`	Site aggregate counts
`03_equipment_labels_testbench.sparql`	Equipment for test-bench site
`03_point_labels_testbench.sparql`	Points for test-bench site
`04_bacnet_devices.sparql`	Recipe 4: BACnet devices
`05_brick_rule_mapping.sparql`	Recipe 5: Brick → rule input
`06_polling_points_brick_type.sparql`	Polling points + Brick type (no BACnet join)
`07_count_triples.sparql`	Triple count

File	Role
`08_bacnet_telemetry_points.sparql`	BACnet telemetry / addressing
`09_graph_db_sync_counts.sparql`	Graph vs DB sync counts
`10_ahus.sparql`	**AHUs** button
`11_zones.sparql`	**Zones** button
`12_building.sparql`	**Building** button
`13_vav_boxes.sparql`	**VAV boxes** button
`14_vavs_per_ahu.sparql`	**VAVs per AHU** button
`15_chillers.sparql`	**Chillers** button
`16_cooling_towers.sparql`	**Cooling towers** button
`17_boilers.sparql`	**Boilers** button
`18_central_plant.sparql`	**Central plant** button
`19_hvac_equipment.sparql`	**HVAC equipment** button
`20_meters.sparql`	**Meters** button
`21_points.sparql`	**Points** button
`22_class_summary.sparql`	**Class summary** button
`23_orphan_external_references.sparql`	Orphan `ref:` hygiene
`24_operator_site_context.sparql`	Operator site context discovery

28.14 Related docs

Topic	Page
Data model flow	Overview
AI-assisted tagging	AI-assisted tagging
Rule inputs and Brick	Expression rule cookbook · Fault rules overview · Test bench rule catalog
Grafana (SQL, not SPARQL)	Grafana SQL cookbook
Dev reference	Technical reference

29 External representations (Brick v1.4)

30 External representations (Brick v1.4)

Open-FDD models Brick points with Brick's **ref schema** so software can discover how each point maps to external systems (BACnet and timeseries storage) using:

- `ref:hasExternalReference`
- `ref:BACnetReference`
- `ref:TimeseriesReference`

Open-FDD keeps legacy `ofdd:*` BACnet fields for compatibility, but `ref:*` is the preferred representation.

30.1 When ref triples appear in the TTL

Brick external references (`ref:hasExternalReference`, etc.) are **materialized when the Brick section of the graph is rebuilt from Postgres** and the graph is serialized (same path as `sync_brick_from_db` / TTL generation). They are **not** invented solely by BACnet discovery in memory: discovery updates **BACnet** RDF in the graph, while `ref:BACnetReference` / `ref:TimeseriesReference` under each point come from **current point rows** in the DB when that Brick sync runs.

You will see `ref:` in TTL (or in **View full data model**) after any of these:

- **CRUD or import** on sites, equipment, or points (the API reserializes Brick to disk on those paths).
- **GET /data-model/ttl** with `save=true` (default in many clients), **POST /data-model/serialize**, or **Data model → Serialize to TTL** in the UI.
- The **background graph sync** on a timer: interval **`graph_sync_interval_min`** (platform config). In the React app this is edited on **OpenFDD Config**, not the Overview page. See Configuration → Platform keys.

So if you only ran BACnet discovery and are watching **`config/data_model.ttl` on disk**, `ref:` may lag until the next serialize (timer or explicit action above). Discovery may also request a TTL write when **write to file** is enabled on that flow—otherwise rely on CRUD/import or an explicit serialize.

30.2 What Open-FDD emits

For each point in `config/data_model.ttl`:

- A **timeseries reference**:
  - `a ref:TimeseriesReference`
  - `ref:hasTimeseriesId "<external_id>"`
  - `ref:storedAt "<postgresql://host:port/db/timeseries_readings>"`
- A **BACnet reference** when BACnet addressing is known (`bacnet_device_id` + `object_identifier`):
  - `a ref:BACnetReference`
  - `bacnet:object-identifier "<object-type,instance>"`
  - optional `bacnet:object-name "<name>"`
  - `bacnet:objectOf <bacnet://<device_instance>>`
  - `brick:BACnetURI "bacnet://<device>/<object>/present-value"`

30.3 Example

```
:pt_123 a brick:Zone_Air_Temperature_Sensor ;
    rdfs:label "ZoneTemp" ;
    ref:hasExternalReference [
        a ref:BACnetReference ;
        bacnet:object-identifier "analog-input,3" ;
        bacnet:object-name "BLDG-Z410-ZATS" ;
        brick:BACnetURI "bacnet://123/analog-input,3/present-
value" ;
        bacnet:objectOf <bacnet://123>
    ] ;
    ref:hasExternalReference [
        a ref:TimeseriesReference ;
        ref:hasTimeseriesId "ZoneTemp" ;
        ref:storedAt "postgresql://localhost:5432/openfdd/
timeseries_readings"
    ] ;
    brick:isPointOf :eq_ahu_1 .
```

30.4 SPARQL discovery pattern

Use this to enumerate all external references and their types:

```
PREFIX ref: <https://brickschema.org/schema/Brick/ref#>
SELECT ?point ?rep ?type WHERE {
  ?point ref:hasExternalReference ?rep .
  ?rep a ?type .
}
```

BACnet-only references:

```
PREFIX ref: <https://brickschema.org/schema/Brick/ref#>
PREFIX bacnet: <http://data.ashrae.org/bacnet/2020#>
SELECT ?point ?oid ?dev WHERE {
  ?point ref:hasExternalReference ?rep .
  ?rep a ref:BACnetReference ;
       bacnet:object-identifier ?oid ;
       bacnet:objectOf ?dev .
}
```

Timeseries-only references:

```
PREFIX ref: <https://brickschema.org/schema/Brick/ref#>
SELECT ?point ?tsid ?store WHERE {
  ?point ref:hasExternalReference ?rep .
  ?rep a ref:TimeseriesReference ;
       ref:hasTimeseriesId ?tsid ;
       ref:storedAt ?store .
}
```

31 Data modeling

32 Data modeling

The Open-FDD data model is built around **sites**, **equipment**, and **points**: the same entities you create, read, update, and delete via the API (**CRUD** = Create, Read, Update, Delete). The REST API (Swagger at `/docs`) and the data-model endpoints (`/data-model/export`, `/data-model/import`, `/data-model/ttl`) are the main way to manage this data. The database is the single source of truth; the Brick TTL file is regenerated on every change so FDD rules and Grafana stay in sync.

Concepts (entities):

- **Sites** — Buildings or facilities. All equipment and points are scoped to a site.
- **Equipment** — Physical devices (AHUs, VAVs, heat pumps). Belong to a site; have points.
- **Points** — Time-series references (sensors, setpoints). Link to equipment and site; store `external_id`, optional `brick_type` and `rule_input` for FDD.
- **External representations (Brick v1.4)** — `ref:hasExternalReference` mappings from points to external systems (metadata / interoperability).
- **Engineering metadata + 223P topology** — equipment-level engineering metadata is stored as JSON and emitted to RDF using `ofdd:*` extension predicates plus `s223:*` topology patterns (`hasConnectionPoint`, connection conduits, mediums). Same graph as Brick and timeseries refs, so you can join **rated capacity** and **topology** to **FDD results** for optimization context or rough energy-penalty sketches (Data model engineering).
- **SPARQL cookbook** — Run SPARQL via POST /data-model/sparql: config, data model, optional BACnet-shaped triples, FDD rule mapping, time-series references. Copy-paste queries for validation and UIs.
- **AI-assisted data modeling** — Export → LLM or human tagging → import (Brick types, rule_input, polling, equipment feeds). External agents (e.g. Open-Claw) can use `GET /model-context/docs` as platform documentation context and OpenAPI (`/openapi.json` or `/docs`) for HTTP discovery; the canonical import schema is the same.
- **LLM workflow (export + rules + validate → import)** — Canonical prompt (fault-first pre-flight for polling), export JSON, faults + rules (prefer YAML); validate with schema or Pydantic; then PUT /data-model/import and run FDD/tests. Published hub.
- **AI-assisted energy calculations** — After the Brick model exists: `GET /energy-calculations/export` → LLM fills predefined `calc_type` / `parameters` → `PUT /energy-calculations/import`. Export embeds full `calc_types` metadata for offline agents.
- **Default FDD energy penalty catalog** — 18 seeded engineering narratives (GL36-style airside, plant, zone), `POST /energy-calculations/seed-default-penalty-catalog`, TTL `ofdd:penaltyCatalogSeq`.

Framework: CRUD is provided by the FastAPI REST API. The data-model API adds bulk export/import and Brick TTL generation; see Overview for the flow (DB → TTL → FDD column_map) and Appendix: API Reference for endpoints.

33 AI-assisted energy calculations

34 AI-assisted energy engineering (export → LLM → import)

After the **Brick / BACnet data model** is in place (AI-assisted data modeling, LLM workflow), you can use the same **export → external LLM → import** pattern for **site-scoped energy / savings calculation specs**. For Open Claw, auth, and **SSH operator** flows (including remote VOLTTRON edge → Central), see **Open Claw integration**. Each building keeps its own rows in Postgres; the API also serializes them into `config/data_model.ttl` as `ofdd:EnergyCalculation` with `brick:isPartOf` the site.

Predefined calculators (labels, parameter keys, units) are versioned in the platform (`openfdd_stack.platform.energy_calc_library`). The UI on **Energy Engineering** mirrors them; the LLM must **only** use `calc_type` values and **parameter keys** that appear in the export bundle.

Default FDD penalty narratives (18) — Engineering descriptions, difficulty ordering, and suggested `calc_type` mappings are in Default FDD energy penalty catalog. You can **materialize** them as disabled rows with **`POST /energy-calculations/seed-default-penalty-catalog?site_id=<uuid>`** (or **Seed default penalty catalog** in the UI), then enable and fill parameters / `point_bindings` per site. Seeded rows use `external_id` values `penalty_default_01` … `penalty_default_18` and may include `_penalty_catalog_seq` in `parameters`; the TTL emitter adds **`ofdd:penaltyCatalogSeq`** for SPARQL.

34.1 Two-phase workflow (data model, then energy)

1. **Phase A — Data model** — `GET /data-model/export` → LLM tagging → `PUT /data-model/import` so equipment, points, `external_id`, and Brick types exist for the site. See AI-assisted data modeling and LLM workflow.
2. **Phase B — Energy** — `GET /energy-calculations/export?site_id=<uuid>` → LLM fills or edits calculation rows → `PUT /`

`energy-calculations/import` with body `{ "site_id": "<same uuid>", "energy_calculations": [ ... ] }`. Optionally seed the 18 penalty templates first (UI or seed endpoint above).

Always pass the **same `site_id`** as in the header site selector (or UUID from `GET /sites`). Calculations are **never** global.

34.2 Export bundle (`GET /energy-calculations/export`)

Query parameter: **`site_id`** (required, UUID).

Response (JSON):

Field	Purpose
`format`	`openfdd_energy_calculations_v1` — version tag for agents.
`site_id`, `site_name`	Scope for the LLM.
`exported_at`	ISO timestamp.
`calc_types`	Same structure as `GET /energy-calculations/calc-types` — **embeds** every allowed calculator with `id`, `label`, `summary`, `category`, and `fields[]` (`key`, `label`, `type`, `min`/`max`, `default`, `options` for enums).
`penalty_catalog`	Same 18 objects as **`GET /energy-calculations/penalty-catalog`** (`seq`, `layer`, `difficulty`, `name`, `fdd_trigger`, `math_summary`, `calc_type`, `default_parameters`) — reference for agents; see energy penalty equations.
`energy_calculations`	Existing rows: `id`, `external_id`, `name`, `description`, `calc_type`, `parameters`, `point_bindings`, `enabled`, `equipment_id`, **`equipment_name`** (resolved for display), timestamps.

Download this JSON from **Energy Engineering** ("Download export JSON") or call the API from an agent.

34.3 Import body (`PUT /energy-calculations/import`)

Top-level keys (**only** these — `extra=forbid` on the body model):

- **`site_id`** (UUID string) — must match the site you are editing.
- **`energy_calculations`** (array) — rows to **create or update** by **`(site_id, external_id)`**.

Each row:

Field	Required	Notes
`external_id`	yes	Stable slug per site; primary upsert key with `site_id`.
`name`	yes	Human label.
`calc_type`	yes	Must be one of the `calc_types[].id` values from the export.
`parameters`	no	Object; keys must match the `fields[].key` for that `calc_type`. Use numeric JSON numbers for floats.
`point_bindings`	no	**Object** mapping **semantic keys** (e.g. `cfm`, `kw`) to **point `external_id`** strings. Must be a JSON object (not `[]` or a primitive); omit or use `{}` if empty. Stored and exported to TTL; preview library may not consume every binding yet.
`enabled`	no	Default `true`.
`equipment_id`	no	UUID of equipment on this site.

Field	Required	Notes
`equipment_name`	no	Alternative to `equipment_id`: resolved under `site_id`. Fails with 400 if the name does not exist — **create equipment on Data Model BRICK first**.

Unknown keys on each row are **ignored** so you can paste rows straight from an export (e.g. `id`, `created_at`) without hand-stripping.

Upsert behavior: If `external_id` already exists for the site, the row is **updated** (including `calc_type`, `parameters`, `enabled`, equipment link). Otherwise a new row is inserted.

34.4 LLM prompt template (copy-paste)

Use as **system** or **developer** instructions when transforming the export JSON into import JSON.

```
You are assisting with Open-FDD energy / savings calculation specs
for ONE building site.

The user will paste JSON from:
GET /energy-calculations/export?site_id=<uuid>

That JSON includes:
- calc_types[] — the ONLY allowed calc_type string ids and the
ONLY valid parameter keys per type.
- penalty_catalog[] (when present) — 18 default FDD penalty
narratives; use for context or to align copy with site-specific
rows (optional for import output).
- energy_calculations[] — existing saved rows (may be empty).

Your job is to return ONLY valid JSON with EXACTLY these two top-
level keys:

{
  "site_id": "<same uuid string as in the export>",
  "energy_calculations": [ ... ]
}

Rules:
- Do not return markdown, comments, or extra top-level keys.
```

```
- Preserve site_id from the export unless the user explicitly asks
to change site (normally never).
- Each energy_calculations[] element must include: external_id,
name, calc_type, parameters, enabled (boolean).
- calc_type MUST be one of calc_types[].id from the export.
- parameters MUST use only keys listed in calc_types[].fields for
that calc_type. Use JSON numbers for numeric fields.
- Use equipment_name (string) to attach a calc to equipment when
the export lists equipment names from the data model; use null
equipment_id/equipment_name only for site-level calcs.
- point_bindings: optional **JSON object** (not an array or null
as the value) mapping semantic keys to point **external_id**
strings from the data model (e.g. `{ "cfm": "OA_FLOW_SCFM" }`).
Use `{}` or omit the key if there are no bindings. Only reference
external_ids that exist in the user's data model export if they
provided one.
- For annualized / M&V style estimates, use hours and rates
consistent with the field labels in calc_types (e.g. hours_fault,
electric_rate_per_kwh).
- If job context is missing (which faults, which equipment),
prefer conservative parameters or ask a short clarifying question
in plain language WITHOUT emitting JSON in that turn.

When returning JSON, the object must parse as Open-FDD PUT /
energy-calculations/import.
```

34.5 Agents and HTTP context

- **`GET /model-context/docs`** — Regenerated doc bundle for RAG-style agents; include updates to this page when rebuilding.
- **OpenAPI** — `/openapi.json` or `/docs` lists export/import paths alongside data-model tools.

34.6 UI: tree and row actions

On **Energy Engineering** (Energy calculations tab), calculations appear under **Site-level** or under each **equipment** node. Use the **row actions** control (⋮) for **Enable**, **Disable**, or **Delete**, or **right-click** the row (same interaction model as **Points**). The **Equipment metadata** tab edits per-equipment engineering fields that feed the penalty catalog and TTL (`equipment.metadata.engineering`); align with Data model engineering.

34.7 See also

- AI-assisted data modeling — Phase A (Brick / points).
- LLM workflow — Canonical data-model prompt and validation habits.
- Default FDD energy penalty catalog — 18 seeded templates, TTL `ofdd:penaltyCatalogSeq`, Open-Meteo / rates notes.
- Data model engineering — Equipment `engineering` metadata and TTL / SPARQL.
- Frontend — Energy Engineering — Tree, seed defaults, row actions / context menu, export/import buttons, Equipment metadata tab.

35 Default FDD energy penalty catalog

36 Default FDD energy penalty equations (platform catalog)

The stack ships **18 engineering narratives** for common HVAC faults, ordered from **easiest** (schedules, booleans, simple setpoint math) to **hardest** (enthalpy, plant COP, network rollups). They are **not** automatic trend integrations: each row is an **`EnergyCalculation`** with a **`calc_type`** from `openfdd_stack.platform.energy_calc_library`, **static parameters** for M&V-style preview, and optional **`point_bindings`** (semantic key → point `external_id`) for documentation and future analytics.

36.1 Where things live (single sources of truth)

Concern	Location
Predefined calculators (`calc_type`, fields, preview math)	Python: `openfdd_stack/platform/energy_calc_library.py`
18 default narratives + default parameters	Python: `openfdd_stack/platform/energy_penalty_catalog.py` Postgres `energy_calculations`; UI: **Energy Engineering** → tree (same

Concern	Location
Per-site enabled/ disabled rows, equipment link	enable/disable/delete pattern as the points tree)
Knowledge graph / SPARQL	`config/data_model.ttl`: `ofdd:EnergyCalculation` (+ optional `ofdd:penaltyCatalogSeq` when `_penalty_catalog_seq` is in parameters)
Weather (Open-Meteo)	`ofdd:platform_config` / site — **one** Open-Meteo config per deployment (see Configuration)
Utility $/kWh, $/therm	Enter in **calc parameters** (defaults in seed rows) or centralize via site/ platform config over time; export bundle carries whatever you saved

36.2 Seeding the 18 defaults

- **UI:** Energy Engineering → **AI-assisted energy calculations** → **Seed default penalty catalog (18)**. Rows are inserted **disabled** (`enabled: false`) with `external_id` `penalty_default_01` … `penalty_default_18`.
- **API:** `POST /energy-calculations/seed-default-penalty-catalog?site_id=<uuid>&replace=false`
  - `replace=true`: delete existing `penalty_default_*` for the site, then insert all 18.
- **Read-only catalog (no DB):** `GET /energy-calculations/penalty-catalog`
- **Export bundle:** `GET /energy-calculations/export?site_id=...` includes `penalty_catalog` (same 18 narratives) next to `calc_types` and `energy_calculations`.

36.3 Equipment metadata (Energy tab)

The **Equipment metadata** tab focuses on **sizing and nameplate fields** that feed these calculators (airflow, motor HP, static pressure, capacities, pump head, etc.). Legacy **controls** and **document provenance** blocks are folded under **Advanced** so existing data is preserved without cluttering the energy workflow.

36.4 Catalog overview (1–18)

1. **Out-of-schedule operation** — `runtime_electric_kw`
2. **HOA in Hand** — `vfd_affinity_cube`

3. **Zone hunting / simultaneous HC** — `zone_simultaneous_sensible`
4. **VAV minimum flow / reheat** — `vav_min_flow_reheat`
5. **AHU SAT not resetting** — `ahu_sat_sensible_waste`
6. **Duct static not resetting** — `pressure_ratio_motor_kw`
7. **Leaking heating valve** — `sensible_coil_leak_kw`
8. **Dirty filter / fouled coil fan kW** — `fan_filter_dp_kw`
9. **Missed air-side economizer** — `missed_economizer_cooling`
10. **ERV / DOAS enthalpy proxy** — `enthalpy_wheel_proxy`
11. **Overcool + VAV reheat** — `ahu_sat_sensible_waste`
12. **Chiller on at no load** — `plant_minimum_stack_kw`
13. **Boiler / HW standby** — `boiler_standby_mix`
14. **Short-cycle financial proxy** — `short_cycle_financial`
15. **CHWST not resetting** — `chwst_reset_penalty_kw`
16. **Pump DP not resetting** — `pressure_ratio_motor_kw`
17. **Cooling tower fan hunting** — `vfd_affinity_cube`
18. **COP degradation** — `cop_gap_electric`

Full trigger text and LaTeX-style math summaries are in the **`description`** field of each seeded row and in `PENALTY_CATALOG` in code.

36.5 Related

- AI-assisted energy calculations — export / LLM / import workflow
- Data model engineering — Brick + 223P + savings context

37 Overview

38 Fault rules

Fault rules are YAML-defined checks run against time-series DataFrames. Each rule produces a boolean fault flag. **Open-FDD is 100% Brick-model driven:** rule inputs refer only to Brick classes; column mapping comes from the Brick TTL via SPARQL and Brick external timeseries references (timeseries storage), which Open-FDD embraces at the heart of FDD.

38.1 Where rules live: config path and how to manage them

Single directory: All FDD rule YAML files live in one directory configured as **`rules_dir`** in platform config (RDF: `ofdd:rulesDir`, e.g. `"stack/rules"`). The FDD loop loads from this path every run; the rules API (list, upload, download, delete) uses the same path. **You still need `rules_dir` in config** — it defines where files are stored. The frontend does not replace it; it lets you manage the files *in* that directory.

Two ways to manage rules:

Method	Use case
React frontend (Faults page)	Upload new YAML, download existing files, delete files, and **Sync definitions** so the fault_definitions table updates without waiting for the next FDD run. Preferred when you have UI access.
Files on disk	Edit or add files directly under the configured path (e.g. `stack/rules/` on the host or in the container). Same outcome: next FDD run (or **Sync definitions** from the UI) picks them up.

Config: `rules_dir: "stack/rules"` (GET/PUT `/config` or `OFDD_RULES_DIR` at bootstrap). If that path does not exist, the loop falls back to `stack/rules`. Default rules ship in `stack/rules/` (e.g. `sensor_bounds.yaml`, `sensor_flatline.yaml`). See the Expression Rule Cookbook to add or adapt rules. For how YAML becomes pandas operations and how telemetry is pivoted into a DataFrame, see YAML rules → Pandas (under the hood).

Reference rule library (not loaded by default): Additional AHU, chiller, heat-pump, and weather YAML examples used for lab automation and docs live under **`openclaw/bench/rules_reference/`** in the repo. See the Test bench rule catalog for a table of every file with GitHub links.

38.2 Hot reload (edit → run → view)

Open-FDD is **AFDD** (Automated Fault Detection and Diagnostics). The project **supports hot reloading of YAML rule files** so the AFDD maintainer can tune faults without restarting. Rule YAML

contains fault logic and **params** (thresholds, tolerances, windows); the FDD loop loads from `rules_dir` every run — no restart.

Fault definitions: Each FDD run syncs the loaded rules into the `fault_definitions` table (fault_id, name, category, equipment_types). When you add or edit a rule (via frontend upload or by editing a file in `rules_dir`), the next run updates the DB and the Faults UI reflects the change. From the frontend you can also click **Sync definitions** to update the definitions table immediately.

1. **Add or edit** rules: use the Faults page (upload/paste YAML, or choose file) or edit files in `stack/rules/*.yaml`. Change `params` (e.g. `tolerance`, `rolling_window`) to tune sensitivity.
2. **Run** FDD: wait for the next scheduled run (per `rule_interval_hours` and `lookback_days` in platform config), or trigger with `touch config/.run_fdd_now` or `POST /run-fdd` (see Appendix: API Reference). Or use **Sync definitions** in the UI to only refresh the definitions table.
3. **View** fault results in the React Faults/Plots views or in Grafana (see Grafana SQL cookbook). Every run reloads all rules from disk — hot reload.

38.3 Rule types

Type	Purpose
`bounds`	Value outside `[low, high]`
`flatline`	Rolling spread < tolerance (stuck sensor)
`hunting`	Excessive state changes (PID hunting)
`expression`	Custom pandas/numpy expression
`oa_fraction`	OA fraction vs design airflow error
`erv_efficiency`	ERV effectiveness out of range

38.4 YAML structure

Inputs use **Brick class names** only; the Brick TTL (SPARQL) resolves each to a DataFrame column via external timeseries references. No `column` in YAML.

```yaml
name: sensor_bounds
type: bounds
flag: bad_sensor
inputs:
  Outside_Air_Temperature_Sensor:
```

```
            brick: Outside_Air_Temperature_Sensor
          Supply_Air_Temperature_Sensor:
            brick: Supply_Air_Temperature_Sensor
        params:
          low: 40
          high: 90
          rolling_window: 6   # optional: require N consecutive True
samples before flagging
```

Per-rule rolling window: Set `params.rolling_window` (e.g. `6`) in a rule to require that many consecutive True samples before the fault is flagged. Omit for "flag on any True." See `sensor_flatline.yaml`, `weather_temp_stuck.yaml`.

38.5 Input resolution (100% Brick)

- **Brick TTL:** Each rule input declares a Brick class (e.g. `brick: Supply_Air_Temperature_Sensor`). The engine resolves Brick points via SPARQL to their external timeseries reference (`ref:TimeseriesReference` / `ref:hasTimeseriesId`), yielding the DataFrame column name. See Brick timeseries storage.
- **Column map:** Built from the Brick model (e.g. `rdfs:label` or external_id) → column names; no `column` in rule YAML.

38.6 Running rules

- **Platform:** FDD loop loads rules from `rules_dir` (default `stack/rules`) each run; edit YAML and trigger a run to see changes in Grafana.
- **Standalone:** `RuleRunner(rules_path=...)` or `RuleRunner(rules=[...])`; call `run(df, ...)`.

38.7 Engineering metadata (223P-style) and energy-style analytics

FDD rules consume **time-series columns** built from the Brick graph (`ref:TimeseriesReference` + `ofdd:mapsToRuleInput`). **Engineering** data (design CFM, cooling tons, heating MBH, `s223:*` connection topology) lives on **equipment** in the DB and in the **same TTL** after import.

That split is intentional: the RuleRunner stays fast and pandas-centric. **Downstream**, you can combine:

- **Fault results** (which equipment, which fault, when), and

- **SPARQL** (or export TTL) for **rated capacity** and **topology** on that equipment, and
- **SQL** on `timeseries_readings` for duty estimates over the fault window

…to approximate **energy penalties** or rank impact (see §5 in `examples/223P_engineering/README.md` and Data model engineering (Brick + 223P MVP)).

38.8 Cookbook

See Expression Rule Cookbook for AHU, chiller, weather, and advanced recipes.

39 Expression Rule Cookbook

40 Expression Rule Cookbook

A reference for building fault detection rules in open-fdd. Rules use **YAML** with **expression** type: when the expression evaluates to **True**, a fault is flagged. open-fdd injects **NumPy** as `np` into expression evaluation, so you can use `np.maximum`, `np.abs`, `np.sqrt`, etc. for vectorized math. **Tuning:** Change `params` in YAML and trigger an FDD run (or wait for the schedule); the platform hot-reloads rules each run, so no restart is needed. See Fault rules overview and Configuration (`rules_dir` is RDF-driven via GET `/config`).

> **`column_map` — platform vs library:** On the **full AFDD stack**, **`fdd-loop`** runs **`run_fdd_loop`**, which builds **`column_map`** from **`config/data_model.ttl`** using **`BrickTtlColumnMapResolver`** in **`openfdd_stack.platform.brick_ttl_resolver`** (**rdflib** is a stack dependency, not an **`open-fdd`** dependency). Recipes below stay **Brick-oriented** and assume that path. When you run **`RuleRunner`** yourself (**`pip install open-fdd`**, notebooks, external IoT pipelines), **you** supply **`column_map`** as a **dict** (logical / Brick-class keys → your DataFrame column names), or plug a custom **`ColumnMapResolver`** if you extend the platform loop. The rule YAML shape does not change; only **who** builds the map changes. See Engine-only deployment and external IoT pipelines and The optional openfdd-engine package.

40.1 100% Brick-model driven (no column in rules)

Rule inputs use **Brick class names** only (e.g. `Supply_Air_Temperature_Sensor`). The **Brick TTL** is the source of truth: SPARQL resolves each Brick point to its external timeseries reference (`ref:TimeseriesReference` / `ref:hasTimeseriesId`), which yields the DataFrame column. Open-FDD embraces Brick timeseries storage at the heart of FDD — **rules never reference `column`**; the data model provides the mapping. See the SPARQL cookbook (Recipe 5) and run similar queries via Data Model Testing or `scripts/automated_testing/sparql/`.

- **Platform (DB):** Points have `brick_type` and `external_id`; TTL is built from DB. PATCH points or use data-model import to set `brick_type` (e.g. `Supply_Air_Temperature_Sensor` for SA-T).
- **Disambiguation:** When multiple points share a Brick class, use `ofdd:mapsToRuleInput` in TTL; the runner resolves `BrickClass|rule_input`.

40.2 How to define expressions

1. **Inputs** — Declare Brick classes only (`brick: Supply_Air_Temperature_Sensor`). The runner resolves these via the Brick TTL (SPARQL) and Brick external timeseries references to DataFrame columns. Do not add `column` in YAML.
2. **Params** — Thresholds and constants go in `params`. Reference by name (e.g. `err_thresh`, `vfd_max`).
3. **Expression** — Must evaluate to a boolean Series (True = fault). Use `&` (AND), `|` (OR), `~` (NOT). Use `.diff()`, `.rolling()`, `.notna()` for time-series logic.

Minimal example (TTL-driven; no column):

```yaml
name: high_temp_check
type: expression
flag: high_temp_flag

inputs:
  Supply_Air_Temperature_Sensor:
    brick: Supply_Air_Temperature_Sensor

params:
  max_temp: 90.0

expression: |
  Supply_Air_Temperature_Sensor > max_temp
```

The Brick TTL (and SPARQL) maps each Brick class to the actual column via external timeseries references. Set `points.brick_type` in the DB and sync TTL so the engine can resolve columns from the Brick model.

40.3 Signal scaling (0–1 fraction vs 0–100 percent)

Many cookbook thresholds (e.g. `drv_hi_frac: 0.93`, comparisons to `0.01`) assume **command signals are in 0–1** (fraction of full scale). In the field, BACnet and some historians often expose **0–100** (percent) for the same semantics.

- **`type: expression` rules** use the DataFrame columns as-is. If your VFD command is 0–100 but the YAML uses `0.93`, either fix the data at the source, document units in the data model, or **scale in the expression**.
- **Built-in** rule types **`hunting`** and **`oa_fraction`** automatically treat signals as percent when **any** sample in the series is `> 1` (they divide by 100). **`expression`** rules do **not** apply that heuristic unless you do it explicitly.

Recommended patterns in expressions

1. **`normalize_cmd(series)`** (open-fdd **2.3+**) — Injected next to `np`. Same heuristic as `hunting`/`oa_fraction`: if any finite value is `> 1`, the whole series is divided by 100; otherwise left as-is. Non-numeric cells become NaN.

   ```
   (normalize_cmd(Supply_Fan_Speed_Command) >= drv_hi_frac -
   drv_near_hi)
   ```

2. **Explicit `np.where`** (works on all open-fdd versions):

   ```
   (np.where(Supply_Fan_Speed_Command > 1,
   Supply_Fan_Speed_Command / 100.0, Supply_Fan_Speed_Command) >=
   0.93)
   ```

Mixed 0–1 and 0–100 in one series is ambiguous—prefer consistent scaling per point in the **data model** or **trend export**.

Engine validation (open-fdd 2.3+)

- Merged rule **`params`** are coerced/validated with **Pydantic** for `schedule` and `weather_band` (string numerics for scalars are best-effort coerced).
- **`RuleRunner.run(..., input_validation='warn'|'strict')`** checks that mapped columns exist and are **largely numeric** after `pd.to_numeric`; **`strict`** raises before evaluating expressions.

- On the **AFDD stack**, set **`OFDD_FDD_STRICT_RULES=true`** to enable strict validation and **`skip_missing_columns=False`** when the installed open-fdd is **≥ 2.3** (older wheels still get fail-fast on missing columns in strict mode, without `input_validation`).

When **`skip_missing_columns=True`** (production default), rules that hit missing names/columns are **skipped** and a **warning** is logged (rule name, flag, exception).

40.4 AHU rules (reference-style)

The following rules follow common industry practice for air-handling fault detection. Rules A through M are adapted from ASHRAE Guideline 36 (GL36) AFDD guidance. Thresholds and logic are tunable; adjust params for your site.

40.4.1 Rule A — Duct static below setpoint at full fan speed

Static pressure under setpoint while supply fan runs near maximum. May indicate duct leakage, undersized fan, or terminal damper issues. *Adapted from GL36 AFDD guidance.*

```yaml
        name: duct_static_low_at_full_speed
        description: Static pressure below setpoint when fan at full speed
(GL36-inspired)
        type: expression
        flag: rule_a_flag
        equipment_type: [VAV_AHU]

        inputs:
          Supply_Air_Static_Pressure_Sensor:
            brick: Supply_Air_Static_Pressure_Sensor
          Supply_Air_Static_Pressure_Setpoint:
            brick: Supply_Air_Static_Pressure_Setpoint
          Supply_Fan_Speed_Command:
            brick: Supply_Fan_Speed_Command

        params:
          sp_margin: 0.12
          drv_hi_frac: 0.93
          drv_near_hi: 0.06

        expression: |
          (Supply_Air_Static_Pressure_Sensor <
Supply_Air_Static_Pressure_Setpoint - sp_margin) &
(Supply_Fan_Speed_Command >= drv_hi_frac - drv_near_hi)
```

40.4.2 Rule B — Blended air temp below expected band

Blended air temp should lie between outdoor and return. If below both (minus tolerance), suspect sensor or mixing fault. *Adapted from GL36 AFDD guidance.*

```
        name: blend_temp_below_band
        description: Blended air temp below expected range (OAT/RAT)
(GL36-inspired)
        type: expression
        flag: rule_b_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Mixed_Air_Temperature_Sensor:
            brick: Mixed_Air_Temperature_Sensor
          Return_Air_Temperature_Sensor:
            brick: Return_Air_Temperature_Sensor
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Supply_Fan_Speed_Command:
            brick: Supply_Fan_Speed_Command

        params:
          blend_tol: 1.15
          rat_tol: 1.15
          oat_tol: 1.15

        expression: |
          (Mixed_Air_Temperature_Sensor - blend_tol <
np.minimum(Return_Air_Temperature_Sensor - rat_tol,
Outside_Air_Temperature_Sensor - oat_tol)) &
(Supply_Fan_Speed_Command > 0.01)
```

40.4.3 Rule C — Blended air temp above expected band

Blended air temp above the higher of OAT and RAT (plus tolerance) indicates mixing or sensor fault. *Adapted from GL36 AFDD guidance.*

```
        name: blend_temp_above_band
        description: Blended air temp above expected range (OAT/RAT)
(GL36-inspired)
        type: expression
        flag: rule_c_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Mixed_Air_Temperature_Sensor:
```

```
            brick: Mixed_Air_Temperature_Sensor
          Return_Air_Temperature_Sensor:
            brick: Return_Air_Temperature_Sensor
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Supply_Fan_Speed_Command:
            brick: Supply_Fan_Speed_Command

        params:
          blend_tol: 1.15
          rat_tol: 1.15
          oat_tol: 1.15

        expression: |
          (Mixed_Air_Temperature_Sensor - blend_tol >
np.maximum(Return_Air_Temperature_Sensor + rat_tol,
Outside_Air_Temperature_Sensor + oat_tol)) &
(Supply_Fan_Speed_Command > 0.01)
```

Hunting/oscillation — see rule types (hunting).

40.4.4 Rule D — Discharge air cold when heating commanded

Discharge air temp below blended air when heating valve is open. Indicates heating coil or valve failure. *Adapted from GL36 AFDD guidance.*

```
        name: discharge_cold_when_heating
        description: Discharge air below blended air when heating active
(GL36-inspired)
        type: expression
        flag: rule_d_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Mixed_Air_Temperature_Sensor:
            brick: Mixed_Air_Temperature_Sensor
          Supply_Air_Temperature_Sensor:
            brick: Supply_Air_Temperature_Sensor
          Valve_Command:
            brick: Valve_Command
          Supply_Fan_Speed_Command:
            brick: Supply_Fan_Speed_Command

        params:
          blend_tol: 1.15
          sat_tol: 1.15
          fan_delta_t: 0.55

        expression: |
```

```
          (Supply_Air_Temperature_Sensor + sat_tol <=
Mixed_Air_Temperature_Sensor - blend_tol + fan_delta_t) &
(Valve_Command > 0.01) & (Supply_Fan_Speed_Command > 0.01)
```

OA fraction — see rule types (oa_fraction).

40.4.5 Rule E — SAT too low with full heating

Heating valve fully open but SAT remains below setpoint. Indicates undersized coil or valve failure. *Adapted from GL36 AFDD guidance.*

```
        name: sat_too_low_full_heating
        description: SAT below setpoint with heating valve fully open
(GL36-inspired)
        type: expression
        flag: rule_e_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Supply_Air_Temperature_Sensor:
            brick: Supply_Air_Temperature_Sensor
          Supply_Air_Temperature_Setpoint:
            brick: Supply_Air_Temperature_Setpoint
          Valve_Command:
            brick: Valve_Command
          Supply_Fan_Speed_Command:
            brick: Supply_Fan_Speed_Command

        params:
          supply_err_thres: 1.0

        expression: |
          (Supply_Air_Temperature_Sensor < Supply_Air_Temperature_Setpoint
- supply_err_thres) & (Valve_Command > 0.9) &
(Supply_Fan_Speed_Command > 0)
```

40.4.6 Rule F — SAT/MAT mismatch in economizer mode

In economizer mode (min mechanical cooling), SAT should approximate MAT. Large deviation suggests coil bypass or sensor error. *Adapted from GL36 AFDD guidance.*

```
        name: discharge_blend_mismatch_econ
        description: Discharge and blended air diverge in economizer mode
(GL36-inspired)
        type: expression
        flag: rule_f_flag
        equipment_type: [AHU, VAV_AHU]
```

```
        inputs:
          Mixed_Air_Temperature_Sensor:
            brick: Mixed_Air_Temperature_Sensor
          Supply_Air_Temperature_Sensor:
            brick: Supply_Air_Temperature_Sensor
          Damper_Position_Command:
            brick: Damper_Position_Command
          Valve_Command:
            brick: Valve_Command

        params:
          fan_delta_t: 0.55
          blend_tol: 1.15
          sat_tol: 1.15
          econ_min_open: 0.12

        expression: |
          (np.abs(Supply_Air_Temperature_Sensor - fan_delta_t -
Mixed_Air_Temperature_Sensor) > np.sqrt(sat_tol**2 +
blend_tol**2)) & (Damper_Position_Command > econ_min_open) &
(Valve_Command < 0.1)
```

40.4.7 Rule G — Ambient too warm for free cooling

Outside air temperature exceeds SAT setpoint while economizer is active and mechanical cooling is off. Economizer should not be providing “free” cooling under these conditions. *Adapted from GL36 AFDD guidance.*

```
        name: ambient_warm_free_cool
        description: Outdoor air above setpoint in free cooling mode
(GL36-inspired)
        type: expression
        flag: rule_g_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Supply_Air_Temperature_Setpoint:
            brick: Supply_Air_Temperature_Setpoint
          Damper_Position_Command:
            brick: Damper_Position_Command
          Valve_Command:
            brick: Valve_Command

        params:
          oat_tol: 1.15
          fan_delta_t: 0.55
          sat_tol: 1.15
          econ_min_open: 0.12
```

```
        expression: |
          (Outside_Air_Temperature_Sensor - oat_tol >
Supply_Air_Temperature_Setpoint - fan_delta_t + sat_tol) &
(Damper_Position_Command > econ_min_open) & (Valve_Command < 0.1)
```

40.4.8 Rule H — Ambient vs blended mismatch (econ + mech cooling)

When both economizer and mechanical cooling are active, MAT should approach OAT. Large deviation suggests inadequate mixing or damper fault. *Adapted from GL36 AFDD guidance.*

```
        name: ambient_blend_mismatch_econ_mech
        description: Outdoor and blended air diverge in econ+mech cooling
(GL36-inspired)
        type: expression
        flag: rule_h_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Mixed_Air_Temperature_Sensor:
            brick: Mixed_Air_Temperature_Sensor
          Valve_Command:
            brick: Valve_Command
          Damper_Position_Command:
            brick: Damper_Position_Command

        params:
          oat_tol: 1.15
          blend_tol: 1.15

        expression: |
          (np.abs(Mixed_Air_Temperature_Sensor -
Outside_Air_Temperature_Sensor) > np.sqrt(blend_tol**2 +
oat_tol**2)) & (Valve_Command > 0.01) & (Damper_Position_Command >
0.9)
```

40.4.9 Rule I — Ambient vs blended mismatch (econ-only)

In economizer-only mode, MAT should match OAT. Deviation indicates damper or mixing fault. *Adapted from GL36 AFDD guidance.*

```
        name: ambient_blend_mismatch_econ
        description: Outdoor and blended air diverge in economizer-only
mode (GL36-inspired)
```

```
        type: expression
        flag: rule_i_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Mixed_Air_Temperature_Sensor:
            brick: Mixed_Air_Temperature_Sensor
          Damper_Position_Command:
            brick: Damper_Position_Command

        params:
          oat_tol: 1.15
          blend_tol: 1.15

        expression: |
          (np.abs(Mixed_Air_Temperature_Sensor -
Outside_Air_Temperature_Sensor) > np.sqrt(blend_tol**2 +
oat_tol**2)) & (Damper_Position_Command > 0.9)
```

40.4.10 Rule J — Discharge above blended in cooling

SAT exceeds MAT when cooling (econ+mech or mech-only) is active. Indicates underperforming cooling coil or valve. *Adapted from GL36 AFDD guidance.*

```
        name: discharge_above_blend_cooling
        description: Discharge air above blended air in cooling modes
(GL36-inspired)
        type: expression
        flag: rule_j_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Supply_Air_Temperature_Sensor:
            brick: Supply_Air_Temperature_Sensor
          Mixed_Air_Temperature_Sensor:
            brick: Mixed_Air_Temperature_Sensor
          Valve_Command:
            brick: Valve_Command
          Damper_Position_Command:
            brick: Damper_Position_Command

        params:
          fan_delta_t: 0.55
          blend_tol: 1.15
          sat_tol: 1.15
          econ_min_open: 0.12
```

```
        expression: |
          (Supply_Air_Temperature_Sensor > Mixed_Air_Temperature_Sensor +
np.sqrt(sat_tol**2 + blend_tol**2) + fan_delta_t) &
(((Damper_Position_Command > 0.9) & (Valve_Command > 0)) |
((Damper_Position_Command <= econ_min_open) & (Valve_Command >
0.9)))
```

40.4.11 Rule K — Discharge above setpoint in full cooling

SAT above setpoint with cooling at full capacity. Suggests undersized coil or plant limits. *Adapted from GL36 AFDD guidance.*

```
        name: discharge_above_sp_full_cool
        description: Discharge air above setpoint in full cooling mode
(GL36-inspired)
        type: expression
        flag: rule_k_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Supply_Air_Temperature_Sensor:
            brick: Supply_Air_Temperature_Sensor
          Supply_Air_Temperature_Setpoint:
            brick: Supply_Air_Temperature_Setpoint
          Valve_Command:
            brick: Valve_Command
          Damper_Position_Command:
            brick: Damper_Position_Command

        params:
          sat_tol: 1.15
          econ_min_open: 0.12

        expression: |
          (Supply_Air_Temperature_Sensor > Supply_Air_Temperature_Setpoint
+ sat_tol) & (((Damper_Position_Command > 0.9) & (Valve_Command >
0.9)) | ((Damper_Position_Command <= econ_min_open) &
(Valve_Command > 0.9)))
```

40.4.12 Rule L — Cooling coil delta-T when inactive

Temperature drop across cooling coil when it should be off. Indicates leaking CHW valve or coil bypass. *Adapted from GL36 AFDD guidance.*

```
        name: clg_coil_drop_when_off
        description: Temperature drop across cooling coil when it should
be off (GL36-inspired)
        type: expression
```

```
        flag: rule_l_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Cooling_Coil_Entering_Air_Temperature_Sensor:
            brick: Cooling_Coil_Entering_Air_Temperature_Sensor
          Cooling_Coil_Leaving_Air_Temperature_Sensor:
            brick: Cooling_Coil_Leaving_Air_Temperature_Sensor
          Heating_Valve_Command:
            brick: Valve_Command
          Cooling_Valve_Command:
            brick: Valve_Command
          Damper_Position_Command:
            brick: Damper_Position_Command

        params:
          enter_tol: 1.15
          leave_tol: 1.15
          econ_min_open: 0.12

        expression: |
          ((Cooling_Coil_Entering_Air_Temperature_Sensor -
Cooling_Coil_Leaving_Air_Temperature_Sensor) >
np.sqrt(enter_tol**2 + leave_tol**2)) & (((Heating_Valve_Command >
0) & (Cooling_Valve_Command == 0) & (Damper_Position_Command <=
econ_min_open)) | ((Heating_Valve_Command == 0) &
(Cooling_Valve_Command == 0) & (Damper_Position_Command >
econ_min_open)))
```

40.4.13 Rule M — Heating coil delta-T when inactive

Temperature rise across heating coil when it should be off. Indicates leaking HW valve. *Adapted from GL36 AFDD guidance.*

```
        name: htg_coil_rise_when_off
        description: Temperature rise across heating coil when it should
be off (GL36-inspired)
        type: expression
        flag: rule_m_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Heating_Coil_Entering_Air_Temperature_Sensor:
            brick: Heating_Coil_Entering_Air_Temperature_Sensor
          Heating_Coil_Leaving_Air_Temperature_Sensor:
            brick: Heating_Coil_Leaving_Air_Temperature_Sensor
          Heating_Valve_Command:
            brick: Valve_Command
          Cooling_Valve_Command:
            brick: Valve_Command
```

```
          Damper_Position_Command:
            brick: Damper_Position_Command

        params:
          enter_tol: 1.15
          leave_tol: 1.15
          fan_delta_t: 0.55
          econ_min_open: 0.12

        expression: |
          ((Heating_Coil_Leaving_Air_Temperature_Sensor -
Heating_Coil_Entering_Air_Temperature_Sensor) >
np.sqrt(enter_tol**2 + leave_tol**2) + fan_delta_t) &
(((Heating_Valve_Command == 0) & (Cooling_Valve_Command == 0) &
(Damper_Position_Command > econ_min_open)) |
((Heating_Valve_Command == 0) & (Cooling_Valve_Command > 0) &
(Damper_Position_Command > 0.9)) | ((Heating_Valve_Command == 0) &
(Cooling_Valve_Command > 0) & (Damper_Position_Command <=
econ_min_open)))
```

Heat exchanger effectiveness — see rule types (erv_efficiency).

40.5 Central plant

40.5.1 Differential pressure at max pump speed

Variable-speed pump cannot meet differential pressure setpoint at full speed. Indicates piping issues, undersized pump, or blocked strainers.

```
        name: dp_below_sp_pump_max
        description: Differential pressure below setpoint with pump at
full speed
        type: expression
        flag: dp_pump_flag

        inputs:
          Differential_Pressure_Sensor:
            brick: Differential_Pressure_Sensor
          Differential_Pressure_Setpoint:
            brick: Differential_Pressure_Setpoint
          Pump_Speed_Command:
            brick: Pump_Speed_Command

        params:
          dp_margin: 2.2
          pmp_hi_frac: 0.93
          pmp_near_hi: 0.06
```

```
        expression: |
          (Differential_Pressure_Sensor < Differential_Pressure_Setpoint -
dp_margin) & (Pump_Speed_Command >= pmp_hi_frac - pmp_near_hi)
```

40.5.2 Plant flow high at max pump

Flow unusually high with pump at high speed. Suggests short circuit or flow meter error.

```
        name: flow_high_pump_max
        description: Water flow unusually high with pump at full speed
        type: expression
        flag: flow_high_flag

        inputs:
          Water_Flow_Sensor:
            brick: Water_Flow_Sensor
          Pump_Speed_Command:
            brick: Pump_Speed_Command

        params:
          flow_hi_limit: 1100.0
          pmp_hi_frac: 0.93
          pmp_near_hi: 0.06

        expression: |
          (Water_Flow_Sensor > flow_hi_limit) & (Pump_Speed_Command >=
pmp_hi_frac - pmp_near_hi)
```

40.5.3 Plant supply temp outside deadband

CHW supply temperature outside deadband while pump runs. Requires coil entering/leaving or plant supply temp sensors.

```
        name: plant_supply_temp_deadband
        description: Supply water temp outside deadband during pump
operation
        type: expression
        flag: chw_temp_fault

        inputs:
          Chilled_Water_Supply_Temperature_Sensor:
            brick: Chilled_Water_Supply_Temperature_Sensor
          Chilled_Water_Supply_Temperature_Setpoint:
            brick: Chilled_Water_Supply_Temperature_Setpoint
          Pump_Speed_Command:
            brick: Pump_Speed_Command

        params:
          sp_band: 2.2
```

```
        expression: |
          (Pump_Speed_Command > 0.01) &
((Chilled_Water_Supply_Temperature_Sensor <
Chilled_Water_Supply_Temperature_Setpoint - sp_band) |
(Chilled_Water_Supply_Temperature_Sensor >
Chilled_Water_Supply_Temperature_Setpoint + sp_band))
```

40.5.4 Chiller runtime over daily limit

Chiller running beyond a daily threshold (e.g. 23 hours). Often indicates over-cooling or schedules that bypass lockout. Use a rolling window sized for your data interval (e.g. 5-min data → 276 samples ≈ 23 h).

```
        name: chiller_excessive_runtime
        description: Chiller runtime exceeds daily threshold (rolling
window)
        type: expression
        flag: chiller_runtime_fault

        inputs:
          Chiller_Status:
            brick: Chiller_Status

        params:
          # For 5-min data: 23 hours ≈ 276 samples; max_runtime = count of
"on" samples in window
          rolling_samples: 276
          max_runtime_samples: 264

        expression: |
          Chiller_Status.rolling(window=rolling_samples).sum() >
max_runtime_samples
```

Note: Adjust `rolling_samples` and `max_runtime_samples` for your data interval. For 5-min data, 264 samples ≈ 22 hours.

40.6 Heat pumps

40.6.1 Discharge cold when heating

Flags when discharge air temperature is below a minimum (e.g. 80°F) while the supply fan is running. Indicates the heat pump is not heating effectively—possible issues with compressor, refrigerant, or reversing valve. Tunable via `min_discharge_temp`. Logic: if zone temp < 69°F (heating mode), the discharge should be warm; a cold discharge with the fan on and a cold zone indicates the heat pump is failing to heat.

```
        name: hp_discharge_cold_when_heating
        description: Discharge air below minimum when fan on and zone is
cold (heating mode)
        type: expression
        flag: hp_discharge_cold_flag
        equipment_type: [Heat_Pump]

        inputs:
          Supply_Air_Temperature_Sensor:
            brick: Supply_Air_Temperature_Sensor
          Zone_Temperature_Sensor:
            brick: Zone_Temperature_Sensor
          Supply_Fan_Status:
            brick: Supply_Fan_Status

        params:
          min_discharge_temp: 85
          zone_cold_threshold: 69.0
          fan_on_threshold: 0.01

        expression: |
          (Supply_Fan_Status > fan_on_threshold) &
(Zone_Temperature_Sensor < zone_cold_threshold) &
(Supply_Air_Temperature_Sensor < min_discharge_temp)
```

40.7 VAV zones

40.7.1 Excessive heating during warm weather

Reheat valve open when outdoor air is warm. Suggests over-cooling or setpoint issues.

```
        name: zone_reheat_warm_ambient
        description: Heating valve open when OAT is high
        type: expression
        flag: excessive_heating_flag
        equipment_type: [VAV]

        inputs:
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Reheat_Valve_Command:
            brick: Valve_Command

        params:
          t_amb_cutoff: 78.0
          reheat_open_min: 0.52
```

```
        expression: |
          (Outside_Air_Temperature_Sensor > t_amb_cutoff) &
(Reheat_Valve_Command > reheat_open_min)
```

40.7.2 Damper or valve at full open

Damper or reheat valve consistently at full open. Indicates override, undersized equipment, or control fault.

```
        name: zone_damper_valve_full_open
        description: Damper or valve at full open for extended period
        type: expression
        flag: damper_100_flag
        equipment_type: [VAV]

        inputs:
          Damper_Position_Command:
            brick: Damper_Position_Command

        params:
          full_open_pct: 97.5
          roll_samples: 105

        expression: |
          (Damper_Position_Command > full_open_pct) &
(Damper_Position_Command.rolling(roll_samples).min() >
full_open_pct)
```

40.7.3 Zone and IAQ bounds

For CO2 and zone temperature out-of-range checks, use the bounds rule type — `co2_bounds` and `zone_temp_bounds` examples.

40.8 Opportunistic rules (economizer & ventilation)

40.8.1 Economizing when outdoor conditions are unfavorable

Economizer active when outdoor air is too warm or humid. Use OAT (or enthalpy if available) vs. threshold.

```
        name: econ_active_warm_ambient
        description: OA damper open when outdoor conditions do not favor
economizing
        type: expression
        flag: econ_when_shouldnt_flag
```

```
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Damper_Position_Command:
            brick: Damper_Position_Command

        params:
          t_amb_econ_cutoff: 63.0
          dpr_econ_min: 0.42

        expression: |
          (Outside_Air_Temperature_Sensor > t_amb_econ_cutoff) &
(Damper_Position_Command > dpr_econ_min)
```

40.8.2 Mechanical cooling when econ could suffice

Cooling valve open when outdoor air is cool enough for economizer. Opportunity to reduce mechanical cooling.

```
        name: mech_cool_when_econ_available
        description: Mechanical cooling active when economizing could
suffice
        type: expression
        flag: cooling_when_econ_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Damper_Position_Command:
            brick: Damper_Position_Command
          Valve_Command:
            brick: Valve_Command

        params:
          t_amb_econ_cutoff: 63.0
          dpr_not_econ_max: 0.32

        expression: |
          (Outside_Air_Temperature_Sensor < t_amb_econ_cutoff) &
(Damper_Position_Command < dpr_not_econ_max) & (Valve_Command >
0.01)
```

40.8.3 Low ventilation (estimated OA fraction)

For units without airflow meters, estimate OA fraction from OAT, MAT, RAT. Flag when below minimum design OA.

```
        name: low_oa_fraction_estimated
        description: Estimated OA fraction below minimum (OAT, MAT, RAT
method)
        type: expression
        flag: low_vent_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Mixed_Air_Temperature_Sensor:
            brick: Mixed_Air_Temperature_Sensor
          Return_Air_Temperature_Sensor:
            brick: Return_Air_Temperature_Sensor
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
          Supply_Fan_Speed_Command:
            brick: Supply_Fan_Speed_Command

        params:
          oa_min_pct: 21.0
          t_rat_oat_min_gap: 2.2

        expression: |
          (Supply_Fan_Speed_Command > 0.01) &
(np.abs(Return_Air_Temperature_Sensor -
Outside_Air_Temperature_Sensor) > t_rat_oat_min_gap) &
(((Mixed_Air_Temperature_Sensor - Return_Air_Temperature_Sensor) /
(Outside_Air_Temperature_Sensor - Return_Air_Temperature_Sensor) *
100) < oa_min_pct)
```

Note: Guard against division-by-zero when outdoor and return temps are close.

40.8.4 Preheat over-conditioning

Preheat coil leaving temp higher than needed (e.g. above OAT when OAT > SAT SP, or above SAT SP when OAT < SAT SP). Indicates wasted heating energy.

```
        name: preheat_excess_temp
        description: Preheat coil leaving temp above required level
        type: expression
        flag: preheat_waste_flag
        equipment_type: [AHU, VAV_AHU]

        inputs:
          Preheat_Coil_Leaving_Air_Temperature_Sensor:
            brick: Preheat_Coil_Leaving_Air_Temperature_Sensor
          Supply_Air_Temperature_Setpoint:
            brick: Supply_Air_Temperature_Setpoint
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor
```

```
          Valve_Command:
            brick: Valve_Command

        params:
          excess_tol: 2.2

        expression: |
          (Valve_Command > 0.01) & (((Outside_Air_Temperature_Sensor >
Supply_Air_Temperature_Setpoint) &
(Preheat_Coil_Leaving_Air_Temperature_Sensor -
Outside_Air_Temperature_Sensor > excess_tol)) |
((Outside_Air_Temperature_Sensor <
Supply_Air_Temperature_Setpoint) &
(Preheat_Coil_Leaving_Air_Temperature_Sensor -
Supply_Air_Temperature_Setpoint > excess_tol)))
```

40.8.5 Blended air damper deviation

Expected MAT (from OAT, RAT, damper positions) differs from measured MAT. Indicates damper leakage or faulty mixing.

Requires airflow or damper position data; logic is more involved. A simplified version compares MAT to a weighted blend of OAT and RAT when dampers suggest significant OA.

40.9 Weather station

weather_temp_stuck (flatline) — see flatline rule type.

40.9.1 Unrealistic temperature spike

Temperature change between consecutive readings exceeds physical limit.

```
        name: weather_temp_spike
        description: Unrealistic temperature change between readings
        type: expression
        flag: fault_temp_spike

        inputs:
          Outside_Air_Temperature_Sensor:
            brick: Outside_Air_Temperature_Sensor

        params:
          spike_limit: 16.0

        expression: |
          Outside_Air_Temperature_Sensor.diff().abs() > spike_limit
```

40.9.2 RH bounds

For relative humidity out-of-range, use the bounds rule type — `rh_bounds` example.

40.9.3 Wind gust vs sustained

```yaml
name: weather_gust_lt_wind
description: Wind gust reported lower than sustained wind
type: expression
flag: fault_gust_lt_wind

inputs:
  Wind_Gust_Speed_Sensor:
    brick: Wind_Gust_Speed_Sensor
  Wind_Speed_Sensor:
    brick: Wind_Speed_Sensor

expression: |
  Wind_Gust_Speed_Sensor.notna() & Wind_Speed_Sensor.notna() &
(Wind_Gust_Speed_Sensor < Wind_Speed_Sensor)
```

40.10 Sensor validation (bounds & flatline)

Use the bounds and flatline rule types for generic sensor checks. Typical bounds:

Sensor type	Min	Max
Zone temp	40	100
Supply air temp	40	150
Air pressure (inH2O)	-5	10
RH	0	100
Chilled water temp	35	100
Hot water temp	50	212
Condenser water	50	110
CO2 (ppm)	400	2000

40.11 Follow-up (RFCs): multi-ontology selectors & cookbook matrix

Multi-ontology rule YAML (e.g. optional `selectors.brick` / `.haystack` / `.dbo` / `.s223` under each input) is a **separate RFC** — it would feed the same **`column_map`** / resolver layer while the **full AFDD stack stays Brick-only by default** for v1. Track design discussion on GitHub **#122** and follow-on issues.

Cookbook matrix / generator: When a selector schema is stable, we can add a **generated appendix** (logical input × Brick × Haystack × DBO × 223P notes) from a single CSV/YAML source so the long cookbook does not drift. Until then, see the **open-fdd** repo's **`examples/column_map_resolver_workshop/simple_ontology_demo.py`** (with **`simple_ontology_rule.yaml`**).

Next: Fake fault schedule / flatline monitoring

41 Test bench rule catalog

42 Test bench rule catalog

Open-FDD ships **two** default rules under `stack/rules/` (`sensor_bounds.yaml`, `sensor_flatline.yaml`). For **extended HVAC, chiller, heat-pump, and weather expression rules** used in lab automation and cookbook examples, the repository also keeps a **reference set** under **`openclaw/bench/rules_reference/`** (not loaded until you copy or upload them).

Each file below links to the copy on **GitHub** (`master` branch) so the published docs site stays in sync with the repo.

42.1 Reference YAML files

File	GitHub
`ahu_fc1.yaml`	link
`ahu_fc2.yaml`	link
`ahu_fc3.yaml`	link
`ahu_fc4.yaml`	link
`ahu_fc5.yaml`	link
`ahu_fc6.yaml`	link

File	GitHub
`ahu_fc7.yaml`	link
`ahu_fc8.yaml`	link
`ahu_fc9.yaml`	link
`ahu_fc10.yaml`	link
`ahu_fc11.yaml`	link
`ahu_fc12.yaml`	link
`ahu_fc13.yaml`	link
`ahu_fc14.yaml`	link
`ahu_fc15.yaml`	link
`ahu_fc16.yaml`	link
`ahu_rule_a.yaml`	link
`chiller_flow_fc2.yaml`	link
`chiller_pump_fc1.yaml`	link
`hp_discharge_cold_when_heating.yaml`	link
`sensor_bounds.yaml`	link
`sensor_flatline.yaml`	link
`weather_gust_lt_wind.yaml`	link
`weather_rh_out_of_range.yaml`	link
`weather_temp_spike.yaml`	link
`weather_temp_stuck.yaml`	link

42.2 Related docs

- Fault rules overview — where live rules live (`stack/rules`) and how to upload or sync.
- Expression rule cookbook — AHU, chiller, weather patterns that match many of these files.
- OpenClaw lab README — bench layout including `rules_reference/` and E2E automation.

The former **open-fdd-automated-testing** repo is superseded by the **`openclaw/`** tree in this repository; do not link automation to a separate rules tree.

43 YAML rules → Pandas (under the hood)

44 YAML rules → Pandas DataFrames (under the hood)

This note is for engineers who want to see **exactly** how Open-FDD turns **rule YAML on disk** into **pandas operations** on **time-series DataFrames**—and where Brick TTL fits in. It complements Fault rules overview and standalone CSV / pandas.

44.1 1. YAML never becomes “one giant DataFrame of rules”

Rule files are **configuration**, not tabular data. Open-FDD:

1. Reads each `*.yaml` in `rules_dir` with PyYAML (`yaml.safe_load`) into a **Python `dict`** per file (`open_fdd.engine.runner.load_rule` / `load_rules_from_dir`).
2. Keeps those dicts in a **`list[dict]`** inside `RuleRunner` (`RuleRunner(rules=...)` or `RuleRunner(rules_path=...)`).
3. At evaluation time, walks that list and, for each rule, computes a **boolean mask** (`pandas.Series`) aligned to the **sensor DataFrame’s index**, then writes a **new column** for the fault flag (e.g. `my_rule_flag`).

So: **rules = list of dicts in memory**; **data = one wide-ish DataFrame** of timestamps and point columns.

44.2 2. From Postgres rows to a site/equipment DataFrame

The continuous loop (`openfdd_stack.platform.loop.run_fdd_loop`) loads telemetry with SQL, then reshapes it for pandas:

1. **Query** `timeseries_readings` joined to `points` for the time window (`load_timeseries_for_site` / `load_timeseries_for_equipment`).
2. **Build a long table**: `pd.DataFrame(rows)` with columns like `ts`, `external_id`, `value`.

3. **Pivot to wide**: `df.pivot_table(index="ts", columns="external_id", values="value")` so each point is a column (one column per `external_id`).
4. **Rename columns** using a **column map** (on the stack: from TTL via **`openfdd_stack.platform.brick_ttl_resolver`**; with **`pip install open-fdd`**: dict or manifest).
5. **Add** `timestamp = pd.to_datetime(df["ts"])` for time-based checks.

Under the hood, pandas is doing **grouped aggregation in the pivot** (duplicate `(ts, external_id)` would aggregate), then **aligning** all series to a common `DatetimeIndex` (implicit via the pivot index).

44.3 3. What `RuleRunner.run` does to that DataFrame

`RuleRunner.run` (`open_fdd.engine.runner`):

1. **`result = df.copy()`** — rules never mutate the caller's frame in place (callers can still hold a reference to the original).
2. For **each** rule dict:
   - Derives **`flag_name`** from `flag` or `{name}_flag`.
   - Calls **`_evaluate_rule`** → returns a **boolean `Series`** (fault mask) aligned to `result.index`.
   - Optionally applies a **rolling window** on the mask: `mask.astype(int).rolling(window=rw).sum() >= rw` so a fault must persist for N consecutive samples.
   - Assigns **`result[flag_name] = ...`** as integer 0/1.

So each rule adds **one column** of flags; the frame grows **width-wise**, not row-wise.

44.4 4. How a YAML rule dict becomes pandas expressions

Inside `_evaluate_rule`, Open-FDD branches on `rule["type"]` (e.g. `bounds`, `flatline`, `expression`, …):

- **`column_map` resolution**: For each logical input key, the runner picks a **DataFrame column name**. If the YAML input has a **`brick`** class, the global `column_map` from TTL is consulted first (`brick` → column label), so the **same YAML** can run against different exports as long as TTL maps Brick classes to the right columns.

- **Bounds / thresholds**: Typical pattern is `(series < low) | (series > high)` using **vectorized** comparisons — these are **numpy ufuncs** under pandas, no Python `for` over rows.
- **Expressions**: String expressions may be evaluated in a **restricted eval** context (`open_fdd.engine.checks.check_expression`) with named series bound to `result[col]` — still vectorized.

If a required column is missing, the runner either **raises** or **skips** the rule (`skip_missing_columns=True`), depending on the call site.

44.5 5. Hot reload: YAML vs DataFrame lifetime

`run_fdd_loop` calls **`load_rules_from_dir(rules_path)` every run**, then filters by equipment types from TTL, then **`RuleRunner(rules=rules)`**. There is **no** long-lived compiled rule object on disk—editing YAML affects the **next** scheduled run (or the next manual `POST /run-fdd`). The **DataFrame** exists only for the duration of that run's Python call stack (load → run → persist results).

44.6 6. Mental model checklist

Artifact	In-memory shape	pandas role
Rule YAML files	N/A (on disk)	None until loaded
Loaded rules	`list[dict]`	None
Telemetry window	`DataFrame` (time × points)	pivot, datetime index, column rename
Rule output	Same index, +flag columns	vectorized masks, optional `rolling`
`fault_results` / DB	Written from row-wise `FDDResult`	After flags are computed

For a **minimal** example of rules + CSV (no DB), see standalone CSV / pandas and unit tests in `open_fdd/tests/engine/test_runner.py`.

45 Fault rules for HVAC

46 Fault rules for HVAC

FDD fault rule types, YAML format, and expression cookbook for HVAC and building systems. Rules are **100% Brick-model driven**: inputs reference Brick classes only; column mapping comes from the Brick TTL and Brick external timeseries references (timeseries storage), which Open-FDD embraces at the heart of FDD.

Page	Description
Overview	Rule types, `stack/rules`, hot reload, Brick inputs
Test bench rule catalog	Links to reference YAML under `openclaw/bench/rules_reference/` (AHU FC, chillers, weather, etc.)

47 Cloud export example

48 Cloud export example

Open-FDD runs **inside the building, behind the firewall**. It does not push data to the cloud itself. Cloud-based FDD providers, MSI (Monitoring, Savings, and Intelligence) vendors, and commissioning or IoT contractors run their **own** export process: a small script or edge gateway on the building or OT network that **pulls** from the Open-FDD API over the LAN and then sends that data to their cloud or analytics platform. See Behind the firewall; cloud export is vendor-led on the docs home.

This page describes the **cloud export example** in the repo: a minimal Python script that demonstrates how vendor X, Y, or Z could use the Open-FDD API to get fault and timeseries data and use it as a starting point for their cloud pipeline.

48.1 What the example does

The script `examples/cloud_export.py` calls standard Open-FDD download and analytics endpoints and prints the results. In a real integration you would replace the print steps with your own logic: POST to your REST API, write to S3, push to Azure IoT Hub or AWS IoT, send to SkySpark or another analytics platform, etc.

Step	Endpoint	Purpose
1	`GET /download/faults?format=json`	Fault results for programmatic ingestion (JSON)
2	`GET /download/faults?format=csv`	Fault CSV (Excel-friendly)
3	`GET /analytics/motor-runtime`	Motor runtime (data-model driven)
4	`GET /download/csv`	Timeseries wide-format CSV
5	`GET /analytics/fault-summary`	Fault counts by fault_id

All requests use date range and optional `site_id`; the API docs at `/docs` list full parameters.

48.2 Run the example

Prerequisites: Open-FDD API running (e.g. `http://localhost:8000`), `pip install httpx`.

```
# From repo root
python examples/cloud_export.py
python examples/cloud_export.py --site default --days 14
API_BASE=http://your-openfdd-host:8000 python examples/cloud_export.py
```

Docker: See `examples/cloud_export/README.md` for building and running the example in a container (e.g. on a vendor edge device that reaches the Open-FDD API over the LAN).

48.3 Use it as your starting point

1. **Clone or copy** `examples/cloud_export.py` into your integration codebase.

2. **Keep** the HTTP calls to `/download/faults`, `/download/csv`, and the analytics endpoints; adjust `API_BASE` (or env) to point at the customer's Open-FDD instance.
3. **Replace** the `print()` / local handling with your cloud send: REST POST, message queue, blob storage, or IoT hub. Add auth (e.g. API key or TLS client cert) if you expose the puller to a wider network.
4. **Schedule** the script (cron, systemd timer, or container restart policy) so you poll the Open-FDD API at the interval that fits your product (e.g. hourly or daily fault sync).

Open-FDD does not initiate outbound cloud connections or manage your data transmission; the example shows how your process can pull data from Open-FDD and then send it to your cloud for deeper insights.

49 Site VOLTTRON and the data plane (ZMQ)

50 Site VOLTTRON and the data plane (ZMQ)

This page is the **default architecture** for how telemetry reaches Open-FDD: **inside each building**, one or more **VOLTTRON** instances own the field layer. The Open-FDD application tier (Postgres, optional FastAPI + React, FDD jobs or agents) **does not speak BACnet, Modbus, or other OT buses** and does not replace the VOLTTRON message bus.

Whether you run Open-FDD **on-prem** (VM or appliance in the plant) or **in a cloud VPC**, the same rule applies: **field protocols and drivers live on site in VOLTTRON**; the app tier consumes **already-normalized** time series and metadata.

50.1 What runs where

Location	Responsibility
Building / site LAN	**VOLTTRON 9**: platform driver, BACnet proxy, Modbus or other drivers as *you* configure upstream—not in this repo. **Inter-agent traffic** uses the VOLTTRON **VIP** path over **ZeroMQ (ZMQ)**. Pub/sub

Location	Responsibility
	between agents uses the platform message bus VOLTTRON documents for your release. **RabbitMQ is not** part of this project's default story; do not assume a separate AMQP bus for core ingest.
Same site or central host	**SQL historian** (or equivalent writers) persisting topics into **Postgres / Timescale** with `tables_def` aligned to your deployment. Open-FDD's **SQL schema** (`sites`, `equipment`, `points`, `timeseries_readings`, faults, …) is the integration surface for rules and UI.
App tier (on-prem or cloud)	**Optional** FastAPI + React for Brick CRUD, SPARQL, export/import; **FDD** via `run_fdd_loop()` or agents reading the **same DSN**. No requirement to colocate the app tier with VOLTTRON; **the database** (and secure networking) is the contract.

50.2 How data actually flows

1. **Devices** → VOLTTRON drivers / proxies on the **OT LAN** (BACnet, Modbus, etc. **only inside VOLTTRON**, per your VOLTTRON config—not Open-FDD code).
2. **Agents** → publish/subscribe on the **ZMQ-backed** VOLTTRON bus; historians **archive** selected topics to SQL.
3. **Open-FDD** → reads **Postgres** (and optional API-driven graph work). **`openfdd_stack.volttron_bridge`** and **`external_id` / point keys** align historian rows with Brick-oriented metadata.

If something is not in **SQL** (or not mapped into Open-FDD tables), the **React** app and **FDD rules** cannot see it—**wire ingest is VOLTTRON's job**.

50.3 Implications for docs and APIs

- **REST `/bacnet/*` helpers** in the optional FastAPI app are **legacy / lab** paths for proxying an external gateway when *you* choose to run one; they are **not** the default product surface and **do not** mean Open-FDD hosts BACnet/IP. Prefer **VOLTTRON** documentation for discovery, writes, and scheduling on the bus.

- **`bacnet_device_id` / `object_identifier`** on points may still appear in the **data model** (imports, migration, RDF) as **opaque metadata**—not as proof that **this service** is polling BACnet.

50.4 See also

- VOLTTRON gateway, FastAPI, and data-model sync — layering Central, SQL, and optional API.
- VOLTTRON Central and AFDD parity (monorepo) — bootstrap and multi-site patterns.
- `afdd_stack/legacy/README.md` — removed Docker services (BACnet scraper, diy-bacnet container, …) for historical reference only.

Upstream: VOLTTRON documentation, SQL Historian.

51 RDF, Brick, and optional BACnet-shaped metadata

52 RDF, Brick, and optional BACnet-shaped metadata

Open-FDD models the building as a **Brick-oriented knowledge graph** in `config/data_model.ttl` when you run **FastAPI**. **CRUD**, **SPARQL**, and **data-model import/export** maintain sites, equipment, and points.

BACnet on the wire is **not** owned by Open-FDD in the default architecture: **site VOLTTRON** performs discovery and reads/writes devices; **historians** persist time series to **Postgres**. Any **BACnet-shaped RDF** in the graph is either **legacy** (from older deployments), **imported**, or produced only if **you** explicitly run **legacy** `/bacnet/point_discovery_to_graph` against a **separate** lab gateway — not something this documentation recommends for new ingest.

Preferred path: align **`external_id`** and SQL rows with VOLTTRON topics (Site VOLTTRON and the data plane (ZMQ)), use **SPARQL** for validation, and keep Brick metadata in sync via CRUD or import.

See also **Edge field buses (VOLTTRON)** (docs index entry under **BACnet** path for historical URLs).

53 Context and recordkeeping

54 Context and recordkeeping

Open-FDD context should be durable, visible, and reusable by both humans and agents.

54.1 Rule

If context affects repeatable operations, store it in versioned docs instead of leaving it in chat-only memory.

54.2 Good context to commit

- Verification strategy and sweep logic.
- BACnet/model assumptions that affect diagnostics.
- Operator playbook updates and recurring workflow improvements.
- Cross-check guidance for frontend/API parity and model-based telemetry.

54.3 Context anti-patterns

- One-machine-only tribal notes with no repo trace.
- Secrets in Markdown.
- Raw transcript dumps without distillation.

55 Operational states

56 Operational states

Open-FDD operations should explicitly identify environment mode before asserting correctness.

56.1 States

- `TEST_BENCH`: deterministic fake-device conditions or lab-only setup.
- `LIVE_HVAC`: production building telemetry and real-world operating behavior.

56.2 Why this matters

The same observed value can be expected in bench mode and suspicious in live mode. Mode-aware interpretation avoids false positives and noisy operations.

57 Big picture gaps and product suggestions

58 Big picture gaps and product suggestions

This is a running list of higher-level product, workflow, and documentation gaps discovered while operating Open-FDD and its automated testing stack.

The goal is to capture the things that matter beyond the immediate bug or test run.

58.1 Current themes

58.1.1 1. AI-assisted data modeling needs a stronger review-before-import flow

The current workflow makes it easy to jump from discovered BACnet points to imported semantic changes.

Suggested improvement:

- stage proposed site/equipment/point modeling changes
- let the user review them clearly before import
- make the before/after delta obvious

58.1.2 2. Graph reset behavior is easy to misunderstand

`POST /data-model/reset` does **not** mean “wipe everything.” It means:

- clear BACnet graph/orphans
- repopulate Brick from DB-only

Suggested improvement:

- clarify this distinction more aggressively in UI/docs
- keep repeating that full empty-state requires deleting all sites via CRUD first

58.1.3 3. Rule-aware minimal polling is a high-impact lever

The testing/modeling flow should emphasize that only points needed by active rules should be polled.

This reduces unnecessary BACnet traffic and makes Open-FDD deployments more realistic for live HVAC systems.

Suggested improvement:

- add rule-aware polling recommendations in product/docs
- highlight minimal polling sets for active rules

58.1.4 4. The gap between discovery and FDD-ready modeling is still wide

Discovery gets BACnet objects into the graph, but that is still far from:

- Brick semantics
- equipment relationships
- rule applicability
- optimization readiness

Suggested improvement:

- describe maturity stages more explicitly:
  - discovered
  - modeled
  - FDD-ready
  - operations-ready

58.1.5 5. Overnight testing still needs stronger generated evidence

The current overnight setup is much better than before, but it should mature into a real evidence pipeline:

- BACnet source evidence
- SPARQL/model evidence
- YAML rule evidence
- fault-result evidence
- docs/link review evidence
- container-log correlation

Suggested improvement:

- generate durable overnight reports by default (see `openclaw/reports/` templates in the repo)

58.1.6 6. Container log viewing should become part of the testing story

The container-log feature is not just an admin convenience. It should support:

- frontend/backend mismatch diagnosis
- scraper troubleshooting
- FDD loop troubleshooting
- overnight evidence correlation

Suggested improvement:

- make container-log review part of normal dev-testing and docs

58.1.7 7. Open-FDD needs a stronger live-HVAC operations mental model

The platform already contains pieces for:

- discovery
- modeling
- rules
- faults
- charts
- logs
- weather

But the product/docs should frame those around the real user contexts:

- controls engineer
- building operator / facilities manager
- AI agent assisting with diagnostics and optimization

58.2 Security / hardening context to remember

Relevant existing issue on the main Open-FDD repo:

- **#73** — Stack security hardening (DB, Caddy, secrets) — https://github.com/bbartling/open-fdd/issues/73

This should remain in the background context for future deployment and production-readiness work.

Also relevant:

- **#44** — `Test Caddy. Integrate and Define Secure Coding Best Practices` — https://github.com/bbartling/open-fdd/issues/44

58.3 Why this doc exists

A lot of the most important insight is not just “what failed,” but “what is still missing from the big picture.”

This file is meant to preserve those bigger observations so future clones, future agents, and future humans can build from them instead of rediscovering them from scratch.

59 VOLTTRON gateway, FastAPI, and data-model sync

60 VOLTTRON gateway, FastAPI, and data-model sync

This page is the **buildable** version of the layered stack: **each site's VOLTTRON owns all field protocols** (BACnet, Modbus, etc., **only inside VOLTTRON**), **ZMQ** VIP / pub-sub on the VOLTTRON bus, and the SQL historian (optional **TimescaleDB**). **Open-FDD does not speak those wires**; it keeps the **semantic layer, CRUD, SPARQL, and the React modeling UI** when you run FastAPI/React, and consumes **SQL** for FDD. It also answers: **can scripts + cron replace "always-on" sync?**

Docker in this repo: `afdd_stack/stack/docker-compose.yml` runs **Postgres/TimescaleDB** (and optional Grafana or Mosquitto profiles) only. It does **not** start the Open-FDD API, Caddy, React, or **any** field-bus gateway; use **site VOLTTRON** and **VOLTTRON Central** (or your own process manager) for ingest and fleet UI.

For a **long-range target** where **Brick is canonical on VOLTTRON or Central** and the **standalone FastAPI service is intentionally retired**, see **Target vision — Brick on VOLTTRON, Central, no FastAPI** (that path replaces FastAPI with explicit alternatives—it does not remove the need for an API surface).

60.1 Layered architecture (what to actually run)

```
  Building LAN (site VOLTTRON)              App tier (same DC,
VPC, or host)
  ----------------------------
---------------------------------

  VOLTTRON 9 + ZMQ VIP / pub-sub  ------>  Postgres / Timescale
  Field drivers (BACnet, Modbus, …        (historian tables +
Open-FDD schema)
   only inside VOLTTRON)

  Historian / ETL agents          ------>  optional FastAPI +
React
```

```
                                                (CRUD, SPARQL, TTL,
jobs)

  Custom agents (alarms, FDD, CSV) ------>  same SQL or HTTP to
API
```

Solid arrows: data flow. **Browsers and LLM tools** should use **FastAPI**, not raw historian SQL.

Rule of thumb: anything a **browser** or **LLM tool** touches should go through **FastAPI** (or a future thin gateway that exposes the same contracts). **VOLTTRON** should not become your public HTTP server for SPARQL and CRUD.

60.2 Where FastAPI is still required (or strongly advised)

Capability	Why not “agents only”
SPARQL (`POST /data-model/sparql`)	Needs the **unified graph** (rdflib), request concurrency, and a stable contract for the React **Data Model Testing** and automation.
CRUD (sites, equipment, points, import or export)	Transactional **Postgres** updates, validation, and orchestration already live here.
Serialize or reset graph (`POST /data-model/serialize`, `POST /data-model/reset`)	The in-memory graph and **TTL file** lifecycle are owned by the API process today.
Auth, websockets, jobs, downloads	Same process boundary; replacing it means reimplementing security and UX contracts.

In a **VOLTTRON-first** deployment, **all** field BACnet/Modbus traffic stays in **VOLTTRON** on the site LAN; keep **FastAPI + React** (from source) when you need **CRUD, SPARQL, and the modeling UI**, and run **FDD** as **agents** or batch jobs against SQL. Default **Compose** in this repo ships **no** field-bus or scraper containers — see Modular architecture, **Site VOLTTRON and the data plane (ZMQ)**, and **`afdd_stack/legacy/README.md`**.

60.3 Can cron + scripts maintain data-model sync?

Yes, for part of the problem. Treat sync as **two different pipelines**:

60.3.1 1) File sync (TTL on disk, backups, GitOps)

The platform already runs a **background thread** that writes `config/data_model.ttl` on an interval (`graph_sync_interval_min` in settings; see System overview). **Cron is redundant** for "write TTL occasionally" unless you want **explicit** checkpoints after batch jobs.

Use **cron or systemd timers** when:

- You want a **guaranteed** serialize after an external batch (ETL, Ansible, OpenClaw) finishes.
- The API process is **always up** and you only need to hit **`POST /data-model/serialize`**.

Repo helper: `afdd_stack/scripts/cron_ttl_serialize.sh` (see script header for env vars).

60.3.2 2) Semantic truth (Brick graph matches Postgres)

If **only** the FastAPI process updates the DB through CRUD, the in-memory graph and TTL stay aligned without cron.

If a **separate writer** (VOLTTRON agent, dbt, script) updates **Postgres** tables that back Brick (sites, equipment, points metadata), the **in-memory RDF graph** in the API process can be **stale** until you:

- **`POST /data-model/reset`** — repopulates **Brick from DB only**, rewrites TTL, and clears **non-Brick** triples that the API had loaded (see the route docstring in `openfdd_stack/platform/api/data_model.py`). **BACnet RDF is not** produced by Open-FDD in the default architecture.
- Or route topology changes through the **existing REST import or CRUD** so the graph updates inside the API.

Cron alone cannot fix stale in-memory state if nothing calls the API or restarts the process; you need at least **one** of: HTTP call to reset or serialize, **shared process** that owns the graph, or a **long-term refactor** (e.g. sidecar triple store).

60.3.3 3) Time-series values (live readings)

Historian topics (`devices/...`) are **not** the same as Brick TTL. Sync **readings** into Open-FDD's `timeseries_readings` (or future merged schema) with a **small ETL agent** on a schedule or on pub/sub. That is **orthogonal** to TTL serialize cron.

60.4 VOLTTRON Central

Use **VOLTTRON Central** when you need **multi-instance** operations and health views. Do **not** treat it as the primary **energy engineering** or **SPARQL** UI; keep those on **Open-FDD's React + FastAPI**.

60.5 Practical deployment profiles

Profile	Edge	App tier
A — Default monorepo compose	VOLTTRON (+ historian) on edge	Postgres/Timescale (+ optional Grafana); FastAPI/React **from source** when needed
B — VOLTTRON field, Open-FDD brain	VOLTTRON historian + ETL to app DB	FastAPI for model/ SPARQL; FDD via agents; cron or HTTP for serialize/reset after bulk DB writes
C — Fleet	Many Pis like profile B	Optional Central + one modeling API per site or shared read models
D — Legacy Docker stack	**Removed** diy-bacnet / scraper compose (fork only)	Do **not** use for new BACnet ingest — **legacy** README

60.6 References

- VOLTTRON **Historian framework** and **platform historian** identity: Historian framework.
- **SQL historian** and **TimescaleDB** connection params: SQLHistorian.

- Lab pattern for VOLTTRON 9 on a Pi: vibe_code_apps_6 in **py-bacnet-stacks-playground**.

61 Target vision — Brick on VOLTTRON, Central as hub, no FastAPI

62 Target vision — Brick on VOLTTRON, Central as hub, no FastAPI

This page captures a **long-range architecture** some deployments want: **Open-FDD runs as a layer across VOLTTRON (and optionally VOLTTRON Central)**, the **canonical model is Brick**, and the **standalone FastAPI service goes away**. Lab folders like **vibe_code_apps_7** stay **experiments and education**; they are not the shipping product, but they inform it.

62.1 What this vision is really asking for

Stated goal	Translation in engineering terms
Layer on **VOLTTRON Central**	One **control plane** (instance registry, health, optional plots) plus **shared storage** (often **PostgreSQL or Timescale** behind historians on each instance). Open-FDD logic becomes **agents + jobs + DB schema**, not a second “platform” beside VOLTTRON.
“Just a Brick model”	**Brick (and friends) is the only semantic schema** you commit to for topology and point identity. You **stop maintaining a parallel RDF universe** (e.g. separate BACnet RDF blobs) **unless** you still need discovery artifacts—in which case they are **import steps** into Brick, not a second permanent graph.

Stated goal	Translation in engineering terms
Get rid of FastAPI	You **remove the current `afdd_stack` FastAPI process** as the public API. **Something else** must still provide: **authenticated CRUD**, **queries** (SPARQL or SQL views over Brick), **file or export contracts**, and **any browser or LLM integration**. That “something” is **new work**—it does not appear for free inside Central’s UI.

62.2 Reality check — VOLTTRON Central is not a Brick server

VOLTTRON Central and the **platform historian** are built around **topics and time series** (`topics`, `data`, metadata)—not around **Brick classes, relationships, and SPARQL**. So “Open-FDD as a layer on top” means:

1. **Brick lives in your own tables** (or in a small triple store) in **PostgreSQL** (or attached to Central’s DB strategy), **designed by you**.
2. **Historian** continues to own **high-volume samples**; you **join** or **map** historian topics to **Brick `Point`** identities via stable IDs you control.
3. **Open-FDD’s rules engine** still needs **pandas-shaped** inputs: you **materialize** frames from historian plus Brick metadata in an **agent or scheduled job**, not by magic from Central alone.

Central is valuable for **many instances** and **operator visibility**; it is **not** a replacement for a **semantic application server** unless you build that server elsewhere.

62.3 If FastAPI goes away, what replaces it?

You are not eliminating **HTTP and business logic**—you are **moving** them. Realistic replacements:

Option	Role of “API”	Pros	Cons
A. VOLTTRON agents + VIP RPC	UIs or scripts call **`vctl`** / RPC from a **thin local gateway**	No FastAPI container; logic colocated with data	**No first-class browser REST** unless you add

Option	Role of “API”	Pros	Cons
			another layer
B. One minimal HTTP agent (FastAPI, Starlette, or Flask **inside** a single agent)	Same routes as today, smaller footprint	Reuse OpenAPI patterns; one process	Still “an HTTP stack”—just not the monorepo’s current layout
C. SQL + PostgREST or Hasura	**Brick tables** exposed as REST or GraphQL	Very little custom server code	SPARQL and complex graph rules need **SQL views** or a **sidecar**
D. Static React + BFF agent	Agent serves static build + a few JSON endpoints	Feels like “no big platform”	Same as B, different packaging

Deleting FastAPI without picking A–D leaves **no supported path** for the React app and OpenClaw-style automation described elsewhere in this repo.

62.4 “Brick only” and the rules engine

The **PyPI `open-fdd` engine** is **YAML + pandas**, not RDF at runtime. A **Brick-only** platform still needs a **deterministic bridge**: Brick metadata + historian samples → **DataFrame columns** the rules expect. That bridge belongs in **VOLTTRON agents** (or DB views + a small runner), not in Central’s stock UI.

62.5 Code landing zone (monorepo)

The first importable slice lives in Python package **`openfdd_stack.volttron_bridge`** (`afdd_stack/openfdd_stack/volttron_bridge/`): **device topic parsing**, **flattening** platform-driver style dicts, **mapping** flattens to Open-FDD `external_id` keys for engine rows, and **DB lookup** of points for an equipment name that matches a VOLTTRON driver device. Agents on the Pi can

depend on this package once it is installed from the monorepo. Unit tests: `afdd_stack/openfdd_stack/tests/platform/test_volttron_bridge.py`.

62.6 Suggested phased path (if the org commits to this)

1. **Freeze the product split** — Edge = VOLTTRON + historian; **Brick + alarms + FDD metadata = Postgres schema** you own (can live next to historian DB with clear boundaries).
2. **Stop growing BACnet-as-RDF** — Treat BACnet discovery as **ingest into Brick + external_id**, align with modular architecture "model" concerns.
3. **Extract** the smallest set of routes you still need (export or import, one SPARQL or SQL path) into **option B or C** above.
4. **Retire** the monorepo FastAPI service only when **feature parity** for your chosen users (modeling, energy, FDD runs) exists on the new layer.
5. **Central last** — Add **VOLTTRON Central** when **multiple** instances justify it; it is **optional** for a single-building Brick + historian deployment.

62.7 How this relates to other docs

- **VOLTTRON gateway, FastAPI, and data-model sync** describes the **practical near-term** split: VOLTTRON on the wire, **FastAPI** for graph and SPARQL until a deliberate cutover.
- This page is the **target-state** counterpoint: **same Brick truth**, **different process boundary**, **no long-lived FastAPI** once replacements exist.

62.8 Bottom line

- **Yes**, you can aim for **Open-FDD conceptual stack** (Brick + engine + alarms) **running across VOLTTRON and Central's ecosystem**, with **vibe_code_apps_7-style work** kept as **education only**.
- **No**, you cannot **delete FastAPI** today without **replacing** its responsibilities; **Central does not subsume** SPARQL, CRUD, and the current React contract by itself.

- **“Just Brick”** is a **schema and product simplification**, not a smaller amount of **software**—it is often **more discipline** in one graph instead of two.

63 Concepts

64 Concepts

Conceptual guides and worked examples for how Open-FDD fits into larger workflows.

Page	Description
Site VOLTTRON and the data plane (ZMQ)	**Start here:** each building runs VOLTTRON (ZMQ VIP / pub-sub); Open-FDD does **not** host BACnet/Modbus; data arrives via historian SQL and identity mapping.
RDF, Brick, and optional BACnet-shaped metadata	How the TTL graph relates to SQL; BACnet-shaped triples are optional/legacy, not default ingest.
Cloud export example	How a cloud vendor or MSI can pull fault and timeseries data from the Open-FDD API and send it to their cloud for deeper insights. Uses the `examples/cloud_export` script as a starting point.
Context and recordkeeping	Durable context policy: what belongs in versioned docs vs private local memory.
Operational states	Mode-aware interpretation between test bench and live HVAC environments.
Big picture gaps and product suggestions	Higher-level product and workflow gaps observed in the field.
VOLTTRON gateway, FastAPI, and data-model sync	VOLTTRON on the edge for **field ingest** and historian time series; optional FastAPI keeps graph, SPARQL, and modeling; cron versus API-owned sync.
Target vision — Brick on VOLTTRON, Central, no FastAPI	Long-range architecture: Brick as canonical model, layer across VOLTTRON or Central, retiring the standalone FastAPI service once replacements exist.

See also Behind the firewall; cloud export is vendor-led on the docs home.

65 Open-Claw Integration

66 Open-Claw Integration (external worker)

Open-FDD does not embed or run an LLM. Instead, you run Open-Claw (or any OpenAI-compatible agent) externally, and use Open-FDD's REST API as the "data + tool" layer. Open-Claw can be local (same host) or hosted remotely; your agent just needs network access to the Open-FDD API.

This page shows how to:

1. Fetch platform documentation as model context (`GET /model-context/docs`)
2. Discover HTTP operations from **OpenAPI** (`GET /openapi.json` or Swagger `/docs`)
3. Automate Brick tagging (`GET /data-model/export` → LLM → `PUT /data-model/import`)
4. Tune FDD faults by updating rule YAML files (`/rules/*`) and syncing definitions

Security note: when **auth is required** (`OFDD_API_KEY` and/or app-user login), API routes (including `GET /model-context/docs`) accept **`Authorization: Bearer`** with either the **API key** or a **JWT access token** issued from **`POST /auth/login`**. External HTTP agents normally use the **API key**; browsers use login + JWT. `/auth/*` is exempt from the middleware so login/refresh/logout work.

66.1 1) Model context endpoint (`GET /model-context/docs`)

External agents should fetch Open-FDD documentation context from:

- `GET /model-context/docs`

Control the output size and retrieval:

- `mode=excerpt` (default): truncated excerpt of `pdf/open-fdd-docs.txt`
- `mode=full`: entire doc file
- `mode=slice&offset=...`: a substring (character offset + `max_chars`)

- `query=...`: keyword-retrieved relevant doc sections (simple lexical matching)

If your agent's context window is running out, re-fetch context using `query=...` (or `mode=slice`) as needed.

66.2 1b) HTTP discovery (OpenAPI)

Use **`GET /openapi.json`** or the **Swagger UI** at **`/docs`** (when enabled) to list paths, methods, and schemas. There is no separate MCP manifest on the core API.

66.3 1c) Edge + central split (VOLTTRON direction)

Field collection is on **VOLTTRON** on the building LAN. **`./afdd_stack/scripts/bootstrap.sh`** prepares **VOLTTRON 9** and can start **optional Timescale** via **`--compose-db`**. **`docker compose -f afdd_stack/stack/docker-compose.yml`** is **SQL-only** (see `afdd_stack/legacy/README.md`).

66.4 1d) Mode-aware orchestration (legacy Docker)

If you still run the **Docker** stack, Compose profiles and services mirror the old collector/model/engine idea; for new bench work prefer **VOLTTRON + historian** on the Pi and keep Open-F-DD for **Brick, SPARQL, and rules** where needed.

66.5 1e) OpenClaw on a different machine than Open-FDD (split setup)

OpenClaw does **not** have to run on the same host as Docker / the git clone. A common pattern is **Open-FDD on a Linux server** and **OpenClaw on a Windows (or other) workstation** on the same **test bench LAN**.

How OpenClaw-style tests talk to Open-FDD (HTTP)

Bench and helper scripts treat Open-FDD like any other REST client: they read **`OFDD_API_KEY`** from the environment (often after loading a copy of **`afdd_stack/stack/.env`** from the Linux host) and send **`Authorization: Bearer <OFDD_API_KEY>`** on each request. They do **not** open a browser, call **`/auth/login`**, or manage HttpOnly cookies — that flow is for the React dashboard only. If the API has **app-user** auth enabled **without** an **`OFDD_API_KEY`**, those scripts would need a JWT instead (cookie + login); **for automation, keep `OFDD_API_KEY` in `afdd_stack/stack/.env` and mirror it into the Windows environment** (or a small `.env` you load before running Python). Product details of the bench live under **`openclaw/`** in the repo; this page only describes how that pattern fits Open-FDD.

What the external agent needs

1. **Shell access to the Linux host** (SSH) if it will run **`./afdd_stack/scripts/bootstrap.sh`**, `pytest`, or edit files in the clone — point OpenClaw's workspace at that path or mount it. If the agent only drives **HTTP tools**, a full clone on Windows is optional.
2. **Reachable base URL** — from the Windows machine, **`http://localhost:8000` does not reach the Linux server.** Use the host's **LAN IP or DNS name**:
   - **Direct API:** `http://<open-fdd-host>:8000/...` (paths as in Swagger, e.g. `/sites`, `/data-model/sparql`). FastAPI must be running (**`uvicorn`**) on that host; the default repo bootstrap does not start it.
   - **Behind a reverse proxy (optional):** If you deploy **Caddy** (or similar) in front of the API, use your operator URL (e.g. `http://<host>/api/...` with a strip-prefix pattern). Auth endpoints stay under **`/auth/...`** per your proxy rules. Bench scripts are usually simplest against **`:8000`** so paths match `/sites` with no prefix. Use **port-forward** or **SSH tunnel** only if you deliberately map remote 8000 to local `localhost`.
3. **Bearer token for REST** — set **`OFDD_API_KEY`** on the bench to the same value as in **`afdd_stack/stack/.env` on the Open-FDD host** (see Security — authentication). Do not commit that file; copy the key out of band. DB passwords and other secrets can stay on the server.
4. **Browser UI from Windows (optional)** — when the **React** dev server or static build is running, open **`http://<open-fdd-host>:5173`** (or your proxy URL). Sign in at **`/login`** when dashboard user auth is configured; **`VITE_API_BASE=/api`** matches a path-based proxy. This is **independent** of the Bearer header OpenClaw uses for REST.

File handoff (`openclaw/issues_log.md`) — `HANDOFF_PROTOCOL.md` assumes both Cursor and OpenClaw can read the **same repo tree**. With a split setup, use **git push/pull** (or SSH to one canonical clone) so the lab trail stays in sync.

66.6 1f) Remote VOLTTRON edge → VOLTTRON Central (Open Claw as the “human operator”)

Open Claw does not replace VOLTTRON. It is an **automation layer** that can **SSH** to the same hosts a human integrator would use, read logs and config, and run the **same** shell commands (`vctl`, `vcfg`, `docker compose`, editors) while you keep a human in the loop for approvals. **Registering a remote edge instance with VOLTTRON Central** is defined and maintained by the **VOLTTRON project**—follow upstream first, then let Open Claw repeat those steps safely.

66.6.1 Official VOLTTRON references (canonical)

Read and bookmark these before automating anything:

- **VOLTTRON Central deployment** — multi-platform, Central UI, instance registration, and the **operator workflow** (including certificates and networking).
- **VOLTTRON 9 / platform docs** — `vctl`, `vcfg`, agents, historians, and the **ZMQ** VIP / pub-sub message bus (not RabbitMQ for core VOLTTRON traffic).
- **volttron-docker** — how this repo’s **`./afdd_stack/scripts/bootstrap.sh --volttron-docker`** clone is meant to be **built and run** (mount **`VOLTTRON_HOME`**, compose layout).

Open-FDD docs that give **context** (not a substitute for the above): Site VOLTTRON and the data plane (ZMQ), VOLTTRON gateway, FastAPI, and data-model sync, VOLTTRON Central and AFDD parity.

66.6.2 What Open Claw should do on the edge box (pattern)

Assume Open Claw has **SSH access** to **`hvac-edge-01`** (or each site gateway) with permission to run **non-destructive** diagnostics first, then **mutating** steps only after you confirm a plan.

1. **Confirm topology** — `docker ps` (if VOLTTRON is containerized), `echo $VOLTTRON_HOME`, `ls` of the mounted config tree, and `vctl status` / `vctl peer list` (exact subcommands depend on VOLTTRON version—**use upstream docs**, not this page, for spelling).
2. **Align addresses** — edge `config` must have a **reachable** `volttron-central-address` / web bind settings matching how Central is published on the LAN or VPN (again: **upstream Central deployment guide**).

3. **Certificates / auth** — follow **VOLTTRON's** certificate and admin-user flows (`vcfg` prompts). Open Claw should **not invent** shortcuts; capture transcripts and errors for a human.
4. **Register / approve in Central** — use the **Central UI** or documented RPC patterns exactly as in **official** docs until the edge instance appears healthy from Central's perspective.

66.6.3 SSH agent forwarding (when people say "forwarding agent to Central")

Two different ideas get conflated:

1. **`ssh -A user@edge`** — forwards your **local SSH agent** to the **edge** session so `git` or **secondary SSH hops** can use your keys **on the edge host**. This is powerful and **risky** (any root-equivalent process on the edge can abuse the agent while connected). Prefer **Deploy keys** or **short-lived host keys** on the edge where possible.
2. **Port forwarding to Central's HTTPS/UI** — e.g. `ssh -L 8443:central-host:8443 user@bastion` lets a workstation (or an Open Claw tool host) open `https://127.0.0.1:8443` while traffic tunnels over SSH. This is **not** "VOLTTRON agent auth"; it is **network reachability** for a human or agent-driven browser step.

VOLTTRON's own credentials (platform CA, admin accounts for Central, instance keys) still come from **VOLTTRON** tooling—Open Claw should read/write only the files and commands **documented upstream**, never guess secrets into chat logs.

66.6.4 Relationship to Open-FDD SQL / OpenAPI

- **Central + edge registration** is **pure VOLTTRON** once you are outside Open-FDD's FastAPI surface.
- **Open-FDD** enters when you want **Brick export/import**, **SPARQL**, **energy penalty catalog**, or **FDD SQL**—use **`OFDD_API_KEY`** and the HTTP patterns in sections **1–3** on this page.

66.7 2) Automated Brick tagging loop

For automated tagging, use the same underlying workflow as manual export/import:

1. Export the current model:
   - `GET /data-model/export` (optionally `?site_id=...`)

2. Get documentation context for the LLM:
   - `GET /model-context/docs?query=data-model+import` (or a broader query)
3. Ask Open-Claw to return **import JSON**:
   - Output must match the Open-FDD import schema: top-level keys `points` and optional `equipment`
4. Validate and import:
   - `PUT /data-model/import` with the validated JSON

Practical prompt pattern for retries:

- If `PUT /data-model/import` returns an error, include the error text in the next LLM prompt and retry (prompt chaining).

66.8 3) "Tune faults" worker loop (rule YAML)

Open-FDD's fault definitions are driven by YAML rule files stored under your configured `rules_dir`.

An Open-Claw worker can update those rules via:

- `GET /rules` (list YAML files and the resolved `rules_dir` path)
- `POST /rules` (upload/overwrite a rule YAML file)
- `POST /rules/sync-definitions` (immediately refresh `fault_definitions` from `rules_dir`)
- `DELETE /rules/{filename}` (delete a YAML file)

Then run FDD so the new rules take effect:

- `POST /run-fdd` (trigger "run now")
- or `POST /jobs/fdd/run` (async job)

If you need to validate improvements before running the loop, use:

- `POST /data-model/sparql` for SPARQL-based checks
- and/or `GET /faults/definitions` / `GET /faults/active` for runtime state.

66.9 4) Example "worker" prompt for Open-Claw

Use this as your agent's system prompt (or as the first instruction in your developer prompt). Replace bracketed placeholders as needed.

```
You are an Open-FDD worker for building operators.

Tools you can call:
- Open-FDD documentation context: GET {{OFDD_BASE_URL}}/model-
context/docs
- Data model export: GET {{OFDD_BASE_URL}}/data-model/export
- Data model import: PUT {{OFDD_BASE_URL}}/data-model/import
- Rule YAML management:
  - GET {{OFDD_BASE_URL}}/rules
  - POST {{OFDD_BASE_URL}}/rules
  - POST {{OFDD_BASE_URL}}/rules/sync-definitions
  - DELETE {{OFDD_BASE_URL}}/rules/{filename}
- Validation helpers:
  - POST {{OFDD_BASE_URL}}/data-model/sparql

Context strategy:
- If you need more Open-FDD details and your context is running
out, re-fetch docs context:
  - GET /model-context/docs?query={{SEARCH_TERMS}}
&mode=excerpt&max_chars=28000

Brick tagging task:
1) Call GET /data-model/export.
2) Call GET /model-context/docs?query=data-model+import.
3) Produce ONLY valid import JSON: top-level keys `points` and
optional `equipment`.
4) Call PUT /data-model/import with the validated JSON.
5) If PUT returns an error, include the error message in your next
attempt prompt and retry (prompt chaining).

Fault tuning task:
1) Read current rules: GET /rules and (optionally) GET /rules/
{filename}.
2) Propose minimal YAML changes to improve the target fault
behavior.
3) Upload updated YAML: POST /rules (overwrite).
4) Sync definitions: POST /rules/sync-definitions.
5) Trigger evaluation: POST /run-fdd (or POST /jobs/fdd/run).
6) Summarize what changed and why.
```

66.10 5) Recommended “context window” behavior

For larger exports and/or multi-step workers:

- Start with `mode=excerpt`
- When you hit a knowledge gap, fetch a smaller targeted subset with `query=...`

- If you still need more, use `mode=slice&offset=...` to walk through the full doc file.

67 How-to Guides

68 How-to Guides

Minimal **`pip install open-fdd`** workflows (no Docker) live on the **engine docs site** — Getting started, Column map & resolvers.

- Open Claw integration — LLM workers over **HTTP** plus **SSH operator** flows for **remote VOLTTRON → Central** (official VOLTTRON doc links on that page).
- MQTT integration (optional) — **Optional** Mosquitto compose profile for **your** MQTT clients; **not** the VOLTTRON bus (VOLTTRON uses **ZMQ**).
- Grafana dashboards (optional) — Optional Grafana; React frontend provides equivalent timeseries and fault views. Datasource and dashboard JSON.
- Grafana SQL cookbook — SQL recipes for time series, faults, weather, system resources (variables, panels, sparklines).
- PyPI releases (open-fdd) — Tags, trusted publishing, local `twine check`.
- The optional openfdd-engine package — `openfdd_engine` vs `open_fdd.engine` vs Docker `fdd-loop`; when to use which install.
- Engine-only deployment and external IoT pipelines — `--mode engine` vs pandas `RuleRunner`; same YAML on DataFrames.
- Data model engineering (Brick + 223P MVP) — Engineering UI, JSON round-trip, `s223/ofdd` RDF, SPARQL examples, and how that ties to FDD + DB for impact-style analytics.
- AI-assisted Brick tagging — `GET /data-model/export` → LLM/human → `PUT /data-model/import` (with LLM workflow).
- AI-assisted energy calculations — penalty catalog, narratives, import/export for energy / impact analytics.
- VOLTTRON Central and AFDD parity (monorepo) — One DB + historian, Central + volttron-docker bootstrap, FDD loop options, multi-site; contrast with the archived all-in-one stack.
- Cloning and porting — Same tools, any building; checklist for OpenClaw clones on a bench.
- OpenClaw subscription setup (Codex OAuth) — ChatGPT subscription path vs API key; stale `openai/...` cleanup.
- Monitor the fake fault schedule — Interpret 180°F spikes on the fake BACnet bench.

69 Data model engineering (Brick + 223P MVP)

70 Data Model Engineering (MVP)

Open-FDD keeps **Brick** as the operational model and adds an **Engineering** layer: schedule-style metadata and **223-style topology** in the database, emitted into the same **knowledge graph** (TTL) you already query for FDD and BACnet.

70.1 What this adds

- Frontend page: **Data Model Engineering**
- Equipment engineering metadata persisted in `equipment.metadata.engineering` (PostgreSQL JSON)
- Round-trip via `PUT /data-model/import` and `GET /data-model/export`
- RDF emission on the unified model (same TTL build as Brick points—see `open_fdd/platform/data_model_ttl.py`, `_append_equipment_engineering`):
  - `ofdd:*` extension predicates for practical fields (design CFM, cooling tons, heating MBH, electrical, feeder panel, source documents, etc.)
  - `s223:*` topology patterns for connection points and conduits (inlet/outlet, duct segments, mediums)
- **Data Model Testing** — SPARQL presets and copy-paste examples for engineering + topology

70.2 223P and what Open-FDD implements

ASHRAE Standard 223 defines a semantic way to describe how building equipment and media connect. In practice, **“223P”** often means tooling and workflows that capture that connectivity and related engineering data alongside operations.

Open-FDD’s MVP is intentionally narrow:

- It **stores** structured engineering and topology in the DB and **emits** them into RDF using `s223:`-namespaced types (e.g. connection points, ducts) plus `ofdd:` for non-223 schedule fields.

- Connection endpoint linkage is currently represented with **`ofdd:connectsFromRef`** / **`ofdd:connectsToRef`** string references (MVP), not full RDF object links via `s223:connectsFrom` / `s223:connectsTo`.
- It does **not** claim full Standard 223 validation, certification, or a complete 223 product data model. The value here is **one graph**: Brick + BACnet + timeseries refs + engineering + topology, all queryable with **SPARQL** and aligned with how integrators already think about **connections** and **submittal data**.

70.3 How this connects to FDD, PostgreSQL, and the knowledge graph

The platform already treats the **database as source of truth** and **regenerates** the Brick TTL on CRUD and data-model import. Engineering metadata follows the same pattern.

Layer	Role
PostgreSQL	Sites, equipment (including `metadata.engineering`), points, **`timeseries_readings`**, fault definitions and fault **results** (per run / equipment / time).
TTL (`config/data_model.ttl` or export)	**Unified RDF**: Brick entities, `ref:BACnetReference` / `ref:TimeseriesReference`, `ofdd:mapsToRuleInput`, BACnet device graph, plus **`ofdd:`** and **`s223:`** from engineering import.
FDD loop	SPARQL over that graph builds the **column map** (timeseries `external_id` → rule input). Rules are still **pandas over pivoted telemetry**; they do not automatically pull engineering scalars into the DataFrame unless you extend the runner or join externally.
SPARQL / integrations	Query **design cooling tons** on an AHU,

Layer	Role
	`brick:feeds` downstream VAVs, and **BACnet-backed** valve points in **one** graph—then join to SQL on site/equipment/point keys or time windows.

So: **yes**, the **data model, knowledge graph, and DB** are wired so you can **reason across** "what the equipment is rated for" (engineering), "what it is doing" (time series), and "what the rules said" (faults). That is the foundation for **optimization playbooks**, **GL36-style context**, and **energy impact sketches**.

70.4 Energy penalty and impact calculations (scope)

Today, core Open-FDD focuses on detection and diagnostics (fault flags, trends, Grafana/React views). It does **not** ship a calibrated **M&V** or automatic **utility-dollar** engine in the main loop.

Architecturally, energy penalty workflows are supported because you can:

1. **SPARQL** equipment for `ofdd:coolingCapacityTons`, `ofdd:heatingCapacityMBH`, `ofdd:designCFM`, and topology (`brick:feeds` / `brick:isFedBy`).
2. **Resolve points** on that equipment via `brick:isPointOf` and `ref:TimeseriesReference` → `external_id` / DB table.
3. **Correlate** with fault episodes from the **fault results** tables (or exported CSV) over a time window.
4. Apply a **simple model** (e.g. design capacity × estimated duty fraction × hours × crude kW/ton)—see §5 in `examples/223P_engineering/README.md`.

That path is appropriate for **notebooks**, **downstream analytics services**, or future product features—not a promise that the FDD binary computes kWh for every fault out of the box.

70.5 Backward compatibility

- Existing Brick / BACnet / time-series / `rule_input` workflows are unchanged
- Unknown metadata keys under `engineering` are preserved on import

- Import/export shape remains backward compatible; engineering fields are **optional**

70.6 JSON pattern

Use `equipment[].engineering` in import payloads. Example: `examples/223P_engineering/engineering_import_example.json`.

70.7 RDF pattern

Examples (offline or diff against live export):

- `examples/223P_engineering/engineering_topology_example.ttl` — `s223` connection points + `ofdd` engineering
- `examples/223P_engineering/engineering_graph_mini.ttl` — adds **`brick:feeds`** / **`brick:isFedBy`**, plus a point with **`ref:BACnetReference`** + **`ref:TimeseriesReference`**

70.8 Query examples

- `examples/223P_engineering/README.md` — §4 SPARQL (copy-paste)
- Broader SPARQL patterns: SPARQL cookbook, External representations

70.9 Related docs

- Data model flow (DB → TTL → FDD)
- Fault rules overview — Brick-driven column map; engineering is complementary metadata for analytics

71 PyPI releases (open-fdd)

72 PyPI releases (`open-fdd`)

Canonical public install for the 2.x rules engine:

```
pip install open-fdd
```

That distribution ships **`open_fdd.engine`** (`RuleRunner`, YAML rules, same stack as the platform FDD loop). **Why does PyPI "Release history" mix `0.1.x` and `2.x`?** That is still **one** project, **`open-fdd`**:

older **0.1.x** uploads were the legacy, hard-coded pandas era; **2.x** is the current config-driven engine + platform-capable tree. Pip resolves the latest **2.x** unless you pin an old version.

72.1 `openfdd-engine` (optional second PyPI name)

The repo also contains **`packages/openfdd-engine/`** (import name **`openfdd_engine`**): a thin re-export that **depends on `open-fdd`**. Its **`version` is kept the same as `open-fdd` 2.x** (e.g. both `2.0.13`) so release numbers stay easy to reason about, even though it is a **separate** PyPI project name.

Publishing uses **`.github/workflows/publish-openfdd-engine.yml`**: push tag **`openfdd-engine-vX.Y.Z`** after the **`openfdd-engine`** PyPI project has a **trusted publisher** for that workflow (same OIDC idea as `open-fdd`). **`workflow_dispatch`** builds only; upload runs on matching tags.

72.2 Release baseline (PyPI)

- Check live version:

```
curl -s https://pypi.org/pypi/open-fdd/json | python3 -c "import sys, json; print(json.load(sys.stdin)['info']['version'])"
```

72.3 1) Before a release of `open-fdd`

1. **Version** — Root **`pyproject.toml`** → `[project] version` (e.g. `2.0.11`).
2. **`openfdd-engine`** — Bump **`packages/openfdd-engine/pyproject.toml`** `version` to the **same `X.Y.Z`** as `open-fdd`, and set **`open-fdd>=X.Y.Z`** in dependencies. Tag **`openfdd-engine-vX.Y.Z`** if you publish that project.

72.3.1 Transition checklist (legacy 0.1.x → 2.x on PyPI)

1. Confirm root `pyproject.toml` sets **`open-fdd`** to the intended **`2.x`** version.
2. Merge/push release commits to **`master`** (or your release branch).
3. Tag **`open-fdd-vX.Y.Z`** and push the tag.
4. Confirm open-fdd on PyPI shows **`X.Y.Z`**.

72.3.2 PyPI upload auth (`open-fdd` only)

CI uses **PyPI Trusted Publishing** (OIDC from GitHub). Configure **one** trusted publisher on the `open-fdd` PyPI project:

1. pypi.org → project `open-fdd` → **Manage project** → **Publishing** (not Settings).
2. Add a **GitHub** trusted publisher:
   - **Owner:** `bbartling`
   - **Repository:** `open-fdd`
   - **Workflow name:** `publish-open-fdd.yml`
3. Save.

Workflow file: `.github/workflows/publish-open-fdd.yml`. Tags `open-fdd-v*` trigger upload. `workflow_dispatch` is build/check only (no upload) if configured that way.

Official guide: Publishing with GitHub Actions OIDC.

72.3.3 Fallback: API token

If you cannot use OIDC, pass a project-scoped token, for example:

```yaml
uses: pypa/gh-action-pypi-publish@release/v1
with:
  packages-dir: dist/
  password: ${{ secrets.PYPI_OPENFDD_TOKEN }}
```

72.3.4 If CI shows `invalid-publisher` or `403`

- Trusted publisher on PyPI doesn't match the **exact** workflow filename or repository.
- Tag points to a commit **before** the OIDC workflow existed — merge fix, retag.
- **Version already on PyPI** — bump version and use a **new** tag.

72.4 2) Publish `open-fdd` (commands)

72.4.1 Local dry run

```bash
cd /path/to/open-fdd   # repo root
python -m pip install --upgrade pip build twine
python -m build
twine check dist/*
```

72.4.2 GitHub Actions

```
git checkout master   # or your release branch
# bump version in pyproject.toml, commit, push
git tag open-fdd-v2.0.11
git push origin open-fdd-v2.0.11
```

72.5 3) Publish `openfdd-engine` (second PyPI project)

1. Create project **`openfdd-engine`** on PyPI (if it does not exist).
2. Add a **trusted publisher** pointing at workflow **`publish-openfdd-engine.yml`** (same repository as `open-fdd`).
3. Tag **`openfdd-engine-vX.Y.Z`** (same numbers as **`open-fdd-vX.Y.Z`**) and push — only after PyPI is configured, or OIDC fails with `invalid-publisher`.

72.6 Scope policy

- **PyPI** ships the **Python packages** (**`open-fdd`** is the supported public name for 2.x).
- **Full edge stack** remains **repo + Docker** (`./scripts/bootstrap.sh`).

See also: Engine-only deployment and external IoT pipelines.

73 Engine-only deployment and external IoT pipelines

74 Engine-only deployment and external IoT pipelines

> **Library-only (`pip install open-fdd`):** See the **engine documentation** — Getting started, Column map & resolvers, and Engine-only & IoT (same topics as below, scoped to pandas integrators).

Some integrators already operate **data collection** (historians, MQTT, proprietary BAS exports) and **modeling / semantics** (warehouse schemas, optional Brick elsewhere). The **pandas YAML engine** (`open_fdd.engine`) and **VOLTTRON agents** let you add **FDD** without adopting a full legacy Docker stack.

> **Package names:** The rules code lives under **`open_fdd.engine`**. The repo's optional **`openfdd-engine`** package (**`openfdd_engine`**) is a thin re-export around the same API — not a different engine. See The optional openfdd-engine package for a comparison table and Docker vs library paths.

74.1 What `--mode engine` starts (Docker)

From the repo root:

```
./scripts/bootstrap.sh --mode engine
```

This brings up **TimescaleDB**, the **`fdd-loop`** service, and **`weather-scraper`** — see Modular architecture for the matrix. There is **no** API, React, or BACnet scraper in this slice by default.

When it fits

- You will **load aligned time-series into Postgres** (same DB the platform uses) and let the existing loop consume **rules + data model** from the normal Open-FDD deployment pattern.
- You want **weather-augmented** or **scheduled** FDD in the same containerized runtime as production Open-FDD.

When something else fits better

- Your data lives in **Snowflake / BigQuery / a lake** and you only need **rule evaluation** on batches or streams: use the **Python library** path below (no Docker).

74.2 Library path — same YAML, any DataFrame

The rule runner is **`open_fdd.engine.runner.RuleRunner`**. It loads the **same** `.yaml` rule files as the platform (`type: bounds|flatline|expression|hunting|oa_fraction|erv_efficiency`, `inputs`, `params`, etc.). Authoring references:

- Expression rule cookbook
- Examples under `examples/my_rules/` in the repo

Minimal integration pattern

1. Build a **pandas** `DataFrame` whose columns are your sensor traces (and optional `timestamp`).
2. Point **`RuleRunner`** at a directory of `.yaml` files **or** pass `rules=[...]` dicts.
3. Call **`run(df, timestamp_col=..., skip_missing_columns=True, column_map={...})`**.
4. Read boolean **`*_flag`** columns (and optional rolling persistence — same parameters as in-platform).

`column_map` — when your modeling layer uses Brick class URIs or tags but dataframe columns are different (e.g. `temp_sa` vs `Supply_Air_Temperature_Sensor`), pass the same **`column_map`** concept the platform uses after SPARQL resolution. See `RuleRunner.run` docstring in `open_fdd/engine/runner.py`.

74.2.1 Bring your own `column_map` or resolver

Integrators own the bridge from **their** naming (warehouse columns, Haystack refs, another graph) into what **`RuleRunner`** expects:

1. **Plain dict (most common)** — Build `column_map: dict[str, str]` (rule input / Brick-class key → **actual DataFrame column name**) and pass it to **`RuleRunner.run(..., column_map=column_map)`**. No TTL required on your side if you already know the columns.

2. **Brick TTL** — The **`open-fdd`** wheel does **not** include **rdflib**. If you need SPARQL over a **`.ttl`** file in **Python code outside the Docker stack**, either install **rdflib** yourself and copy the small resolver from **`openfdd_stack.platform.brick_ttl_resolver`** in **`afdd_stack/openfdd_stack/`** (same monorepo), or build **`column_map`** another way (manifest / dict).

3. **Custom `ColumnMapResolver`** — Implement the **`ColumnMapResolver`** protocol (`build_column_map(*, ttl_path: Path) -> dict[str, str]`) with your own lookup (REST, SQL, manifest file, etc.). You may ignore **`ttl_path`** if your source is elsewhere. For **forked** platform code, **`run_fdd_loop(..., column_map_resolver=your_resolver)`** swaps mapping for the DB loop; the **stock Docker `fdd-loop`** does **not** set this — it keeps **`BrickTtlColumnMapResolver`** from **`openfdd_stack.platform.brick_ttl_resolver`**.

Priority / policy: There is no automatic “ontology priority” in the engine — you supply **one** `column_map` per run (or one resolver that returns it). Ambiguity (e.g. multiple Haystack matches) should be resolved **before** calling **`RuleRunner`** with a strict dict.

Types live in **`open_fdd.engine.column_map_resolver`** and are re-exported from **`openfdd_engine`** (shim only). More context: The optional openfdd-engine package, GitHub **#122** (resolver RFC).

74.2.2 Manifest file + composite priority (workshop / gap-fill)

- **`load_column_map_manifest(path)`** — reads **`.json`** or **`.yaml`** / **`.yml`**. Accepts either a flat `str → str` object or **`column_map:`** nested mapping.
- **`ManifestColumnMapResolver(path)`** — same map via the **`ColumnMapResolver`** protocol; **`build_column_map`** ignores **`ttl_path`** (manifest is the source of truth for that resolver).
- **`FirstWinsCompositeResolver(r1, r2, ...)`** — runs each resolver in order; **the first resolver to define a key wins** (on the stack: **`BrickTtlColumnMapResolver`** then **`ManifestColumnMapResolver("extras.yaml")`**; in library-only code, two manifests or a dict + manifest). This is the supported pattern for **ontology-style priority** without unsafe dynamic imports.

Config-driven resolver class names (e.g. loading a Python import path from env) are **not** supported on purpose — easy to turn into **import injection**. Compose resolvers in code or a thin startup script.

Examples: `examples/column_map_resolver_workshop/` in the **open-fdd** repo — **`simple_ontology_demo.py`** + **`simple_ontology_rule.yaml`**.

Install

```
pip install -e .          # from open-fdd clone, or
pip install open-fdd      # PyPI engine
```

The repo also contains an optional **`openfdd-engine`** tree (`packages/openfdd-engine/`) that re-exports the same API; **`pip install open-fdd`** is the supported public install — use **`open_fdd.engine`** after install. More detail: The optional openfdd-engine package.

74.3 Standalone playground (optional)

An **in-repo** example folder is included for workshops and quick starts:

- `examples/engine_iot_playground/` with `README.md`, `rules/*.yaml`, `data/RTU11.csv`, `run_demo.py`, and `RTU11_engine_tutorial.ipynb`.

That pattern is **not** a second engine — it is the **same** code path as production rules, without Docker.

74.4 Summary

Approach	You bring	Open-FDD provides
VOLTTRON + SQL / agents	Historian + agent packaging	FDD against DB time series
`RuleRunner` in Python	DataFrame + YAML directory	Identical rule YAML semantics on pandas
Full platform (custom)	BACnet / UI / REST	Restore legacy compose or run API + React from source + VOLTTRON on edge

For mode overview and service list, start at Modular architecture.

75 The optional openfdd-engine package

76 The optional `openfdd-engine` package (`openfdd_engine`)

Open-FDD has **one real rules implementation**: YAML files evaluated on **pandas** via `RuleRunner` in the `open_fdd.engine` package. Everything else is either that same code in a different wrapper, or documentation about how to deploy it.

This page explains how the **in-repo** package `packages/openfdd-engine/` (PyPI name `openfdd-engine`, import `openfdd_engine`) fits next to `open_fdd.engine` and the **Docker** `fdd-loop` service.

76.1 Re-exports only — no duplicated types

`openfdd_engine` **does not define its own rule engine, Pydantic models, or resolver types.** The shim `__init__.py` imports from `open_fdd.engine` (e.g. `RuleRunner`, `ManifestColumnMapResolver`, `ColumnMapResolver`) and lists them in `__all__`. Brick TTL mapping lives in `openfdd_stack.platform.brick_ttl_resolver` (stack repo). Any

new public engine API should be added **once** under **`open_fdd.engine`**; the shim only gains new names via **import + re-export**.

- **Why:** One source of truth for PyPI **`open-fdd`** and optional **`openfdd-engine`**; no drift between packages.
- **Versioning:** **`open-fdd`** on PyPI is authoritative; **`openfdd-engine`** pins **`open-fdd>=…`** in its **`pyproject.toml`** when published.

The same policy applies to the repo-root **`openfdd_engine/`** namespace used in some dev installs.

76.2 Mental model (three layers)

Layer	What it is	Typical import / entrypoint
Core engine	Loads YAML rules, runs checks on a `DataFrame`, produces fault columns	`from open_fdd.engine.runner import RuleRunner`
Optional shim	Same classes, different package name (depends on **`open-fdd`**)	`from openfdd_engine import RuleRunner`
Platform loop	Pulls Timescale data + TTL, builds frames per site, calls **`RuleRunner`**, writes **`fault_results`**	Docker: `python -m openfdd_stack.platform.drivers.run_rule_loop --loop` → `openfdd_stack.platform.loop.run_fdd_loop`

There is **no second rule engine** inside `openfdd_engine`: its modules **re-export** symbols from **`open_fdd.engine`** (see `packages/openfdd-engine/src/openfdd_engine/runner.py`).

76.3 When to use `pip install open-fdd` vs `openfdd-engine`

Default for everyone (PyPI, notebooks, CI, integrators):

```
pip install open-fdd

from open_fdd.engine.runner import RuleRunner, load_rule,
load_rules_from_dir
from open_fdd.engine.column_map_resolver import
load_column_map_manifest
```

Optional `openfdd-engine` distribution — use only if you **intentionally** want:

- A **separate PyPI project name** (`openfdd-engine`) for procurement or a minimal dependency line that still pulls **`open-fdd`** underneath, or
- An **editable install** of only the shim while developing the monorepo:

```
cd packages/openfdd-engine
pip install -e .

from openfdd_engine import (
    RuleRunner,
    load_rule,
    bounds_map_from_rule,
    ColumnMapResolver,
    ManifestColumnMapResolver,
    load_column_map_manifest,
)
```

Public story: `pip install open-fdd` is the supported install; **`openfdd-engine`** may or may not be published to PyPI as a second project (maintainer setup). See PyPI releases (`open-fdd`).

76.4 How the full stack uses the engine (so IoT-only users know what they are skipping)

In Docker, **`fdd-loop`** runs:

```
python -m openfdd_stack.platform.drivers.run_rule_loop --loop
```

That driver:

1. Optionally refreshes Open-Meteo for the lookback window.

2. Calls **`run_fdd_loop()`** in **`openfdd_stack.platform.loop`**, which loads YAML from **`rules_dir`**, loads timeseries from Postgres, builds **`column_map`** via **`BrickTtlColumnMapResolver`** in **`openfdd_stack.platform.brick_ttl_resolver`** from **`config/data_model.ttl`**, and constructs **`RuleRunner`** from **`open_fdd.engine.runner`**. Advanced use: pass **`column_map_resolver=...`** into **`run_fdd_loop`** to swap mapping logic; the default stack does not. Manifest-style helpers live in **`open_fdd.engine.column_map_resolver`**: **`ManifestColumnMapResolver`**, **`FirstWinsCompositeResolver`**, **`load_column_map_manifest`** (also re-exported from **`openfdd_engine`**).
3. Writes results to **`fault_results`** (and run log for Grafana).

So: **same YAML and same `RuleRunner` semantics** as a standalone script; the platform adds **DB, schedule, TTL, and weather**.

Operational triggers (touch file, one-shot exec): Operations — FDD loop.

76.5 Library-only / external data (no Docker engine mode)

If your data is already in a warehouse or a CSV pipeline, you do **not** need `openfdd_engine` — use **`open_fdd.engine`** on a **`DataFrame`** and the same rule files as the platform. Step-by-step and **`column_map`**: Engine-only deployment and external IoT pipelines.

Worked example under the repo: `examples/engine_iot_playground/`.

76.6 Summary

Goal	Install	Import / run
Normal development & PyPI users	`pip install open-fdd`	`open_fdd.engine.*`
Optional second package name	`pip install -e packages/openfdd-engine` (or published `openfdd-engine`)	`openfdd_engine.*`
Scheduled FDD against platform DB	Docker **`--mode engine`** / full stack	`fdd-loop` → `run_rule_loop` → `run_fdd_loop` → `RuleRunner`

76.7 See also

- Engine-only deployment and external IoT pipelines — `--mode engine` vs pandas `RuleRunner`
- PyPI releases (`open-fdd`) — tags, publishing, `openfdd-engine` scope
- Modular architecture — which services run in **engine** mode
- Expression rule cookbook — YAML rule types and patterns
- Package README: `packages/openfdd-engine/README.md`

77 Grafana SQL cookbook

78 Grafana SQL cookbook —

78.1 BACnet

- Dashboard Variables (Required for Dropdowns)

Before creating the panels below, you must set up the dashboard variables so the drop-down menus work.

In Grafana, go to **Dashboard Settings (gear icon) > Variables > Add variable**. Set the **Type** to `Query`, select your TimescaleDB datasource, and use the following queries:

78.1.1 1. Site Variable

- **Name:** `site`
- **Query:**

```
SELECT s.name AS __text, s.id::text AS __value
FROM sites s
ORDER BY s.name;
```

78.1.2 2. Device Variable

- **Name:** `device`
- **Query:**

```
SELECT DISTINCT p.bacnet_device_id::text AS __text,
p.bacnet_device_id::text AS __value
FROM points p
WHERE p.site_id::text = '$site'
```

```
  AND p.bacnet_device_id IS NOT NULL
ORDER BY 1;
```

78.1.3 3. Point Variable

- **Name:** `point`
- **Query:**

```
SELECT p.external_id AS __text, p.external_id AS __value
FROM points p
WHERE p.site_id::text = '$site'
  AND p.bacnet_device_id::text = '$device'
ORDER BY 1;
```

78.1.4 BACnet only data

Insert Snip here

- This dashboard uses `point`, `device`, and `site` variables.

```
{
  "id": 1,
  "type": "timeseries",
  "title": "BACnet",
  "gridPos": {
    "x": 0,
    "y": 0,
    "h": 16,
    "w": 24
  },
  "fieldConfig": {
    "defaults": {
      "custom": {
        "drawStyle": "line",
        "lineInterpolation": "linear",
        "barAlignment": 0,
        "barWidthFactor": 0.6,
        "lineWidth": 1,
        "fillOpacity": 0,
        "gradientMode": "none",
        "spanNulls": false,
        "insertNulls": false,
        "showPoints": "auto",
        "showValues": false,
        "pointSize": 5,
        "stacking": {
          "mode": "none",
          "group": "A"
        },
        "axisPlacement": "auto",
```

```
                "axisLabel": "",
                "axisColorMode": "text",
                "axisBorderShow": false,
                "scaleDistribution": {
                  "type": "linear"
                },
                "axisCenteredZero": false,
                "hideFrom": {
                  "tooltip": false,
                  "viz": false,
                  "legend": false
                },
                "thresholdsStyle": {
                  "mode": "off"
                }
              },
              "color": {
                "mode": "palette-classic"
              },
              "mappings": [],
              "thresholds": {
                "mode": "absolute",
                "steps": [
                  {
                    "color": "green",
                    "value": null
                  },
                  {
                    "color": "red",
                    "value": 80
                  }
                ]
              }
            },
            "overrides": []
          },
          "pluginVersion": "12.4.0",
          "targets": [
            {
              "editorMode": "code",
              "format": "time_series",
              "rawQuery": true,
              "rawSql": "SELECT\n
time_bucket(\n    make_interval(secs => ($__interval_ms /
1000)::int),\n    tr.ts\n  ) AS \"time\",\n  avg(tr.value) AS
\"value\",\n  p.external_id AS \"metric\"\nFROM
timeseries_readings tr\nJOIN points p ON p.id =
tr.point_id\nWHERE $__timeFilter(tr.ts)\n  AND p.site_id::text IN
(${site:sqlstring})\n  AND p.bacnet_device_id::text IN ($
{device:sqlstring})\n  AND p.external_id IN (${point:sqlstring})
\nGROUP BY 1, 3\nORDER BY 1, 3;",
              "refId": "A",
```

```
        "sql": {
          "columns": [
            {
              "parameters": [],
              "type": "function"
            }
          ],
          "groupBy": [
            {
              "property": {
                "type": "string"
              },
              "type": "groupBy"
            }
          ],
          "limit": 50
        }
      }
    ],
    "datasource": {
      "uid": "openfdd_timescale"
    },
    "options": {
      "tooltip": {
        "mode": "single",
        "sort": "none",
        "hideZeros": false
      },
      "legend": {
        "showLegend": true,
        "displayMode": "list",
        "placement": "bottom",
        "calcs": []
      }
    }
  }
```

78.2 BACnet Plus Fault Data

```
  {
    "id": null,
    "type": "timeseries",
    "title": "BACnet Telemetry w/ Fault Overlays",
    "gridPos": {
      "x": 0,
      "y": 16,
      "h": 12,
      "w": 24
    },
    "fieldConfig": {
```

```
"defaults": {
  "custom": {
    "drawStyle": "line",
    "lineInterpolation": "linear",
    "barAlignment": 0,
    "barWidthFactor": 0.6,
    "lineWidth": 2,
    "fillOpacity": 10,
    "gradientMode": "none",
    "spanNulls": false,
    "insertNulls": false,
    "showPoints": "never",
    "showValues": false,
    "pointSize": 5,
    "stacking": {
      "mode": "none",
      "group": "A"
    },
    "axisPlacement": "left",
    "axisLabel": "",
    "axisColorMode": "text",
    "axisBorderShow": false,
    "scaleDistribution": {
      "type": "linear"
    },
    "axisCenteredZero": false,
    "hideFrom": {
      "tooltip": false,
      "viz": false,
      "legend": false
    },
    "thresholdsStyle": {
      "mode": "off"
    }
  },
  "color": {
    "mode": "palette-classic"
  },
  "mappings": [],
  "thresholds": {
    "mode": "absolute",
    "steps": [
      {
        "value": null,
        "color": "green"
      },
      {
        "value": 80,
        "color": "red"
      }
    ]
  }
```

```
            },
            "overrides": [
              {
                "matcher": {
                  "id": "byFrameRefID",
                  "options": "B"
                },
                "properties": [
                  {
                    "id": "custom.axisPlacement",
                    "value": "right"
                  },
                  {
                    "id": "custom.drawStyle",
                    "value": "bars"
                  },
                  {
                    "id": "custom.fillOpacity",
                    "value": 30
                  },
                  {
                    "id": "min",
                    "value": 0
                  },
                  {
                    "id": "max",
                    "value": 1.2
                  },
                  {
                    "id": "custom.axisLabel",
                    "value": "Fault Active (1 = Yes)"
                  }
                ]
              }
            ]
          },
          "pluginVersion": "12.4.0",
          "targets": [
            {
              "refId": "A",
              "format": "time_series",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT\n  time_bucket(make_interval(secs =>
($__interval_ms / 1000)::int), tr.ts) AS \"time\",\n
avg(tr.value) AS \"value\",\n  p.external_id AS \"metric\"\nFROM
timeseries_readings tr\nJOIN points p ON p.id =
tr.point_id\nWHERE $__timeFilter(tr.ts)\n  AND p.site_id::text IN
(${site:sqlstring})\n  AND p.bacnet_device_id::text IN ($
{device:sqlstring})\n  AND p.external_id IN (${point:sqlstring})
\nGROUP BY 1, 3\nORDER BY 1, 3;"
            },
```

```
            {
              "refId": "B",
              "format": "time_series",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT\n  time_bucket(make_interval(secs =>
($__interval_ms / 1000)::int), fr.ts) AS \"time\",\n
max(fr.flag_value) AS \"value\",\n  fd.name AS \"metric\"\nFROM
fault_results fr\nJOIN fault_definitions fd ON fd.fault_id =
fr.fault_id\nWHERE $__timeFilter(fr.ts)\n  AND (fr.site_id IN ($
{site:sqlstring}) OR fr.site_id IN (SELECT name FROM sites WHERE
id::text IN (${site:sqlstring})))\nGROUP BY 1, 3\nORDER BY 1, 3;"
            }
          ],
          "datasource": {
            "uid": "openfdd_timescale"
          },
          "options": {
            "tooltip": {
              "mode": "multi",
              "sort": "none",
              "hideZeros": false
            },
            "legend": {
              "showLegend": true,
              "displayMode": "list",
              "placement": "bottom",
              "calcs": []
            }
          }
        }
```

78.3 Host stats (stack-host-stats)

The **stack-host-stats** container scrapes host memory, load, swap, disk usage, and Docker container stats into `host_metrics`, `disk_metrics`, and `container_metrics`. Use the same TimescaleDB datasource (`openfdd_timescale`).

78.3.1 Dashboard variable (Host)

- **Name:** `hostname`
- **Type:** Query
- **Datasource:** openfdd_timescale
- **Query:**

```
SELECT DISTINCT hostname AS __text, hostname AS __value
FROM host_metrics
```

```
        WHERE $__timeFilter(ts)
        ORDER BY 1;
```

78.3.2 Host — Memory (used / available / total)

```
        {
          "id": null,
          "type": "timeseries",
          "title": "Host — Memory (used / available / total)",
          "gridPos": { "x": 0, "y": 0, "h": 10, "w": 24 },
          "fieldConfig": {
            "defaults": {
              "custom": {
                "drawStyle": "line",
                "lineInterpolation": "linear",
                "lineWidth": 2,
                "fillOpacity": 12,
                "axisPlacement": "left",
                "scaleDistribution": { "type": "linear" }
              },
              "unit": "bytes",
              "color": { "mode": "palette-classic" },
              "thresholds": { "mode": "absolute", "steps": [{"color":
"green", "value": null}] }
            },
            "overrides": [
              {
                "matcher": { "id": "byName", "options":
"mem_used_bytes" },
                "properties": [{ "id": "displayName", "value": "Used" }]
              },
              {
                "matcher": { "id": "byName", "options":
"mem_available_bytes" },
                "properties": [{ "id": "displayName", "value":
"Available" }]
              },
              {
                "matcher": { "id": "byName", "options":
"mem_total_bytes" },
                "properties": [{ "id": "displayName", "value": "Total" }]
              }
            ]
          },
          "targets": [
            {
              "refId": "A",
              "format": "time_series",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT ts AS \"time\", mem_used_bytes AS
```

```
\"mem_used_bytes\", mem_available_bytes AS
\"mem_available_bytes\", mem_total_bytes AS \"mem_total_bytes\"
FROM host_metrics WHERE $__timeFilter(ts) AND hostname IN ($
{hostname:sqlstring}) ORDER BY ts;"
            }
          ],
          "datasource": { "uid": "openfdd_timescale" },
          "options": { "legend": { "showLegend": true, "placement":
"bottom" } }
        }
```

78.3.3 Host — Load (1 / 5 / 15 min)

```
        {
          "id": null,
          "type": "timeseries",
          "title": "Host — Load (1 / 5 / 15 min)",
          "gridPos": { "x": 0, "y": 10, "h": 8, "w": 24 },
          "fieldConfig": {
            "defaults": {
              "custom": { "drawStyle": "line", "lineWidth": 2,
"fillOpacity": 8, "axisPlacement": "left" },
              "color": { "mode": "palette-classic" },
              "thresholds": { "mode": "absolute", "steps": [{"color":
"green", "value": null}] }
            },
            "overrides": [
              { "matcher": { "id": "byName", "options": "load_1" },
"properties": [{ "id": "displayName", "value": "Load 1m" }] },
              { "matcher": { "id": "byName", "options": "load_5" },
"properties": [{ "id": "displayName", "value": "Load 5m" }] },
              { "matcher": { "id": "byName", "options": "load_15" },
"properties": [{ "id": "displayName", "value": "Load 15m" }] }
            ]
          },
          "targets": [
            {
              "refId": "A",
              "format": "time_series",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT ts AS \"time\", load_1 AS \"load_1\",
load_5 AS \"load_5\", load_15 AS \"load_15\" FROM host_metrics
WHERE $__timeFilter(ts) AND hostname IN (${hostname:sqlstring})
ORDER BY ts;"
            }
          ],
          "datasource": { "uid": "openfdd_timescale" },
          "options": { "legend": { "showLegend": true, "placement":
```

```
"bottom" } }
        }
```

78.3.4 Host — Swap (used / total)

```
        {
          "id": null,
          "type": "timeseries",
          "title": "Host — Swap (used / total)",
          "gridPos": { "x": 0, "y": 18, "h": 6, "w": 24 },
          "fieldConfig": {
            "defaults": {
              "custom": { "drawStyle": "line", "lineWidth": 2,
"fillOpacity": 10, "axisPlacement": "left" },
              "unit": "bytes",
              "color": { "mode": "palette-classic" },
              "thresholds": { "mode": "absolute", "steps": [{"color":
"green", "value": null}] }
            },
            "overrides": [
              { "matcher": { "id": "byName", "options":
"swap_used_bytes" }, "properties": [{ "id": "displayName",
"value": "Swap used" }] },
              { "matcher": { "id": "byName", "options":
"swap_total_bytes" }, "properties": [{ "id": "displayName",
"value": "Swap total" }] }
            ]
          },
          "targets": [
            {
              "refId": "A",
              "format": "time_series",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT ts AS \"time\", swap_used_bytes AS
\"swap_used_bytes\", swap_total_bytes AS
\"swap_total_bytes\" FROM host_metrics WHERE $__timeFilter(ts) AND
hostname IN (${hostname:sqlstring}) ORDER BY ts;"
            }
          ],
          "datasource": { "uid": "openfdd_timescale" },
          "options": { "legend": { "showLegend": true, "placement":
"bottom" } }
        }
```

78.3.5 Host — Disk space (free / used / total)

Disk metrics come from `disk_metrics` (mount path from env `OFDD_DISK_MOUNT_PATHS`, default `/`). The table is created by the same migration as `host_metrics` (`stack/sql/006_host_metrics.sql`); re-run migrations if you added host-stats after an older deploy.

```
        {
          "id": null,
          "type": "timeseries",
          "title": "Host — Disk space (free / used / total)",
          "gridPos": { "x": 0, "y": 24, "h": 10, "w": 24 },
          "fieldConfig": {
            "defaults": {
              "custom": { "drawStyle": "line", "lineWidth": 2,
"fillOpacity": 12, "axisPlacement": "left" },
              "unit": "bytes",
              "color": { "mode": "palette-classic" },
              "thresholds": { "mode": "absolute", "steps": [{"color":
"green", "value": null}] }
            },
            "overrides": [
              { "matcher": { "id": "byName", "options": "free_bytes" },
"properties": [{ "id": "displayName", "value": "Free" }] },
              { "matcher": { "id": "byName", "options": "used_bytes" },
"properties": [{ "id": "displayName", "value": "Used" }] },
              { "matcher": { "id": "byName", "options": "total_bytes" },
"properties": [{ "id": "displayName", "value": "Total" }] }
            ]
          },
          "targets": [
            {
              "refId": "A",
              "format": "time_series",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT ts AS \"time\", free_bytes AS
\"free_bytes\", used_bytes AS \"used_bytes\", total_bytes AS
\"total_bytes\" FROM disk_metrics WHERE $__timeFilter(ts) AND
hostname IN (${hostname:sqlstring}) AND mount_path = '/' ORDER BY
ts;"
            }
          ],
          "datasource": { "uid": "openfdd_timescale" },
          "options": { "legend": { "showLegend": true, "placement":
"bottom" } }
        }
```

78.3.6 Host — Disk available (gauge, latest)

```
        {
          "id": null,
          "type": "gauge",
          "title": "Host — Disk space available (latest)",
          "gridPos": { "x": 0, "y": 34, "h": 6, "w": 8 },
          "fieldConfig": {
            "defaults": {
              "unit": "bytes",
              "min": 0,
              "color": { "mode": "thresholds" },
              "thresholds": { "mode": "absolute", "steps": [{"color":
"red", "value": null}, {"color": "yellow", "value": 5368709120},
{"color": "green", "value": 10737418240}] }
            }
          },
          "targets": [
            {
              "refId": "A",
              "format": "time_series",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT ts AS \"time\", free_bytes AS \"value\"
FROM disk_metrics WHERE hostname IN (${hostname:sqlstring}) AND
mount_path = '/' ORDER BY ts DESC LIMIT 1;"
            }
          ],
          "datasource": { "uid": "openfdd_timescale" },
          "options": { "showThresholdLabels": false }
        }
```

78.3.7 Containers — CPU %

```
        {
          "id": null,
          "type": "timeseries",
          "title": "Containers — CPU %",
          "gridPos": { "x": 0, "y": 40, "h": 8, "w": 24 },
          "fieldConfig": {
            "defaults": {
              "custom": { "drawStyle": "line", "lineWidth": 1,
"fillOpacity": 5, "axisPlacement": "left" },
              "unit": "percent",
              "min": 0,
              "color": { "mode": "palette-classic" },
              "thresholds": { "mode": "absolute", "steps": [{"color":
"green", "value": null}] }
            }
          },
          "targets": [
```

```
            {
              "refId": "A",
              "format": "time_series",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT ts AS \"time\", cpu_pct AS \"value\",
container_name AS \"metric\" FROM container_metrics WHERE
$__timeFilter(ts) ORDER BY ts, container_name;"
            }
          ],
          "datasource": { "uid": "openfdd_timescale" },
          "options": { "legend": { "showLegend": true, "placement":
"bottom" } }
        }
```

78.3.8 Containers — Memory %

```
        {
          "id": null,
          "type": "timeseries",
          "title": "Containers — Memory %",
          "gridPos": { "x": 0, "y": 48, "h": 8, "w": 24 },
          "fieldConfig": {
            "defaults": {
              "custom": { "drawStyle": "line", "lineWidth": 1,
"fillOpacity": 5, "axisPlacement": "left" },
              "unit": "percent",
              "min": 0,
              "max": 100,
              "color": { "mode": "palette-classic" },
              "thresholds": { "mode": "absolute", "steps": [{"color":
"green", "value": null}] }
            }
          },
          "targets": [
            {
              "refId": "A",
              "format": "time_series",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT ts AS \"time\", COALESCE(mem_pct, 0) AS
\"value\", container_name AS \"metric\" FROM container_metrics
WHERE $__timeFilter(ts) ORDER BY ts, container_name;"
            }
          ],
          "datasource": { "uid": "openfdd_timescale" },
          "options": { "legend": { "showLegend": true, "placement":
"bottom" } }
        }
```

78.3.9 Containers — Table (latest)

```
        {
          "id": null,
          "type": "table",
          "title": "Containers — Latest stats",
          "gridPos": { "x": 0, "y": 56, "h": 12, "w": 24 },
          "fieldConfig": {
            "defaults": { "custom": { "align": "auto", "displayMode":
"auto" }, "mappings": [], "thresholds": { "mode": "absolute",
"steps": [{"color": "green", "value": null}] } },
            "overrides": [
              { "matcher": { "id": "byName", "options": "cpu_pct" },
"properties": [{ "id": "unit", "value": "percent" }] },
              { "matcher": { "id": "byName", "options": "mem_pct" },
"properties": [{ "id": "unit", "value": "percent" }] },
              { "matcher": { "id": "byName", "options":
"mem_usage_bytes" }, "properties": [{ "id": "unit", "value":
"bytes" }] }
            ]
          },
          "targets": [
            {
              "refId": "A",
              "format": "table",
              "rawQuery": true,
              "editorMode": "code",
              "rawSql": "SELECT container_name, ts, cpu_pct, mem_pct,
mem_usage_bytes, pids FROM (SELECT *, row_number() OVER (PARTITION
BY container_name ORDER BY ts DESC) AS rn FROM container_metrics
WHERE $__timeFilter(ts)) sub WHERE rn = 1 ORDER BY
container_name;"
            }
          ],
          "datasource": { "uid": "openfdd_timescale" },
          "options": { "showHeader": true }
        }
```

78.4 Weather

78.4.1 Temp RH Dewpoint

```
        {
          "id": 21,
          "type": "timeseries",
          "title": "Weather — Temp / RH / Dewpoint",
          "gridPos": {
            "x": 0,
            "y": 0,
            "h": 10,
```

```
      "w": 24
    },
    "fieldConfig": {
      "defaults": {
        "custom": {
          "drawStyle": "line",
          "lineInterpolation": "linear",
          "barAlignment": 0,
          "barWidthFactor": 0.6,
          "lineWidth": 2,
          "fillOpacity": 12,
          "gradientMode": "none",
          "spanNulls": false,
          "insertNulls": false,
          "showPoints": "never",
          "showValues": false,
          "pointSize": 4,
          "stacking": {
            "mode": "none",
            "group": "A"
          },
          "axisPlacement": "left",
          "axisLabel": "°F",
          "axisColorMode": "text",
          "axisBorderShow": false,
          "scaleDistribution": {
            "type": "linear"
          },
          "axisCenteredZero": false,
          "hideFrom": {
            "tooltip": false,
            "viz": false,
            "legend": false
          },
          "thresholdsStyle": {
            "mode": "off"
          }
        },
        "color": {
          "mode": "palette-classic"
        },
        "mappings": [],
        "thresholds": {
          "mode": "absolute",
          "steps": [
            {
              "color": "green",
              "value": null
            },
            {
              "color": "red",
              "value": 80
```

```
            }
          ]
        }
      },
      "overrides": [
        {
          "matcher": {
            "id": "byName",
            "options": "temp_f"
          },
          "properties": [
            {
              "id": "unit",
              "value": "fahrenheit"
            },
            {
              "id": "custom.axisPlacement",
              "value": "left"
            },
            {
              "id": "custom.axisLabel",
              "value": "°F"
            }
          ]
        },
        {
          "matcher": {
            "id": "byName",
            "options": "dewpoint_f"
          },
          "properties": [
            {
              "id": "unit",
              "value": "fahrenheit"
            },
            {
              "id": "custom.axisPlacement",
              "value": "left"
            }
          ]
        },
        {
          "matcher": {
            "id": "byName",
            "options": "rh_pct"
          },
          "properties": [
            {
              "id": "unit",
              "value": "percent"
            },
            {
```

```
                    "id": "custom.axisPlacement",
                    "value": "right"
                  },
                  {
                    "id": "custom.axisLabel",
                    "value": "RH (%)"
                  },
                  {
                    "id": "min",
                    "value": 0
                  },
                  {
                    "id": "max",
                    "value": 100
                  }
                ]
              }
            ]
          },
          "pluginVersion": "12.4.0",
          "targets": [
            {
              "refId": "A",
              "editorMode": "code",
              "rawQuery": true,
              "format": "time_series",
              "rawSql": "SELECT\n  time_bucket(make_interval(secs =>
($__interval_ms / 1000)::int), tr.ts) AS \"time\",\n
avg(tr.value) AS \"value\",\n  p.external_id AS \"metric\"\nFROM
timeseries_readings tr\nJOIN points p ON p.id =
tr.point_id\nWHERE $__timeFilter(tr.ts)\n  AND p.site_id::text IN
(${site:sqlstring})\n  AND p.external_id IN
('temp_f','rh_pct','dewpoint_f')\nGROUP BY 1, 3\nORDER BY 1, 3;"
            }
          ],
          "datasource": {
            "uid": "openfdd_timescale"
          },
          "options": {
            "tooltip": {
              "mode": "single",
              "sort": "none",
              "hideZeros": false
            },
            "legend": {
              "showLegend": true,
              "displayMode": "list",
              "placement": "bottom",
              "calcs": []
            }
          }
        }
```

78.4.2 Wind Speed Gusts Direction

```
{
  "id": 22,
  "type": "timeseries",
  "title": "Weather — Wind (Speed / Gust / Direction)",
  "gridPos": {
    "x": 0,
    "y": 10,
    "h": 9,
    "w": 24
  },
  "fieldConfig": {
    "defaults": {
      "custom": {
        "drawStyle": "line",
        "lineInterpolation": "linear",
        "barAlignment": 0,
        "barWidthFactor": 0.6,
        "lineWidth": 2,
        "fillOpacity": 10,
        "gradientMode": "none",
        "spanNulls": false,
        "insertNulls": false,
        "showPoints": "never",
        "showValues": false,
        "pointSize": 4,
        "stacking": {
          "mode": "none",
          "group": "A"
        },
        "axisPlacement": "left",
        "axisLabel": "mph",
        "axisColorMode": "text",
        "axisBorderShow": false,
        "scaleDistribution": {
          "type": "linear"
        },
        "axisCenteredZero": false,
        "hideFrom": {
          "tooltip": false,
          "viz": false,
          "legend": false
        },
        "thresholdsStyle": {
          "mode": "off"
        }
      },
      "color": {
        "mode": "palette-classic"
```

```
        },
        "mappings": [],
        "thresholds": {
          "mode": "absolute",
          "steps": [
            {
              "color": "green",
              "value": null
            },
            {
              "color": "red",
              "value": 80
            }
          ]
        }
      },
      "overrides": [
        {
          "matcher": {
            "id": "byName",
            "options": "wind_mph"
          },
          "properties": [
            {
              "id": "unit",
              "value": "mph"
            },
            {
              "id": "custom.axisPlacement",
              "value": "left"
            },
            {
              "id": "custom.axisLabel",
              "value": "mph"
            },
            {
              "id": "min",
              "value": 0
            }
          ]
        },
        {
          "matcher": {
            "id": "byName",
            "options": "gust_mph"
          },
          "properties": [
            {
              "id": "unit",
              "value": "mph"
            },
            {
```

```
                "id": "custom.axisPlacement",
                "value": "left"
              },
              {
                "id": "min",
                "value": 0
              }
            ]
          },
          {
            "matcher": {
              "id": "byName",
              "options": "wind_dir_deg"
            },
            "properties": [
              {
                "id": "unit",
                "value": "degrees"
              },
              {
                "id": "custom.axisPlacement",
                "value": "right"
              },
              {
                "id": "custom.axisLabel",
                "value": "°"
              },
              {
                "id": "min",
                "value": 0
              },
              {
                "id": "max",
                "value": 360
              }
            ]
          }
        ]
      },
      "pluginVersion": "12.4.0",
      "targets": [
        {
          "refId": "A",
          "editorMode": "code",
          "rawQuery": true,
          "format": "time_series",
          "rawSql": "SELECT\n  time_bucket(make_interval(secs =>
($__interval_ms / 1000)::int), tr.ts) AS \"time\",\n
avg(tr.value) AS \"value\",\n  p.external_id AS \"metric\"\nFROM
timeseries_readings tr\nJOIN points p ON p.id =
tr.point_id\nWHERE $__timeFilter(tr.ts)\n  AND p.site_id::text IN
(${site:sqlstring})\n  AND p.external_id IN
```

```
('wind_mph','gust_mph','wind_dir_deg')\nGROUP BY 1,
3\nORDER BY 1, 3;"
            }
          ],
          "datasource": {
            "uid": "openfdd_timescale"
          },
          "options": {
            "tooltip": {
              "mode": "single",
              "sort": "none",
              "hideZeros": false
            },
            "legend": {
              "showLegend": true,
              "displayMode": "list",
              "placement": "bottom",
              "calcs": []
            }
          }
        }
```

78.4.3 Wind Speed Gust Direction

```
        {
          "id": 22,
          "type": "timeseries",
          "title": "Weather — Wind (Speed / Gust / Direction)",
          "gridPos": {
            "x": 0,
            "y": 10,
            "h": 9,
            "w": 24
          },
          "fieldConfig": {
            "defaults": {
              "custom": {
                "drawStyle": "line",
                "lineInterpolation": "linear",
                "barAlignment": 0,
                "barWidthFactor": 0.6,
                "lineWidth": 2,
                "fillOpacity": 10,
                "gradientMode": "none",
                "spanNulls": false,
                "insertNulls": false,
                "showPoints": "never",
                "showValues": false,
                "pointSize": 4,
                "stacking": {
                  "mode": "none",
```

```
            "group": "A"
          },
          "axisPlacement": "left",
          "axisLabel": "mph",
          "axisColorMode": "text",
          "axisBorderShow": false,
          "scaleDistribution": {
            "type": "linear"
          },
          "axisCenteredZero": false,
          "hideFrom": {
            "tooltip": false,
            "viz": false,
            "legend": false
          },
          "thresholdsStyle": {
            "mode": "off"
          }
        },
        "color": {
          "mode": "palette-classic"
        },
        "mappings": [],
        "thresholds": {
          "mode": "absolute",
          "steps": [
            {
              "color": "green",
              "value": null
            },
            {
              "color": "red",
              "value": 80
            }
          ]
        }
      },
      "overrides": [
        {
          "matcher": {
            "id": "byName",
            "options": "wind_mph"
          },
          "properties": [
            {
              "id": "unit",
              "value": "mph"
            },
            {
              "id": "custom.axisPlacement",
              "value": "left"
            },
```

```
        {
          "id": "custom.axisLabel",
          "value": "mph"
        },
        {
          "id": "min",
          "value": 0
        }
      ]
    },
    {
      "matcher": {
        "id": "byName",
        "options": "gust_mph"
      },
      "properties": [
        {
          "id": "unit",
          "value": "mph"
        },
        {
          "id": "custom.axisPlacement",
          "value": "left"
        },
        {
          "id": "min",
          "value": 0
        }
      ]
    },
    {
      "matcher": {
        "id": "byName",
        "options": "wind_dir_deg"
      },
      "properties": [
        {
          "id": "unit",
          "value": "degrees"
        },
        {
          "id": "custom.axisPlacement",
          "value": "right"
        },
        {
          "id": "custom.axisLabel",
          "value": "°"
        },
        {
          "id": "min",
          "value": 0
        },
```

```
                {
                  "id": "max",
                  "value": 360
                }
              ]
            }
          ]
        },
        "pluginVersion": "12.4.0",
        "targets": [
          {
            "refId": "A",
            "editorMode": "code",
            "rawQuery": true,
            "format": "time_series",
            "rawSql": "SELECT\n  time_bucket(make_interval(secs =>
($__interval_ms / 1000)::int), tr.ts) AS \"time\",\n
avg(tr.value) AS \"value\",\n  p.external_id AS \"metric\"\nFROM
timeseries_readings tr\nJOIN points p ON p.id =
tr.point_id\nWHERE $__timeFilter(tr.ts)\n  AND p.site_id::text IN
(${site:sqlstring})\n  AND p.external_id IN
('wind_mph','gust_mph','wind_dir_deg')\nGROUP BY 1,
3\nORDER BY 1, 3;"
          }
        ],
        "datasource": {
          "uid": "openfdd_timescale"
        },
        "options": {
          "tooltip": {
            "mode": "single",
            "sort": "none",
            "hideZeros": false
          },
          "legend": {
            "showLegend": true,
            "displayMode": "list",
            "placement": "bottom",
            "calcs": []
          }
        }
      }
```


78.4.4 Solar Radiation


```
      {
        "id": 23,
        "type": "timeseries",
        "title": "Weather — Solar / Radiation (W/m²)",
        "gridPos": {
          "x": 0,
```

```
        "y": 19,
        "h": 9,
        "w": 24
      },
      "fieldConfig": {
        "defaults": {
          "custom": {
            "drawStyle": "line",
            "lineInterpolation": "linear",
            "barAlignment": 0,
            "barWidthFactor": 0.6,
            "lineWidth": 2,
            "fillOpacity": 10,
            "gradientMode": "none",
            "spanNulls": false,
            "insertNulls": false,
            "showPoints": "never",
            "showValues": false,
            "pointSize": 4,
            "stacking": {
              "mode": "none",
              "group": "A"
            },
            "axisPlacement": "left",
            "axisLabel": "W/m²",
            "axisColorMode": "text",
            "axisBorderShow": false,
            "scaleDistribution": {
              "type": "linear"
            },
            "axisCenteredZero": true,
            "hideFrom": {
              "tooltip": false,
              "viz": false,
              "legend": false
            },
            "thresholdsStyle": {
              "mode": "off"
            }
          },
          "color": {
            "mode": "palette-classic"
          },
          "mappings": [],
          "thresholds": {
            "mode": "absolute",
            "steps": [
              {
                "color": "green",
                "value": null
              },
              {
```

```
            "color": "red",
            "value": 80
          }
        ]
      },
      "unit": "wattperm2",
      "min": 0
    },
    "overrides": [
      {
        "matcher": {
          "id": "byName",
          "options": "shortwave_wm2"
        },
        "properties": [
          {
            "id": "min",
            "value": 0
          }
        ]
      },
      {
        "matcher": {
          "id": "byName",
          "options": "direct_wm2"
        },
        "properties": [
          {
            "id": "min",
            "value": 0
          }
        ]
      },
      {
        "matcher": {
          "id": "byName",
          "options": "diffuse_wm2"
        },
        "properties": [
          {
            "id": "min",
            "value": 0
          }
        ]
      },
      {
        "matcher": {
          "id": "byName",
          "options": "gti_wm2"
        },
        "properties": [
          {
```

```
                "id": "min",
                "value": 0
              }
            ]
          }
        ]
      },
      "pluginVersion": "12.4.0",
      "targets": [
        {
          "refId": "A",
          "editorMode": "code",
          "rawQuery": true,
          "format": "time_series",
          "rawSql": "SELECT\n  time_bucket(make_interval(secs =>
($__interval_ms / 1000)::int), tr.ts) AS \"time\",\n
avg(tr.value) AS \"value\",\n  p.external_id AS \"metric\"\nFROM
timeseries_readings tr\nJOIN points p ON p.id =
tr.point_id\nWHERE $__timeFilter(tr.ts)\n  AND p.site_id::text IN
(${site:sqlstring})\n  AND p.external_id IN
('shortwave_wm2','direct_wm2','diffuse_wm2','gti_wm2')\nGROUP BY
1, 3\nORDER BY 1, 3;"
        }
      ],
      "datasource": {
        "uid": "openfdd_timescale"
      },
      "options": {
        "tooltip": {
          "mode": "single",
          "sort": "none",
          "hideZeros": false
        },
        "legend": {
          "showLegend": true,
          "displayMode": "list",
          "placement": "bottom",
          "calcs": []
        }
      }
    }
```


78.4.5 Cloud Cover


```
    {
      "id": 24,
      "type": "timeseries",
      "title": "Weather — Cloud Cover (%)",
      "gridPos": {
        "x": 0,
```

```
        "y": 28,
        "h": 7,
        "w": 24
      },
      "fieldConfig": {
        "defaults": {
          "custom": {
            "drawStyle": "line",
            "lineInterpolation": "linear",
            "barAlignment": 0,
            "barWidthFactor": 0.6,
            "lineWidth": 2,
            "fillOpacity": 35,
            "gradientMode": "opacity",
            "spanNulls": false,
            "insertNulls": false,
            "showPoints": "never",
            "showValues": false,
            "pointSize": 4,
            "stacking": {
              "mode": "none",
              "group": "A"
            },
            "axisPlacement": "left",
            "axisLabel": "Cloud (%)",
            "axisColorMode": "text",
            "axisBorderShow": false,
            "scaleDistribution": {
              "type": "linear"
            },
            "axisCenteredZero": false,
            "hideFrom": {
              "tooltip": false,
              "viz": false,
              "legend": false
            },
            "thresholdsStyle": {
              "mode": "off"
            }
          },
          "color": {
            "mode": "palette-classic"
          },
          "mappings": [],
          "thresholds": {
            "mode": "absolute",
            "steps": [
              {
                "color": "green",
                "value": null
              },
              {
```

```
                "color": "red",
                "value": 80
              }
            ]
          },
          "unit": "percent",
          "min": 0,
          "max": 100
        },
        "overrides": []
      },
      "pluginVersion": "12.4.0",
      "targets": [
        {
          "refId": "A",
          "editorMode": "code",
          "rawQuery": true,
          "format": "time_series",
          "rawSql": "SELECT\n  time_bucket(make_interval(secs =>
($__interval_ms / 1000)::int), tr.ts) AS \"time\",\n
avg(tr.value) AS \"value\",\n  p.external_id AS \"metric\"\nFROM
timeseries_readings tr\nJOIN points p ON p.id =
tr.point_id\nWHERE $__timeFilter(tr.ts)\n  AND p.site_id::text IN
(${site:sqlstring})\n  AND p.external_id IN ('cloud_pct')\nGROUP
BY 1, 3\nORDER BY 1, 3;"
        }
      ],
      "datasource": {
        "uid": "openfdd_timescale"
      },
      "options": {
        "tooltip": {
          "mode": "single",
          "sort": "none",
          "hideZeros": false
        },
        "legend": {
          "showLegend": true,
          "displayMode": "list",
          "placement": "bottom",
          "calcs": []
        }
      }
    }
```

78.5 Faults

78.5.1 Fault Definitions


```
{
  "id": 1,
  "type": "table",
  "title": "Fault Definitions",
  "gridPos": {
    "x": 0,
    "y": 0,
    "h": 10,
    "w": 14
  },
  "fieldConfig": {
    "defaults": {
      "custom": {
        "align": "auto",
        "footer": {
          "reducers": []
        },
        "cellOptions": {
          "type": "auto"
        },
        "inspect": false,
        "hideFrom": {
          "viz": false
        }
      },
      "mappings": [],
      "thresholds": {
        "mode": "absolute",
        "steps": [
          {
            "color": "green",
            "value": null
          },
          {
            "color": "red",
            "value": 80
          }
        ]
      },
      "color": {
        "mode": "thresholds"
      }
    },
    "overrides": []
  },
  "pluginVersion": "12.4.0",
  "targets": [
```

```
            {
              "dataset": "openfdd",
              "datasource": {
                "type": "grafana-postgresql-datasource",
                "uid": "openfdd_timescale"
              },
              "editorMode": "code",
              "format": "table",
              "rawQuery": true,
              "rawSql": "SELECT \r\n  fault_id, \r\n  name, \r\n
category, \r\n  severity,
\r\n  -- Shows what Brick equipment this rule binds to (e.g., AHU,
VAV)\r\n  array_to_string(equipment_types, ', ') AS
target_equipment,
\r\n  -- Shows the raw JSON of the specific Brick points it needs
(e.g., sat, mat)\r\n  inputs AS required_points,\r\n
updated_at\r\nFROM fault_definitions\r\nWHERE
$__timeFilter(updated_at)\r\nORDER BY updated_at DESC;",
              "refId": "A",
              "sql": {
                "columns": [
                  {
                    "parameters": [],
                    "type": "function"
                  }
                ],
                "groupBy": [
                  {
                    "property": {
                      "type": "string"
                    },
                    "type": "groupBy"
                  }
                ],
                "limit": 50
              }
            }
          ],
          "datasource": {
            "type": "grafana-postgresql-datasource",
            "uid": "openfdd_timescale"
          },
          "options": {
            "showHeader": true,
            "cellHeight": "sm"
          }
        }
```

78.5.2 Fault Definition Count


```
        {
          "id": 2,
          "type": "stat",
          "title": "Fault Definition Count",
          "gridPos": {
            "x": 0,
            "y": 15,
            "h": 15,
            "w": 24
          },
          "fieldConfig": {
            "defaults": {
              "mappings": [],
              "thresholds": {
                "mode": "absolute",
                "steps": [
                  {
                    "color": "green",
                    "value": null
                  },
                  {
                    "color": "red",
                    "value": 80
                  }
                ]
              },
              "color": {
                "mode": "thresholds"
              }
            },
            "overrides": []
          },
          "pluginVersion": "12.4.0",
          "targets": [
            {
              "dataset": "openfdd",
              "datasource": {
                "type": "grafana-postgresql-datasource",
                "uid": "openfdd_timescale"
              },
              "editorMode": "code",
              "format": "table",
              "rawQuery": true,
              "rawSql": "SELECT count(*)::int AS \"Fault
definitions\"\r\nFROM fault_definitions;",
              "refId": "A",
              "sql": {
                "columns": [
```

```
            {
              "parameters": [],
              "type": "function"
            }
          ],
          "groupBy": [
            {
              "property": {
                "type": "string"
              },
              "type": "groupBy"
            }
          ],
          "limit": 50
        }
      }
    ],
    "datasource": {
      "type": "grafana-postgresql-datasource",
      "uid": "openfdd_timescale"
    },
    "options": {
      "reduceOptions": {
        "values": false,
        "calcs": [
          "lastNotNull"
        ],
        "fields": ""
      },
      "orientation": "auto",
      "textMode": "auto",
      "wideLayout": true,
      "colorMode": "value",
      "graphMode": "area",
      "justifyMode": "auto",
      "showPercentChange": false,
      "percentChangeColorMode": "standard"
    }
  }
```

79 Grafana dashboards

80 Grafana dashboards (optional)

Grafana is optional. Install with `./scripts/bootstrap.sh --with-grafana`. The React frontend provides equivalent timeseries, fault, and system-resource views. Use Grafana when you want custom SQL dashboards or existing Grafana workflows.

No dashboards are provisioned by default. Only the **datasource** (TimescaleDB, uid: `openfdd_timescale`) is prebuilt.

To build your own dashboards using SQL and the Open-FDD database, see the **Grafana SQL cookbook**.

You can also place dashboard JSON files in `stack/grafana/dashboards/` (or the directory mounted at `/var/lib/grafana/dashboards` in the Grafana container) and restart Grafana to load them.

80.1 React frontend parity

The React UI provides equivalent time-series and stat views for FDD workflows:

- **Trending** — Multi-point time-series (site/equipment/point picker, date range, dual Y-axis). Uses the same `timeseries_readings` data as the Grafana BACnet timeseries dashboard.
- **Faults** — Active fault table (with **Device** and **Sensor/point** columns so you can see which bad sensor was found on which device), fault definitions, fault count (period) stat, and fault flags over time chart (from `fault_results` via `GET /analytics/fault-timeseries` and `GET /analytics/fault-summary`).
- **Plots (trend + fault overlay)** — Same idea as the Grafana “BACnet Plus Fault Data” panel: one chart with point time-series (left axis) and fault data overlaid. In the React UI, select a site, add points and **Add faults**; fault activity is drawn as **vertical colored bands** on the same plot (data from `GET /analytics/fault-timeseries`, i.e. `fault_results`). So fault data does show on the plot in parity with the Grafana version.
- **Overview** — Site cards with fault counts, equipment/points stats, and FDD status.
- **System resources** — A dedicated sidebar page that mirrors the Grafana “System Resources (Host & Containers)”

dashboard: host memory used/available, load average (1m/5m/15m), swap used (stat cards with threshold colors), host memory and load time-series charts, container memory (MB) and CPU % charts, **Containers (latest)** table (CPU %, Mem MB, Mem %, PIDs), and **disk usage** per mount (used/free/total GB, % used, progress bar). Data comes from `host_metrics`, `container_metrics`, and `disk_metrics` via `GET /analytics/system/*`. Requires the **host-stats** service to be running so those tables are populated.

If you prefer building custom SQL dashboards in Grafana, use the cookbook and the same tables (`host_metrics`, `container_metrics`, `disk_metrics`, `timeseries_readings`, `fault_results`, etc.).

81 Open-FDD integrity sweep

82 Open-FDD integrity sweep

The integrity sweep is a recurring, operator-style check that verifies Open-FDD platform health, graph integrity, and representative telemetry trustworthiness.

82.1 Cadence

- Daytime recurring sweep target: every 20 minutes.
- Overnight window (`18:00-06:00` local): reduce duplicate low-signal chatter and prioritize richer overnight review artifacts.

82.2 Source-of-truth order

1. Live backend data model (`/data-model/check`, `/data-model/sparql`, `/data-model/export`)
2. Current launch/auth/runtime context
3. Repo docs and playbooks

82.3 Sweep checklist

1. Verify authenticated backend reachability.
2. Validate data model integrity and SPARQL query path.
3. Discover representative points from the current model (not hard-coded point lists).
4. Validate representative BACnet reads for model-derived devices/points.

5. Classify failures cleanly.

82.4 Failure classes

- `auth_config_drift`
- `graph_model_drift`
- `bacnet_device_state_drift`
- `testbench_limitation`
- `likely_openfdd_product_behavior`
- `likely_ui_api_parity_bug`

82.5 Mode detection

- `TEST_BENCH`: fake device semantics and deterministic schedule artifacts present.
- `LIVE_HVAC`: occupied-building semantics and live operational telemetry.

The sweep should always report mode + basis, then reason accordingly.

83 Overnight review

84 Overnight review

Overnight review is the deeper process window for quality and drift detection. It complements daytime integrity sweeps.

84.1 Window

- Recommended local window: `18:00-06:00`.

84.2 Goals

- Catch regressions that only appear over longer runtime windows.
- Distinguish product defects from auth/env/testbench drift.
- Create durable context updates for future OpenClaw clones.

84.3 Expected outputs

- Environment mode and mode basis.
- High-signal changes in BACnet behavior, graph behavior, and fault outputs.
- Candidate docs/playbook updates when operating process improves.
- If durable bench/operator lessons emerge, capture them in versioned notes under `openclaw/references/` so they can later be curated into broader docs.

84.4 Alerting discipline

- Suppress repeated low-signal notices.
- Escalate only meaningful drift or new high-impact findings.
- Convert durable lessons into versioned docs, not chat-only memory.

85 OpenClaw context bootstrap

86 OpenClaw context bootstrap

This page defines how to keep context durable and portable for future OpenClaw sessions and new machines.

86.1 Store in Git (durable)

- Operating philosophy and sweep cadence.
- Failure classification standards.
- SPARQL/query patterns that generalize.
- Operator playbooks and runbooks.
- API workflow contracts and tool usage patterns.

86.2 Keep local/private (do not commit)

- API keys, bearer tokens, `.env` secrets.
- Raw auth stores and private local databases.

- Full raw transcript exports with sensitive state.

86.3 Portability principle

Same tools, any building: repo stores reusable process, while site-specific truth lives in the Open-FDD live model.

86.4 Minimal bootstrap read list for fresh clones

OpenClaw + lab (files in repo — read in order on first session):

1. `openclaw/HANDOFF_PROTOCOL.md` — mailbox handoff with `issues_log.md` and log files.
2. `openclaw/SKILL.md` — agent behavior, bootstrap modes, MCP, security scope.
3. `openclaw/references/testing_layers.md` — where pytest vs bench vs `bootstrap.sh` live.
4. `openclaw/references/legacy_automated_testing.md` — redirect from deprecated **open-fdd-automated-testing** if anything still points there.
5. `openclaw/references/session_status_summary.md` — **5-bullet** lab snapshot format when agents must not dump logs into provider chat.

Product and operations (published docs paths):

1. OpenClaw integration
2. Open-FDD integrity sweep
3. Operator framework
4. AI PR review playbook

AI data modeling (when the stack includes model/API): LLM workflow, AI-assisted tagging, plus `GET /data-model/export` and `PUT /data-model/import` as in OpenClaw integration.

87 Operator framework

88 Operator framework

The canonical machine-readable framework lives at `config/ai/operator_framework.yaml`.

This page explains how to use it.

88.1 Intent

Provide a portable operator reasoning contract for Open-FDD automation:

- source-of-truth ordering
- mode detection (`TEST_BENCH` vs `LIVE_HVAC`)
- seasonal reasoning bias
- failure classification
- overnight improvement loop

88.2 Usage

- Humans: review this page + YAML before changing automation behavior.
- Agents: treat YAML as policy contract and derive site specifics from live Open-FDD model/data.
- Repos: keep this file generic; do not hard-code one-site tribal assumptions.

89 MCP RAG service

90 MCP RAG service

Deprecated: the **`mcp-rag`** container, **`/mcp/manifest`**, and **`--with-mcp-rag`** were **removed** from the monorepo. Use **`GET /model-context/docs`**, **`GET /openapi.json`**, and Swagger **`/docs`** for agent discovery. The text below is kept only as historical context.

90.1 Canonical vs derived context

- Canonical human docs: `docs/` and generated `pdf/open-fdd-docs.txt`.
- Derived AI index: `stack/mcp-rag/index/rag_index.json`.

Never edit index artifacts as source-of-truth documentation.

90.2 Bootstrap (historical only)

The following **no longer applies** to the default monorepo; the **`mcp-rag`** profile and **`--with-mcp-rag`** flag were removed.

```
# REMOVED — do not run; kept for archaeology
# ./scripts/bootstrap.sh --with-mcp-rag
# ./scripts/bootstrap.sh --mode model --with-mcp-rag
```

Use **`GET /model-context/docs`**, **`GET /openapi.json`**, and **`/docs`** on a running FastAPI instance instead.

90.3 Service endpoints

- `GET /health`
- `GET /manifest`
- `POST /tools/search_docs`
- `POST /tools/get_doc_section`
- `POST /tools/search_api_capabilities`
- `POST /tools/get_operator_playbook`

Optional guarded action tools are present but disabled by default via `OFDD_MCP_ENABLE_ACTION_TOOLS=false`.

91 Mode-aware runbooks

92 Mode-aware runbooks

The historical **`./scripts/bootstrap.sh --mode {collector|model|engine|full}`** flow matched a **full Docker Compose** stack (BACnet scraper, API, Caddy, `fdd-loop`, etc.). That stack is **no longer** the default in this monorepo.

92.1 Current commands

```
        ./afdd_stack/scripts/bootstrap.sh --doctor
        ./afdd_stack/scripts/bootstrap.sh --central-lab
        ./afdd_stack/scripts/bootstrap.sh --compose-db
# optional local Postgres/Timescale
        ./afdd_stack/scripts/bootstrap.sh --print-paths
# PYTHONPATH for openfdd_stack agents
```

Then follow **VOLTTRON** and **VOLTTRON Central** docs on the edge host. Use **FastAPI** (`uvicorn`) and the **React** app from source when you need REST/SPARQL/modeling.

92.2 Tests (monorepo)

From the repo root with a dev venv:

```
pip install -e ".[dev]"
python -m pytest
```

92.3 Legacy Docker modes

If you run a **fork** or custom compose that still exposes `--mode`, treat those runbooks as **deployment-specific**. Upstream removal details: `afdd_stack/legacy/README.md`.

93 Testing plan

94 Testing plan

This is the evolving engineering plan for Open-FDD automated testing and the optional **OpenClaw lab** bench under `openclaw/README.md`.

94.1 Near-term priorities

94.1.1 0. Continuous PR and CI review

The OpenClaw workflow should continuously watch active PRs in the same spirit as CodeRabbit:

- detect new commits quickly
- re-check CI and review state
- inspect changed files directly
- run targeted local checks where possible
- write down risks, limitations, and next tests instead of relying on chat memory

See AI PR review playbook.

94.1.2 Daytime short-run counterpart

The overnight suite should have a daytime counterpart that exercises the same major paths without taking the full 12-hour window.

Current recommended shape:

- keep E2E, SPARQL/frontend parity, hot-reload, and BACnet/FDD checks
- use a shorter BACnet scrape profile that stays under 2 hours total
- use this as a daytime confidence pass before the real overnight run

Example:

```
python openclaw/bench/e2e/automated_suite.py \
  --api-url http://HOST:8000 \
  --frontend-url http://HOST \
  --long-run-check-faults \
  --long-run-short-day
```

Or on Windows: `openclaw/windows/run_daytime_short_suite.cmd` (edit URLs first).

94.1.3 1. Keep authenticated backend graph checks healthy

Latest status:

- direct authenticated backend access has been restored on the current bench context
- `GET /data-model/check` and authenticated `POST /data-model/sparql` are now working again

Ongoing risk:

- auth still depends on the real launch context, not just a remembered file location
- future overnight runs can regress back into auth/config drift if the bench shell or Python context loses `OFDD_API_KEY`
- the current `/data-model/sparql` response shape may be `{"bindings": [...]}` in some environments, so tooling should not assume strict SPARQL JSON `results.bindings`

Action:

- keep verifying that `OFDD_API_KEY` is available to the actual automated testing environment before overnight runs
- make SPARQL-parsing code tolerant to both `bindings` and `results.bindings`

- keep treating backend auth failure as a pre-overnight readiness issue, not a product regression by default

Why it matters:

- without authenticated SPARQL/API access, BACnet graph validation is incomplete
- without response-shape tolerance, a healthy graph can be misclassified as empty

94.1.4 2. Promote BACnet addressing to a first-class validation target

We need to explicitly validate:

- BACnet devices in the graph
- device instance and address visibility
- object identifiers for polling points
- semantic equipment type for those points

This is no longer optional background metadata. It is core operational context.

94.1.5 3. Prove fault calculation from end to end

The target standard is:

- fake BACnet device fault schedule is known
- DIY BACnet server RPC confirms source values
- Open-FDD data-model SPARQL queries confirm BACnet devices and point addressing
- Open-FDD scrape path receives those values
- YAML rules + rolling windows predict an expected fault
- Open-FDD fault outputs show that exact fault
- the overnight process writes a durable report

See Site VOLTTRON and the data plane (ZMQ) and the report template at `openclaw/reports/overnight-bacnet-verification-template.md` in the repo (not on GitHub Pages) if you still run a bench that mentions BACnet in the filename.

94.1.6 4. Preserve reusable context for future clones

The repo should keep visible documentation and machine-readable operator policy for:

- the operational states
- overnight review discipline
- BACnet graph context
- portability assumptions

- future optimization intent
- documentation guidance that works for both humans and AI agents

94.1.7 5. Add nightly docs and link review

The overnight workflow should also:

- validate important README and docs links
- make sure link checking is done against the correct target branches
- treat `master` as the primary target branch
- optionally check one active development branch when it is the intended docs destination for unreleased fixes
- avoid mixing findings from unrelated feature branches
- identify missing docs pages or thin areas
- suggest documentation improvements that make the system easier for both humans and AI agents to understand
- record those suggestions in durable repo docs or review notes

94.2 Future role in live HVAC systems

In a live HVAC deployment, the same testing and validation assets should support:

- confidence in FDD outputs
- confidence in model/rule applicability
- future optimization and supervisory logic
- operator- or facility-manager-facing monitoring summaries

The repo is not only a test harness. It is becoming a reproducible engineering context pack.

95 Continuous context backup loop

96 Continuous context backup loop

This is the non-site-specific process for making the virtual operator improve over time without turning the repo into one building’s notebook.

96.1 What gets backed up

Back up only **durable, reusable process knowledge**:

- operator heuristics
- failure classification rules
- dashboard expectations
- overnight review discipline
- SPARQL/query patterns that discover site context from the model
- how to separate product bugs from setup/auth drift

96.2 What does not get backed up

Do not back up:

- bearer keys or API keys
- raw `.env` contents
- device pairing secrets
- local SQLite conversation history
- one-off private notes that should stay local
- hard-coded site assumptions that belong in the Open-FDD model instead

96.3 Trigger for an update

During overnight review, update the repo when any of these are true:

- the process changed materially
- the operator framework got smarter
- a failure pattern should be remembered by future clones
- a new query pattern or heuristic is now part of normal operation
- the dashboard or alert contract changed

96.4 Update sequence

1. Distill the lesson from local context
2. Put it into the best durable home:
   - docs
   - query template under `openclaw/bench/sparql/README.md`
   - YAML policy file (e.g. `config/ai/operator_framework.yaml`)
3. Commit the change professionally
4. Push to GitHub
5. Rebuild/publish the docs PDF when the docs set changed materially

96.5 Why this matters

This makes the OpenClaw/Open-FDD operator behave more like a continuously improving engineering system and less like a chat session with amnesia.

The repo becomes the transferable brainstem. The live Open-FDD model remains the site-specific truth.

97 Usage budget and throttling

98 Usage budget and throttling

Open-FDD automation and OpenClaw-assisted workflows should behave responsibly when model/API budget is tight.

That applies to:

- OpenClaw chat usage
- Codex / OpenAI model budget
- overnight review loops
- daytime smoke runs
- every-20-minute integrity sweeps
- weather/support API calls

98.1 Goal

Do not let automation burn expensive model budget on low-value repetition.

When usage pressure is high, the system should degrade gracefully:

- keep the most operator-relevant checks
- reduce frequency of expensive AI/browser work
- preserve durable findings to GitHub/docs
- stay useful without going silent too early

98.2 Practical throttle ladder

98.2.1 Level 0 — Normal

Use the full intended workflow:

- 20-minute integrity sweep
- daytime smoke when requested
- full overnight review
- browser parity when it adds real value
- docs/PDF rebuilds on meaningful changes

98.2.2 Level 1 — Moderate conservation

Use when model budget is healthy but burn rate is rising.

Actions:

- keep the 20-minute sweep concise
- avoid repeating large narrative summaries in chat
- batch docs updates into fewer larger commits
- prefer one focused rerun over repeated full-suite reruns
- avoid unnecessary weather fetches when test-bench mode is already obvious

98.2.3 Level 2 — High conservation

Use when account/session budget is tight or approaching a hard cap.

Actions:

- keep the 20-minute sweep to auth + graph + one or two model-derived BACnet checks
- suppress low-signal chatter
- avoid broad browser parity passes unless specifically needed for release confidence
- prefer direct backend/API checks before UI replays
- do not rerun full daytime smoke or overnight review repeatedly for the same known failure mode
- write the failure classification once, then work the fix

98.2.4 Level 3 — Survival mode

Use when usage is critically constrained or a weekly cap is effectively exhausted.

Actions:

- only run high-signal checks

- prefer local/backend checks over browser/UI checks
- skip optional weather/history enrichments
- skip long narrative output
- skip docs/PDF rebuilds unless the change is critical
- notify the human clearly that budget pressure is affecting coverage

98.3 Priority order under throttling

If budget is tight, preserve this order:

1. auth verification
2. backend reachability
3. data-model / SPARQL integrity
4. BACnet readability
5. high-signal product regressions
6. browser parity extras
7. weather/context enrichments
8. long prose summaries

98.4 Applying this to the Open-FDD workflows

98.4.1 Every-20-minute integrity sweep

Under pressure:

- keep it short
- use model-derived checks only
- do not repeat the same alert every cycle
- if a failure is already known and unchanged, stay quiet unless the state materially worsens or recovers

98.4.2 Daytime smoke

Under pressure:

- prefer targeted reruns instead of full reruns
- fix the first blocker before repeating the whole suite
- use the short-day BACnet profile (`openclaw/bench/e2e/README.md`)

98.4.3 Overnight review

Under pressure:

- keep the overnight run focused on evidence gathering and high-signal failures

- avoid chat spam
- distill durable lessons into docs once instead of narrating every small step

98.5 What can be inferred from usage tools

Useful signals include:

- session/model usage status
- visible Codex/OpenAI usage dashboard state
- repeated provider failures or hard-cap symptoms
- evidence that a task is re-spending tokens on the same known issue

The system should use those signals to reduce repeated model-heavy work.

98.6 Important limitation

OpenClaw can adapt its own behavior, but it does not directly control provider billing/quotas.

So the right design is:

- observe usage pressure
- throttle the workflow behavior
- preserve the most valuable engineering checks
- keep the human informed when coverage is reduced by budget pressure

99 Operations

100 Operations

Operational playbooks for recurring Open-FDD verification, OpenClaw context durability, and operator-style review loops.

Page	Description
Open-FDD integrity sweep	20-minute operator-style health and model integrity checks.
Overnight review	

Page	Description
	6 PM-6 AM deeper review pattern for logs, behavior drift, and durable context updates.
OpenClaw context bootstrap	What durable context should be stored in-repo vs kept local/private.
Operator framework	Canonical operator reasoning contract and source-of-truth order.
MCP RAG service	**Deprecated** — removed; see page for migration to OpenAPI/model-context.
Mode-aware runbooks	Collector/model/engine/full orchestration flows for humans and OpenClaw.
Testing plan	Engineering plan for bench testing, auth/SPARQL checks, and overnight evidence.
Continuous context backup loop	What to commit to the repo vs keep local (no secrets).
Usage budget and throttling	Throttle ladder when model/API budget is tight.

101 Configuration

102 Configuration

102.1 Platform config (RDF + CRUD)

Platform config lives in the **same RDF graph** as Brick (`config/data_model.ttl`). Optional legacy BACnet-related keys may still appear in RDF from older imports; **ingestion** is from **VOLTTRON → SQL**, not from Open-FDD scrapers. No YAML file. **Where:** Bootstrap seeds via PUT /config; API GET/PUT /config and POST /data-model/sparql use the graph. The **entire app is bootstrapped from this data model** (sites, equipment, points, and platform config such as `ofdd:rulesDir`, `ofdd:bacnetScrapeIntervalMin`, etc.). **`rules_dir` (ofdd:rulesDir) remains required**: it is the path where FDD rule YAML files are stored; the frontend upload/download UI manages files *in* that path and does not replace the need for the path itself. **Individual**

rule YAML files are not stored in the data model—only the single directory path (e.g. `ofdd:rulesDir "stack/rules"`) is; the files themselves live on disk under that path.

- **Bootstrap:** `./scripts/bootstrap.sh` seeds config via PUT /config (defaults or `OFDD_*` from `stack/.env`).
- **API:** GET /config, PUT /config; query via POST /data-model/sparql.
- **Env at seed:** Set `OFDD_*` in `stack/.env` before first bootstrap to customize the seed. **Docker:** Set env in `stack/docker-compose.yml` or `stack/.env`; bootstrap uses these when calling PUT /config.

Rename / multiple configs: Use different env files (e.g. `platform-prod.env`, `platform-building-a.env`) and start with `docker compose --env-file platform-building-a.env up -d`, or set `OFDD_*` in that file. No built-in "config name" selector; use env files or env vars per deployment.

102.2 Platform keys (RDF / GET-PUT / config)

Example keys (GET/PUT /config or OFDD_* at bootstrap seed):

Key	Default	Description
`rule_interval_hours`	3	FDD loop run interval (when no trigger file: run every N hours)
`lookback_days`	3	Days of data loaded per run (each run pulls last N days into the rule engine)
`fdd_strict_rules`	false	Env: **`OFDD_FDD_STRICT_RULES`**. When true, the FDD loop uses stricter open-fdd **RuleRunner** behavior (fail fast on bad column maps / non-numeric inputs when open-fdd **≥ 2.3**; see Expression Rule Cookbook). Intended for dev/CI, not typical production.
`rules_dir`	stack/rules	**Single directory for your rules** (hot reload)
`brick_ttl_dir`	—	Optional. Directory containing Brick model TTL (e.g. `config/`); platform uses first `.ttl` or brick_ttl path

Key	Default	Description
		for FDD column mapping. Optional if using points `brick_type`/fdd_input. See Data modeling.
`bacnet_enabled`	true	**Legacy:** BACnet scraper toggle (removed from default Compose). Ignored when **only VOLTTRON** writes SQL.
`graph_sync_interval_min`	5	Minutes between serializing the **full** in-memory graph to `data_model.ttl` (API background thread). That write runs `sync_brick_from_db` first, so Brick **`ref:`** external references reflect the DB at serialize time. Env: `OFDD_GRAPH_SYNC_INTERVAL_MIN`. Edit in the React app under **OpenFDD Config** (not Overview). For an immediate refresh after discovery or DB edits, use **Data model → Serialize to TTL** or GET `/data-model/ttl` with save.
`bacnet_scrape_interval_min`	5	**Legacy:** scraper poll interval (minutes); not used in VOLTTRON-only ingest.
`bacnet_site_id`	default	**Legacy:** site tag for removed scraper / remote-gateway pattern. Prefer **VOLTTRON** historian site keys in SQL.
`bacnet_gateways`	—	**Legacy:** central-aggregator JSON for diy-bacnet gateways. **Not** the default architecture.
`open_meteo_enabled`	true	Enable weather; when true, **FDD loop runs a weather fetch at the start of each run** (same cadence as rules, every `rule_interval_hours`).
`open_meteo_interval_hours`	24	**Standalone weather-scraper only.** Poll interval (hours) when using the optional weather-scraper

Key	Default	Description
		container. Ignored when weather is run from the FDD loop.
`open_meteo_latitude`	41.88	Site latitude
`open_meteo_longitude`	-87.63	Site longitude
`open_meteo_timezone`	America/Chicago	Timezone
`open_meteo_days_back`	3	**Standalone weather-scraper only.** Days of archive to fetch per run. When weather runs with FDD, a 1-day lookback is used (no separate config).
`open_meteo_site_id`	default	Site ID for weather points

Weather fetch: Weather is normally fetched **with each FDD run** (same interval as fault rules, e.g. every 3 h), using a **1-day lookback** so the API is not over-used. A standalone weather scraper is only for setups that do not run the FDD loop; do not run both to avoid redundant fetches. The standalone scraper reads config from **GET /config** when available.

Where rules live: Platform config includes **`rules_dir`** (e.g. `stack/rules`). This path is **required**: the FDD loop and the rules API both use it. Rule YAML files are stored there; you can manage them in two ways. (1) **React frontend (Faults page):** upload, download, delete YAML and **Sync definitions** so the fault_definitions table updates without waiting for the next FDD run. (2) **Files on disk:** edit or add files under the configured path (e.g. `stack/rules/`). Both target the same `rules_dir`; the frontend is the preferred path when you have UI access. See Fault rules overview and the Expression Rule Cookbook.

Hot reload (AFDD tuning): The FDD loop loads rules from `rules_dir` on every run (no cache). Edit YAML on disk or upload via the frontend → trigger a run (or wait for the schedule) or click **Sync definitions** in the UI → results and the definitions table reflect the new rules. The `rules_dir` path is **RDF-driven** (GET/PUT `/config`, same graph as the rest of platform config). Unit tests: `open_fdd/tests/platform/test_rules_loader.py`, `open_fdd/tests/platform_api/test_rules.py`.

Rolling window (per rule): Set `params.rolling_window` in each rule YAML; see Fault rules for HVAC.

102.3 Model context endpoint (external agents)

Open-FDD can serve its own documentation as plain-text model context for external AI agents (for example an OpenAI-compatible tool like Open-Claw).

The endpoint is `GET /model-context/docs`.

By default it returns a truncated excerpt of `pdf/open-fdd-docs.txt` (or the file pointed to by `OFDD_DOCS_PATH` if set). If you need specific sections, pass `query=...` for keyword retrieval and control output size with `max_chars`.

Open-FDD does not embed or run an LLM; you supply the LLM provider externally.

Security note: when `OFDD_API_KEY` is enabled, this endpoint requires Bearer auth like other API routes.

102.4 Config consumers (today)

Default: VOLTTRON historians and agents write **SQL** using deployment-specific config; they do **not** need Open-FDD's **GET /config** for BACnet scrape intervals.

Optional FastAPI process: When you run the API, **GET /config** still serves platform RDF keys for the React **Config** UI (rules dir, graph sync, Open-Meteo toggles, **legacy** BACnet keys for forks). **Standalone weather scraper** (if you still run one from `legacy`) can read **GET /config** for `open_meteo_*` fields.

Legacy BACnet scraper / diy-bacnet central aggregator: Removed from upstream default Compose. If you maintain a **private fork** that restores those services, consult **`afdd_stack/legacy/README.md`** and git history — do **not** use that pattern for new BACnet ingest; use **Site VOLTTRON and the data plane (ZMQ)**.

102.5 Open-Meteo weather points

These keys still apply when weather is fetched **with the FDD loop** or a standalone scraper. Values are stored in `timeseries_readings` by `external_id`:

Point	Description
`temp_f`	Temperature (°F)
`rh_pct`	Relative humidity (%)
`dewpoint_f`	Dew point (°F)
`wind_mph`	Wind speed (mph)
`gust_mph`	Wind gusts (mph)
`wind_dir_deg`	Wind direction (degrees)
`shortwave_wm2`	Shortwave radiation (W/m²)
`direct_wm2`	Direct radiation (W/m²)
`diffuse_wm2`	Diffuse radiation (W/m²)
`gti_wm2`	Global tilted irradiance (W/m²)
`cloud_pct`	Cloud cover (%)

In the **data model** and **Points** UI these appear under equipment **Open-Meteo** (type **Weather_Service**); the RDF graph marks that equipment with `ofdd:dataSource "open_meteo"`.

102.6 Edge / resource limits

For edge deployments with limited disk, set these at bootstrap (or in `stack/.env`). See Getting Started — Bootstrap options.

Setting	Default	Bootstrap arg	Env (stack/.env)
Data retention	365 days	`--retention-days N`	`OFDD_RETENTION_DAYS`
Docker log size	100m per file	`--log-max-size SIZE`	`OFDD_LOG_MAX_SIZE`
Docker log files	3	`--log-max-files N`	`OFDD_LOG_MAX_FILES`

Data retention: TimescaleDB drops chunks older than the configured interval from `timeseries_readings`, `fault_results`, `host_metrics`, `container_metrics`. Set at first run, e.g. `./scripts/bootstrap.sh --retention-days 180`, or set `OFDD_RETENTION_DAYS` in `stack/.env` before running bootstrap.

Log rotation: Containers use `json-file` driver; size and file count come from the table above. Restart the stack after changing so new values apply.

Accessing logs: `docker logs <container> --tail 50` for any **optional** containers you run (e.g. legacy weather scraper names in a fork). Default compose is **db-only** (+ optional Grafana/Mosquitto).

102.7 Environment

`OFDD_`-prefixed vars override YAML:

Variable	Description
`OFDD_DB_HOST`	TimescaleDB host
`OFDD_DB_PORT`	TimescaleDB port
`OFDD_DB_NAME`	Database name
`OFDD_DB_USER`	Database user
`OFDD_DB_PASSWORD`	Database password
`OFDD_BACNET_SERVER_URL`	**Legacy / lab only:** diy-bacnet-server base URL if you enable FastAPI proxy routes
`OFDD_BACNET_SITE_ID`	**Legacy:** site tag for removed scraper patterns
`OFDD_BACNET_GATEWAYS`	**Legacy:** JSON array for removed central-aggregator scraper
`OFDD_RULES_DIR`	Rules directory (default: stack/rules)

103 Contributing

104 Contributing to Open-FDD

First off, thanks for taking the time to contribute!

Open-FDD is in **Alpha**. The most valuable contributions right now are **bug reports** and **FDD rules**—especially from mechanical engineers and building professionals who can add and refine fault-detection rules using the Expression Rule Cookbook. All types of contributions are encouraged.

Phase focus: Alpha emphasizes platform stability, driver implementation beyond BACnet, and API changes for specific integration needs. Beta (planned) will focus more on Brick data modeling (e.g. ASHRAE 223P), mechanical engineering and consulting input into the expression rule cookbook, and better default Grafana dashboards for HVAC analytics. See the Table of Contents for different ways to help and how this project handles them. Please read the relevant section before contributing; it helps maintainers and keeps things smooth for everyone.

> If you like the project but don't have time to contribute, that's fine. Other ways to support it: - Star the repository - Share it with colleagues or at meetups - Refer to Open-FDD in your project's readme or documentation

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104.1 Table of Contents

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104.2 Code of Conduct

This project expects everyone to be respectful and constructive. By participating, you agree to uphold a positive environment. Please report unacceptable behavior by opening an issue or contacting the repository maintainers.

104.3 I Have a Question

Before asking, check **this site's docs** (platform) and the **engine docs** (`pip install open-fdd`), and search stack issues / engine issues.

If you still need help:

- Open an issue.
- Provide as much context as you can (what you're trying to do, what you ran, what happened).
- Include relevant versions: Python, Docker, OS (e.g. Ubuntu, Linux Mint), and Open-FDD commit or release if known.

We'll respond as soon as we can.

104.4 I Want To Contribute

104.4.1 Legal notice

When contributing, you agree that you have authored 100% of the content, have the necessary rights to it, and that your contribution may be provided under the project license (MIT).

104.5 Reporting Bugs

Bug testing is a top priority in Alpha. Good bug reports help us fix issues quickly.

104.5.1 Before submitting a bug report

- Use the latest version (main branch or latest release).
- Confirm the bug is in Open-FDD and not in your environment (e.g. wrong Python version, missing config). Check the platform docs or engine docs and I Have a Question first.
- Search issues to see if the bug is already reported.
- Collect:
  - **Stack trace** (Traceback) if applicable
  - **OS and platform** (e.g. Linux, macOS, Windows; x86, ARM)
  - **Versions**: Python, Docker, Docker Compose, and how you ran Open-FDD (pip, Docker Compose, etc.)
  - **Steps to reproduce** and, if possible, sample input/config
  - Whether you can reproduce it every time and on an older version

104.5.2 How to submit a good bug report

- **Security issues:** Do **not** report security vulnerabilities in public issues. Email the repository owner or open a private security advisory on GitHub.
- Open a new issue. Don't assume it's a bug yet—avoid using the word "bug" in the title until it's confirmed.

- Describe **expected behavior** vs **actual behavior**.
- Provide **reproduction steps** so someone else can recreate the issue. Isolate the problem when possible (e.g. minimal rule YAML, minimal config).
- Paste the information you collected above.

After you file:

- Maintainers will label the issue. If we can't reproduce it, we may ask for more steps and mark it `needs-repro`. Reproducible issues may be marked `needs-fix` and opened for implementation.

104.6 Suggesting Enhancements

Enhancements are tracked as GitHub issues.

104.6.1 Before submitting

- Use the latest version and read the platform or engine documentation to see if the behavior already exists or can be configured.
- Search issues to see if the enhancement was already suggested; if so, add to that discussion.
- Consider whether the idea fits Open-FDD's scope (edge AFDD, rules, API, BACnet, Grafana). Make a clear case for why it would help most users.

104.6.2 How to submit a good enhancement suggestion

- Use a **clear, descriptive title**.
- Describe the **current behavior** and what you **want instead**, and why.
- Provide **step-by-step detail** where useful; screenshots or example YAML/config can help.
- Explain why this would be useful to the community.

104.7 Contributing Rules (Expression Rule Cookbook)

We especially welcome contributions from mechanical engineers and building professionals who can add or improve FDD rules.

- **Where rules live:** Fault rules overview — put project rules in `stack/rules/` (YAML). The FDD loop reloads them every run; no restart needed.
- **How to write rules:** Expression Rule Cookbook — expression-type rules use YAML with BRICK-style inputs, params, and pandas/NumPy expressions. The cookbook includes AHU-style rules (e.g. GL36-inspired) and patterns you can adapt.
- **What to contribute:**
  - **New rule YAMLs** for common faults (AHU, VAV, plant, sensors) that others can reuse.
  - **Improvements to existing cookbook rules** (thresholds, logic, descriptions).
  - **New cookbook sections or examples** (e.g. for a specific equipment type or standard).

To contribute a rule or cookbook change:

1. Add or edit files under `stack/rules/` for new/updated rules, or under `docs/expression_rule_cookbook.md` (and related docs) for cookbook content.
2. Follow the existing YAML style and BRICK naming used in the cookbook.
3. In a PR, describe the fault the rule targets and any assumptions (e.g. Brick types, units). If it's from a standard (e.g. ASHRAE), note the reference.

104.8 Your First Code Contribution

- Look for issues labeled `good first issue` or `help wanted` (if we've added them).
- The codebase is Python 3.11+, with FastAPI (API), pandas/NumPy (rules), and Docker for the platform. See Getting Started and the README for setup.
- For rule changes, run the test suite and, if possible, the FDD loop locally with sample data before submitting a PR.

104.9 Improving the Documentation

Documentation for **this monorepo** lives in `docs/` and is published at **bbartling.github.io/open-fdd** (Just the Docs — engine + AFDD stack). Improvements are welcome:

- Fix typos, clarify wording, or update steps.
- Add examples or how-tos (e.g. for the Expression Rule Cookbook, How-to Guides, or BACnet).
- Keep code blocks and links in sync with the codebase.

Building a PDF: To generate a single PDF of the docs (e.g. for offline use or to push to the repo), run `python3 scripts/build_docs_pdf.py`. Requires pandoc and either **weasyprint** (`pip install weasyprint`) or LaTeX. Output is `pdf/open-fdd-docs.pdf` by default; use `--output path` to change it. You can commit and push the PDF if you want it in the repo.

Open a PR with your changes; for large edits, an issue first can help align with maintainers.

After changing README or high-traffic doc links, spot-check the live site (GitHub Pages builds extensionless URLs like `/modeling/llm_workflow`; the same path with a **trailing slash** often 404s because Jekyll emits `page.html`, not `page/index.html`):

```
BASE=https://bbartling.github.io/open-fdd
curl -sS -o /dev/null -w "%{http_code}" "$BASE/modeling/llm_workflow"    # expect 200
curl -sS -o /dev/null -w "%{http_code}" "$BASE/modeling/llm_workflow/"   # often 404 — avoid trailing slash in external links
curl -sS -o /dev/null -w "%{http_code}" "https://github.com/bbartling/open-fdd/blob/master/pdf/open-fdd-docs.pdf"  # expect 302 or 200
```

104.10 Styleguides

- **Python:** We use Black for formatting. Run `black` on changed files before committing.
- **YAML:** Match the style in `stack/rules/` and the expression rule cookbook (indentation, key order).
- **Markdown:** Use clear headings and lists; link to existing docs where relevant.

104.11 Git workflow (branch → PR → sync)

Create the branch **before** committing so that after the PR is merged, `git pull` on `main`/`master` fast-forwards and you avoid divergent-branch errors (protected `main`/`master`).

1. **Create a branch**

   ```
   git checkout -b feature/short-name
   ```

2. **Stage, commit, and push**

   ```
   git add .
   git commit -m 'short note describing the change'
   git push -u origin feature/short-name
   ```

3. **Open a PR manually on GitHub** from `feature/short-name` into `main` (or `master`). Create the PR in the GitHub UI; we do not use a CLI for this step.

4. **After the PR is merged** — fetch prune, switch to default branch, and pull to sync:

   ```
   git fetch --prune
   git checkout main
   git pull
   ```

   Use `master` instead of `main` if that is your default branch. If you ever committed on `main`/`master` before creating the branch, local and remote can diverge after the merge; run `git reset --hard origin/main` (or `origin/master`) to match the remote.

104.12 Commit Messages

- Use a clear, short summary line. Optionally add a body with details.
- Reference issues/PRs when relevant (e.g. `Fix bounds rule when input is all NaN (#123)`).

104.13 Join the Project Team

If you're interested in ongoing maintenance or a larger role, say so in an issue or reach out to the maintainers. We're especially interested in folks who can triage bugs, review rule contributions, or improve docs and the cookbook.

104.14 Attribution

This contributing guide was inspired by the contributing.md project. You don't need to pay to create or use a CONTRIBUTING file—it's just a markdown file in your repo that helps contributors and maintainers.

105 API Reference

106 API Reference

The Open-FDD **FastAPI** app exposes a **REST API** (default port **8000**) for CRUD, config, data model, bulk download, analytics, **legacy** BACnet proxy routes (if you enable a lab gateway), faults, and jobs. It is **not** started by the default **`bootstrap.sh`** or slim **Compose** — run it with **`uvicorn`** from a dev install when you need REST/SPARQL/modeling. **Use the React frontend** (e.g. http://localhost:5173 with `npm run dev` under `afdd_stack/frontend/`) for workflows such as sites, points, config, faults, and plots. Use the API for scripts, Open-Claw, and cloud export.

Interactive docs: When the API is running, open **Swagger UI** at http://localhost:8000/docs or **ReDoc** at http://localhost:8000/redoc. Full **OpenAPI 3.1** spec: http://localhost:8000/openapi.json. When `OFDD_API_KEY` is set (e.g. in `afdd_stack/stack/.env`), click **Authorize** in Swagger and paste the key so **Try it out** works.

106.1 High-level overview

Area	Purpose
CRUD	**Sites** — list, create, get, update, delete (cascade to equipment, points, timeseries, fault results). **Equipment** — list, create, get, update, delete (cascade to points). **Points** — list, create, get, update, delete (cascade to timeseries). **`external_id`** aligns with **VOLTTRON** historian / topic naming; optional `bacnet_*` columns are **metadata**, not live wire ingest.
Config	**GET/PUT /config** — platform config (rule interval, **legacy** BACnet keys, Open-Meteo, etc.) stored in the RDF

Area	Purpose
	graph. The React Config page uses these.
Data model	**GET /data-model/export** — DB points + graph snapshot (for LLM tagging). **PUT /data-model/import** — bulk create/update points and equipment. **GET /data-model/ttl** — Brick TTL. **POST /data-model/sparql** — run SPARQL. **POST /data-model/reset** — reset graph to DB-only.
Energy calculations	**GET /energy-calculations/export?site_id=** — LLM bundle: `calc_types`, **`penalty_catalog`** (18 defaults), `energy_calculations`. **GET /energy-calculations/penalty-catalog** — narratives only. **POST /energy-calculations/seed-default-penalty-catalog?site_id=** (`replace=true` optional) — insert disabled `penalty_default_*` rows. **PUT /energy-calculations/import** — upsert by `(site_id, external_id)`. CRUD: **GET/POST/PATCH/DELETE /energy-calculations**, **POST /energy-calculations/preview**. See AI-assisted energy calculations and energy penalty catalog.
Download	**GET/POST /download/csv** — timeseries CSV (wide or long). **GET /download/faults** — fault results (CSV or JSON). Excel-friendly; timestamps in UTC (Z).
Analytics	**GET /analytics/fault-summary**, **GET /analytics/fault-timeseries** — fault counts and time-series for charts. **GET /analytics/motor-runtime** — fan/VFD runtime (data-model driven). **GET /analytics/system/*** — host/container/disk metrics.
BACnet (legacy lab proxy)	**GET /bacnet/gateways**, **POST /bacnet/*** — only when **you** run a separate JSON-RPC gateway; **not** default ingest. Prefer **VOLTTRON** on site.
Faults	

Area	Purpose
	GET /faults/active, **GET /faults/state**, **GET /faults/definitions** — active state, full state, definitions.
Jobs	**POST /jobs/bacnet/discovery** (legacy), **POST /jobs/fdd/run** — async discovery (if enabled) and FDD run. **GET /jobs/{job_id}** — status.
Run FDD	**POST /run-fdd** — trigger FDD run now. **GET /run-fdd/status** — last run.
Model context	**GET /model-context/docs** — serve Open-FDD documentation as plain-text model context for external agents (supports excerpt/full/slice and keyword retrieval via `query`).

Base URL: `http://localhost:8000` (use your host or IP when remote). **Auth:** When `OFDD_API_KEY` is set, send `Authorization: Bearer <key>`; the React frontend does this when built with `VITE_OFDD_API_KEY`.

106.2 Model context docs endpoint

Open-FDD can serve its documentation as plain text model context via:

- `GET /model-context/docs`

This endpoint is designed for external LLM agents (OpenAI-compatible providers like Open-Claw). Open-FDD does not embed or run an LLM; it only provides documentation context.

By default, `mode=excerpt` returns a truncated excerpt of `pdf/open-fdd-docs.txt` (or `OFDD_DOCS_PATH` if set). For more control:

- `mode=full` returns the entire file.
- `mode=slice&offset=...` returns a substring.
- `query=...` returns keyword-retrieved relevant doc sections (simple lexical matching) to help keep prompts smaller.

When `OFDD_API_KEY` auth is enabled, this endpoint requires Bearer auth like other API routes.

107 AI PR review playbook

108 AI PR review playbook

Use this checklist when an AI agent reviews Open-FDD pull requests.

108.1 Priority order

1. Behavior regressions and runtime risks.
2. Auth/security exposure or secret handling mistakes.
3. Data-model/API contract breaks.
4. Missing tests for changed behavior.
5. Documentation drift for operationally meaningful changes.

108.2 Review output contract

- Findings first, ordered by severity.
- Include concrete file references and reproducible rationale.
- Keep summary brief after findings.
- If no findings, state residual risks/testing gaps explicitly.

109 Appendix

110 Appendix

Technical deep dives and maintainer reference material.

Page	Description
API Reference	REST API at a glance: CRUD, config, data model, download, analytics, BACnet, faults, jobs. Full spec at Swagger /docs and /openapi.json.
Technical reference	Directory structure, environment variables, unit tests, BACnet scrape, bootstrap, database schema, LLM tagging workflow.
Developer guide	Front-end development, database schema (migrations, tables, cascade deletes), where to go next.

Page	Description
AI PR review playbook	Review checklist focused on regressions, risks, and missing validation before merge.

111 MQTT integration (optional)

112 MQTT integration (optional)

VOLTTRON's default platform bus is ZeroMQ (ZMQ) VIP and pub/sub — **not** MQTT and **not** RabbitMQ. Open-FDD documentation assumes **site VOLTTRON** handles BACnet/Modbus and historians write **SQL**. See **Site VOLTTRON and the data plane (ZMQ)**.

This page covers an **optional Mosquitto** broker from `afdd_stack/stack/docker-compose.yml` when you enable the **`mosquitto`** / MQTT-related **compose profile**. Use it only if **you** add MQTT clients (Home Assistant, custom bridges, cloud shadow topics, etc.). It is **not** required for FDD, for the React UI, or for VOLTTRON ingest.

112.1 What you can do

1. **Start Mosquitto** from `afdd_stack/stack/` using the compose file's **MQTT profile** (see comments in `docker-compose.yml` and Quick reference).
2. **Publish or subscribe** from your own services at the edge or in the data center. Keep ACLs and TLS aligned with your security model.
3. **Do not confuse** this broker with **VOLTTRON's internal message bus** — that remains **ZMQ** per upstream VOLTTRON docs.

112.2 Related docs

- Getting started — bootstrap and compose
- Quick reference — ports and checks

- Edge field buses (VOLTTRON) — field protocols live in VOLTTRON, not in Open-FDD

113 VOLTTRON Central and AFDD parity (monorepo)

114 VOLTTRON Central and AFDD parity (monorepo)

This guide is the **recommended path back to parity** with the historical all-in-one Docker AFDD stack, but on **VOLTTRON Central** and this **monorepo** (bbartling/open-fdd). The old standalone stack repository is **archived** and retained for history only — do not start new deployments there. See the deprecation notice in bbartling/open-fdd-afdd-stack; active development lives under **`afdd_stack/`** here.

Mix-and-match rule: combine **VOLTTRON** (wire protocol, drivers, historian) + **Open-FDD SQL schema** + **`run_fdd_loop()`** (thin agent, systemd timer, or CronJob) + **optional FastAPI + React from source** for Brick/SPARQL/CRUD. **Avoid** running two parallel "stacks" (archived compose + this monorepo) both owning the same database and rules.

114.1 Phase 1 — Data plane (one Postgres / Timescale home)

- Start the platform database from the monorepo: `./afdd_stack/scripts/bootstrap.sh --compose-db` (or **`--central-lab`**, which includes compose + stubs + schema check + **volttron-docker** clone). See Getting started.
- Use **one** Postgres (or Timescale) instance for:
  - Open-FDD schema (sites, equipment, points, faults, readings) initialized from `afdd_stack/stack/sql/`.
  - **VOLTTRON SQL historian** tables when you colocate telemetry — configure the historian's **`tables_def`** (and connection string) so time series land in the same server/database you pass as **`OFDD_DB_DSN`**. Upstream reference: SQLHistorian.
- **Edges** register to **VOLTTRON Central** per upstream deployment docs; **Central** holds roster and web UI. The

database remains the integration point for FDD SQL joins (metadata + historian tables).

114.2 Phase 2 — Mapping (topics → Open-FDD)

- Install or mount this repo where FDD code runs (host venv with `pip install -e ".[stack]"` or **`PYTHONPATH`** from `./afdd_stack/scripts/bootstrap.sh --print-paths`; same layout inside a custom agent container if you bake an image).
- Use **`openfdd_stack.volttron_bridge`** so device / topic naming lines up with **`external_id`**, equipment, and point naming in SQL. Conceptual background: VOLTTRON gateway and sync, Brick and VOLTTRON Central target.

114.3 Phase 3 — FDD execution (pick one first)

A) Thin VOLTTRON agent on the Central host (or a dedicated "compute" instance): periodic RPC or scheduled behavior calls **`run_fdd_loop()`** with **`OFDD_DB_DSN`** pointing at the shared database. Fits teams that want everything as agents and consistent ops with the rest of the platform.

B) Alternative: a **`systemd`** **timer** or **Kubernetes CronJob** invoking the same Python entrypoint (same loop, same DSN). Fewer moving parts if you do not need VIP for FDD.

Rule authoring and expression patterns: Expression rule cookbook.

114.4 Phase 4 — Modeling / Brick editing (optional UI)

When operators need **UI + SPARQL + CRUD**, run **FastAPI** and **React from source** in `afdd_stack/` (as the gateway and security docs describe), not a full resurrection of the old all-in-one compose unless you deliberately need that packaging. See Getting started, Data model engineering, and SPARQL cookbook.

114.5 Phase 5 — Multi-site

- **Central** = roster + web + optionally **one** FDD runner that reads all sites from the database, **or** one FDD runner per edge for fault isolation.
- The **shared database** (and clear site / equipment keys) stays the integration contract between edges, historian, and FDD.

114.6 Bootstrap alignment (this repo)

From the **repo root**:

Goal	Command
Doctor (git, Python, Docker, compose, paths)	`./afdd_stack/scripts/bootstrap.sh --doctor`
One-shot lab: DB + `VOLTTRON_HOME` stubs + schema verify + clone **volttron-docker**	`./afdd_stack/scripts/bootstrap.sh --central-lab`
Clone/update official Docker layout	`./afdd_stack/scripts/bootstrap.sh --volttron-docker`
Local test suite (pytest; optional frontend via env)	`./afdd_stack/scripts/bootstrap.sh --test`

Then build and run **volttron-docker** per VOLTTRON/volttron-docker: mount the host directory you prepared with **`--volttron-config-stub`** / **`--write-env-defaults`** as **`VOLTTRON_HOME`** so Central state and Open-FDD env survive restarts.

`--test` runs **`pytest`** on `open_fdd/tests` and `afdd_stack/openfdd_stack/tests`. If **`pytest`** is missing, use a venv with **`pip install -e ".[dev]"`** or run once with **`OFDD_BOOTSTRAP_INSTALL_DEV=1 ./afdd_stack/scripts/bootstrap.sh --test`**. To also run **eslint**, **`npm run build`** (includes `tsc`), and **Vitest** in `afdd_stack/frontend`, set **`OFDD_BOOTSTRAP_FRONTEND_TEST=1`**. Extra pytest CLI flags: **`OFDD_PYTEST_ARGS`** (space-separated; same caveats as any shell-expanded variable).

114.7 See also

- Getting started — prerequisites and bootstrap options
- Mode-aware runbooks — collector / model / engine / interface framing

- openfdd_central_ui README — static Open-FDD UI under `/openfdd/` on Central

115 Cloning and porting

116 Cloning and porting

The **open-fdd** repo should be portable to another lab, another workstation, or another Open-FDD deployment with minimal conceptual changes. Optional lab/bench assets may live under **`afdd_stack/openclaw/`** when that subtree is present in your checkout.

116.1 Core portability idea

Same tools, any building — only the knowledge graph changes.

That means the repo carries the reusable process, while the live Open-FDD model carries the site-specific truth.

116.2 What should transfer cleanly

- automated tests (`python -m pytest` from the monorepo root with dev deps, plus CI in `.github/workflows/ci.yml`)
- the BACnet fake-device approach (`afdd_stack/scripts/fake_bacnet_devices/README.md`)
- the overnight review discipline
- the SPARQL validation patterns (modeling/sparql_cookbook)
- the operator framework (`afdd_stack/config/ai/operator_framework.yaml`)
- the continuous context-backup loop
- the idea of proving telemetry-to-fault correctness rather than only checking page loads

116.3 What usually changes per environment

- frontend URL
- API URL
- API auth setup
- site IDs / names
- BACnet gateway hostnames or IPs

- active Open-FDD rules directory (under `afdd_stack/stack/rules/`, mounted into containers)
- Docker/container naming
- LAN / OT network topology
- the actual HVAC system, naming conventions, and semantic model shape
- SPARQL queries or filters needed for that environment

116.4 What to do when deploying to another site

1. Resolve the target frontend/backend/BACnet endpoints from the real launch context.
2. Confirm auth works from the shell or runtime that will actually run the checks.
3. Query the Open-FDD model first:
    - sites
    - equipment
    - BACnet devices
    - representative outdoor / plant / air / zone points
4. Let the model decide what should be checked at that site.
5. Keep repo docs generic; put site-specific truth into the Open-FDD model instead of hard-coding it into Markdown.

116.5 Recommended first-pass deployment flow for a new building

Use this order on a fresh site:

- verify backend auth and reachability
- run SPARQL/model sanity checks
- discover representative operator-relevant points from the model
- run the daytime smoke suite first (`openclaw/bench/e2e/README.md`)
- fix auth/model/BACnet issues found there before trusting the overnight 12-hour run
- only then move into recurring integrity sweeps and overnight review

116.6 Same-bench OpenClaw clone checklist

If OpenClaw is cloned onto another machine for the **same current test bench**, the new clone should read these first:

1. Root `README.md`

2. `openclaw/README.md`
3. OpenClaw context bootstrap
4. Open-FDD integrity sweep
5. Site VOLTTRON and the data plane (ZMQ) — verify historian → SQL → FDD chain
6. `openclaw/bench/fake_bacnet_devices/README.md`
7. Monitor the fake fault schedule

And it should know these durable facts immediately:

- the fake devices intentionally inject faults on a **UTC** schedule
- the 180°F spike is expected only during the shared out-of-bounds window
- the correct way to judge that spike is to compare live BACnet RPC reads against `openclaw/bench/fake_bacnet_devices/fault_schedule.py`
- the integrity sweep should classify graph drift, auth drift, BACnet drift, and product behavior separately
- durable reasoning belongs in this repo, not only in local OpenClaw chat memory

116.7 Portability goal

A clone of this repo should make it easy for another engineer to answer:

- Check whether Open-FDD is healthy.
- Confirm BACnet scraping is working.
- Verify the building model is usable.
- Are faults being computed here?
- Are regressions visible here before they affect a real deployment?

116.8 Engineering principle

Keep environment-specific values configurable and keep the verification logic reusable.

In practice, deployment to another site usually looks like this:

- Open-FDD runs on some other server (often a Linux box on the OT LAN)
- tooling is cloned onto another machine
- the tooling is pointed at the target Open-FDD URL, auth, BACnet gateway, and rule/model context for that environment

The tooling should therefore be robust to:

- different LAN IP schemes
- different Open-FDD hosts

- different HVAC systems and point naming
- different site/equipment modeling shapes
- different SPARQL needs per deployment

The goal is portability with context, not a one-off lab setup.

117 OpenClaw subscription setup (Codex OAuth)

118 OpenClaw setup: ChatGPT subscription / Codex OAuth (not API key)

This note captures a setup path that worked for using **OpenClaw with the OpenAI Codex OAuth / ChatGPT subscription path** instead of the plain OpenAI API-key provider.

118.1 Why this matters

A normal ChatGPT subscription and a normal OpenAI API key are **not** the same thing.

That means OpenClaw can appear partly configured, but still fail with errors like:

```
401 Incorrect API key provided: sk-proj-...
```

if stale `openai/...` API-key settings are still present.

118.2 The key distinction

In OpenClaw, these are separate provider paths:

- **Subscription / OAuth path:** `openai-codex/...`
- **API-key path:** `openai/...`

For the subscription/OAuth setup, the model should look like:

```
openai-codex/gpt-5.4
```

Not:

```
openai/gpt-5.4
openai/gpt-5.1-codex
```

118.3 First commands to run

Use this basic command ladder first:

```
openclaw status
openclaw models status --probe
openclaw config get agents.defaults.model
openclaw config get agents.defaults.models
```

What you want to see is the default model set to something like:

```
{
  "primary": "openai-codex/gpt-5.4"
}
```

118.4 If OpenClaw is still trying to use the API-key provider

If you still see `openai/...` models or a 401 against `sk-proj-...`, the most likely cause is stale config.

118.5 What worked

118.5.1 1. Re-auth the OAuth provider

```
openclaw models auth login --provider openai-codex
```

That should complete the browser-based OAuth flow for the subscription path.

118.5.2 2. Make sure the default model points to `openai-codex/...`

```
openclaw models set openai-codex/gpt-5.4
```

118.5.3 3. Clean `openclaw.json`

The important cleanup is to remove all stale `openai/...` references and keep only `openai-codex/...`.

A working pattern looked like this:

```
{
  "auth": {
    "profiles": {
      "openai-codex:default": {
        "provider": "openai-codex",
        "mode": "oauth"
      }
    }
  },
  "agents": {
    "defaults": {
      "model": {
        "primary": "openai-codex/gpt-5.4"
      },
      "models": {
        "openai-codex/gpt-5.4": {}
      }
    },
    "list": [
      {
        "id": "main",
        "model": "openai-codex/gpt-5.4"
      }
    ]
  }
}
```

118.5.4 4. Remove stale API-key provider entries

Problematic leftovers included things like:

- `openai:default`
- `openai/gpt-5.1-codex`
- aliases like `GPT -> openai/gpt-5.1-codex`
- per-agent model overrides pinned to `openai/...`

Those kept the old API-key path alive even though OAuth was configured.

118.5.5 5. Check the agent auth file too

Also inspect:

```
notepad $HOME\.openclaw\agents\main\agent\auth-profiles.json
```

If that file still contains a stale `"openai:default"`, remove it and keep only `"openai-codex:default"`.

118.5.6 6. Restart the gateway

```
openclaw gateway stop
openclaw gateway
```

118.5.7 7. Verify again

```
openclaw models status --probe
```

What you want to see:

- default = `openai-codex/gpt-5.4`
- no `openai:default (api_key)`
- no `openai/gpt-5.1-codex` in configured models
- healthy `openai-codex:default (oauth)`

118.6 One more gotcha: old sessions

Even after config is corrected, an older session/thread can still behave like it is using the old provider.

So after restarting:

```
openclaw dashboard
```

start a **fresh** chat thread and ask:

```
Tell me exactly which provider and model you are using right now.
```

You want the answer to be:

```
openai-codex/gpt-5.4
```

118.7 Practical summary

The OAuth login can work **and** the system can still fail if stale `openai/...` API-key config remains in:

- `openclaw.json`
- `agents.list[].model`
- `agents.defaults.models`
- `auth.profiles`
- `agents/main/agent/auth-profiles.json`

The successful fix was:

1. keep only `openai-codex/...`
2. remove stale `openai/...` references
3. restart the gateway

4. test from a fresh session

118.8 Operational context to save for future setups

When OpenClaw is used with Open-FDD in the field, the agent should expect to ask for and record at least:

- the **Open-FDD base URL** the system is running at
- the **OFDD API key / Bearer token** if API auth is enabled
- whether the Open-FDD host is a **Linux box on the OT LAN** (this is the likely default deployment shape)

Those values should be treated as setup context, not tribal knowledge. They should be written into durable repo notes when appropriate, with secrets handled carefully.

118.9 Why this doc exists

This was learned through actual setup/debugging and should not live only in chat history. It is the kind of practical onboarding/context that future humans and agents will need again.

119 Monitor the fake fault schedule

120 Monitor the fake fault schedule

This page explains the right way to interpret spikes from the Open-FDD **fake BACnet bench** (`openclaw/bench/fake_bacnet_devices/README.md`).

120.1 Why this matters

On this bench, a value like **SA-T = 180°F** is not automatically a product bug.

The fake BACnet devices intentionally inject deterministic faults so Open-FDD can be verified end to end. That behavior lives in:

- `openclaw/bench/fake_bacnet_devices/fault_schedule.py`
- `openclaw/bench/fake_bacnet_devices/fake_ahu_faults.py`
- `openclaw/bench/fake_bacnet_devices/fake_vav_faults.py`

120.2 The shared UTC fault schedule

The fake AHU and fake VAV both follow the same repeating **UTC minute-of-hour** schedule:

Minute (UTC)	Mode	Expected behavior
0-9	normal	scheduled points should stay in their rough normal band
10-49	flatline	scheduled points should hold constant
50-54	bounds	scheduled points should jump to **180.0°F**
55-59	normal	scheduled points should return to normal variability

120.2.1 Scheduled points

The current fake devices intentionally schedule faults on these points:

- fake AHU: `SA-T`, `RA-T`, `MA-T`
- fake VAV: `ZoneTemp`

That means the 180°F spike should be treated as:

- **expected bench behavior** during UTC minutes **50-54**
- **unexpected drift or misalignment** outside that window

120.3 Recommended monitor

Use the repo helper:

```
python openclaw/bench/scripts/monitor_fake_fault_schedule.py
```

What it does:

- loads the same `fault_schedule.py` module used by the fake devices
- determines the current expected mode from UTC minute-of-hour
- reads the scheduled BACnet points directly through the DIY BACnet JSON-RPC API
- explains whether the observed values match the expected schedule

120.3.1 Example output

```
UTC time: 2026-03-24T12:52:10+00:00
Scheduled mode: bounds (minute 52 UTC)
Expected windows: normal 0-9, flatline 10-49, bounds 50-54, normal
55-59
- SA-T: PASS value=180.000 | matches scheduled bounds value 180.0
F
- RA-T: PASS value=180.000 | matches scheduled bounds value 180.0
F
- MA-T: PASS value=180.000 | matches scheduled bounds value 180.0
F
- ZoneTemp: PASS value=180.000 | matches scheduled bounds value
180.0 F
```

Configure the DIY BACnet gateway URL in the script or environment if it is not the default bench host.

120.4 How to reason about anomalies

120.4.1 Case 1: 180°F during UTC minutes 50-54

Treat this as **expected testbench fault injection**, not an unexpected product failure.

Next question:

- did Open-FDD surface the expected `bad_sensor_flag` in the corresponding fault window?

120.4.2 Case 2: 180°F outside UTC minutes 50-54

Treat this as a likely anomaly in one of:

- fake-device schedule alignment
- fake-device runtime state
- testbench orchestration / timebase drift

- Open-FDD ingest lag or stale-value behavior

120.4.3 Case 3: flatline window but value keeps changing

During UTC minutes 10-49, scheduled points should stop moving. If they do not, that points more toward:

- fake-device scheduling drift
- device restart / process interruption
- wrong host timezone / no UTC alignment on the Pi

120.5 How this connects to Open-FDD verification

The better verification chain is:

1. use this monitor to confirm the fake BACnet source state
2. use SPARQL to confirm the points are modeled and addressable
3. use Open-FDD fault APIs or frontend fault views to verify the expected fault flags
4. compare the observed fault windows against `fault_schedule.py`

That is much stronger than treating a single spike as mysterious.

120.6 Clone note

If OpenClaw is cloned onto another machine on the same bench, this page plus:

- `openclaw/bench/fake_bacnet_devices/README.md`
- Site VOLTTRON and the data plane (ZMQ)
- OpenClaw context bootstrap
- `openclaw/bench/scripts/monitor_fake_fault_schedule.py`

should be enough to explain why the 180°F spike exists and how to monitor it correctly.

121 Frontend Evaluation

122 React frontend evaluation

This document summarizes the quality and feature coverage of the Open-FDD React frontend (CRUD UI), its alignment with the API and the Grafana cookbook, and recommendations for WebSockets and the legacy static UI.

122.1 Quality and stack

- **Stack:** React 19, TypeScript, Vite, TanStack Query, React Router, Recharts, Tailwind CSS, shadcn-style UI (Card, Table, Badge, Skeleton). Fits a modern SPA setup.
- **Structure:** Clear separation of pages, hooks (data + WebSocket), contexts (site, theme), and shared UI. API types in `src/types/api.ts` match the OpenAPI schemas (Site, Equipment, Point, FaultState, FaultDefinition, etc.).
- **Auth:** Bearer token from `VITE_OFDD_API_KEY` is sent on all API requests and as `?token=` on the WebSocket; no auth UI (key is build-time env).
- **Missing pieces (added in this pass):** `src/lib/api.ts` (apiFetch with Bearer), `src/lib/csv.ts` (fetchCsv, parseLongCsv, pivotForChart for trending), and `src/lib/utils.ts` (cn, timeAgo, severityVariant) were not in the PR and are required for the app to build and run.

122.2 Feature coverage vs OpenAPI CRUD

API area	Frontend coverage
Sites	✅ List, select site; overview cards (equipment/points/faults per site). Create/update/delete not in UI (API-only or future).
Equipment	✅ List by site or all; table with point count and fault count.
Points	

API area	Frontend coverage
	✅ List by site or all; table with external_id, equipment, brick_type, fdd_input, unit.
Faults	✅ Active faults list; fault definitions table + count (Grafana cookbook parity).
Trending	✅ Site-scoped point picker, date presets (24h, 7d, 30d, custom), line chart via POST /download/csv (long format) and pivot.
FDD status	✅ Last run in banner; run FDD is API-only (could add button).
Config, data-model, BACnet, jobs	❌ Not in UI; use Swagger or API.

So the frontend covers the main "operator" workflows: sites, equipment, points, active faults, fault definitions, and trending. Advanced flows (config, data-model export/import, BACnet discovery, jobs) stay in Swagger or the API.

122.3 Grafana cookbook parity

The Grafana SQL cookbook describes:

- **BACnet:** Variables (site, device, point), timeseries panel, BACnet + fault overlays.
  Frontend: Site selector + point picker + trending chart (same data via `/download/csv`). No separate "device" (bacnet_device_id) filter in the UI; points are chosen by site/equipment. Fault overlays on the chart could be a future enhancement.
- **Host stats:** Memory, load, swap, disk, containers.
  Frontend: No host-stats views; those remain in Grafana (or a future "System" page).
- **Weather:** Temp/RH/Dewpoint, wind, solar, cloud.
  Frontend: Covered indirectly: weather points (e.g. temp_f, rh_pct) appear in the point list and can be trended like any other point.
- **Faults:** Fault definitions table, fault definition count.
  Frontend: Implemented: "Fault definitions (N)" table (fault_id, name, category, severity, target equipment) and active faults table.

So the frontend is capable of the main cookbook workflows that are data-model/CRUD driven (sites, points, faults, trending). Grafana remains the place for raw SQL, host metrics, and pre-built dashboard JSON.

122.4 WebSockets: keep or remove?

- **Current use:** The React app uses `/ws/events` with token auth; on `fault.*`, `fdd.run`, `crud.*` it invalidates TanStack Query so lists and banners refresh. The HA integration optionally uses the same WebSocket for coordinator refresh.
- **Recommendation: keep WebSockets.** They give the React UI live updates without polling. If you later add MQTT for some integrations, you can keep WebSockets for the browser and use MQTT elsewhere; they are complementary (WS for real-time UI, MQTT for pub/sub or constrained clients). No need to remove the WebSocket feature.

122.5 Legacy static UI and BACnet tree

- The old config UI in `open_fdd/platform/static` (data-model tree Site → Equipment → Points, BACnet test/whois/discovery) was removed so the stack uses a single frontend: the React app, served from its own container and via Caddy.
- **BACnet tree / BAS-style tree:** The static UI had a tree that mirrored the data model and building automation. To get that in the React app, add a “Data model” or “Tree” page that:
  - Fetches `/sites`, then for each site `/equipment?site_id=`, then for each equipment `/points?equipment_id=`,
  - Renders a collapsible tree (sites → equipment → points) with the same behavior (expand/collapse, optional right-click delete).
  - Optionally add a “BACnet” section: server URL, Test (server_hello), Who-Is range, Point discovery, using the same API endpoints the static UI used (`/bacnet/server_hello`, `/bacnet/whois_range`, `/bacnet/point_discovery`). That would bring the legacy config + BACnet flows into the React app without serving static files from FastAPI.

122.6 Caddy and security

- With the React frontend in its own container, **Caddy** is the single entry point (e.g. :8088): basic auth, then proxy to API (and `/ws/*`) and to the frontend (`/*`). Optional Grafana at `/grafana` when started with `--with-grafana`. See Security and Caddy.

122.7 Summary

- **Quality:** Solid React/TS/Vite setup; missing `lib` (api, csv, utils) was added so the app builds and runs.
- **Features:** Good coverage of sites, equipment, points, faults, definitions, and trending; config/data-model/BACnet/jobs stay in API/Swagger.
- **Grafana cookbook:** Parity on fault definitions and trending; host stats and raw SQL stay in Grafana; optional to add fault overlays and a data-model/BACnet tree page later.
- **WebSockets:** Keep; they support real-time UI and HA. MQTT can be added alongside.
- **Static UI:** Removed; React is the primary UI, served by Caddy. BACnet tree and discovery can be reimplemented as a React page if desired.

123 Plan Rules Crud And Hot Reload

124 Plan: Default Rules, Fix Phantom Definitions, Rules CRUD + Frontend (Hot Reload)

124.1 Goals

1. **Fix phantom "AHU Short Cycling"** — Definitions table and Faults matrix show only what exists in `rules_dir`; remove stale rows when a rule file is gone.
2. **Default rules** — Ship with only `sensor_bounds.yaml` and `sensor_flatline.yaml` in `stack/rules`; no extra files that create phantom definitions.
3. **Rules CRUD + frontend** — Users can upload, download, and delete rule YAML files via the UI; changes hot-reload (next FDD run picks them up).

124.2 1. Fix phantom definitions (backend)

Problem: `fault_definitions` is upserted from YAML on each FDD run but never pruned. If a rule file is removed (or never existed on this server), its row remains and the UI shows e.g. “AHU Short Cycling”.

Change: In `open_fdd/platform/loop.py`, extend `_sync_fault_definitions_from_rules(rules)` so that after upserting from the current rule list it **prunes** definitions that no longer have a corresponding rule:

- Collect `fault_id` for every rule in `rules` (same logic: `r.get("flag") or f"{r.get('name', 'rule')}_flag"`).
- After the upsert loop, run: `DELETE FROM fault_definitions WHERE fault_id != ALL(%s)` (or equivalent) with the list of current fault_ids.
- No FK from `fault_results` / `fault_state` to `fault_definitions` (they store `fault_id` as text), so deleting definitions is safe; historical results keep their fault_id for display/analytics.

Result: Definitions list and Faults matrix only show faults that have a YAML file in `rules_dir`. One-time fix for existing DBs: next FDD run will prune “AHU Short Cycling” (and any other orphan).

124.3 2. Default rules (no code change)

- **Repo:** `stack/rules` already contains only `sensor_bounds.yaml` and `sensor_flatline.yaml`. Keep it that way; do not add an AHU Short Cycling YAML unless you want that rule.
- **Server:** If the Linux server has extra files in `stack/rules` that you don’t want, remove them; after the prune logic above and one FDD run, definitions will match the remaining files.
- **New deployments:** Same two files as defaults; anything else is added by users via the new CRUD (or manually).

124.4 3. Backend: Rules CRUD (upload / delete)

Existing:

- `GET /rules` — list `.yaml` filenames and `rules_dir` (from config).
- `GET /rules/{filename}` — return file content (plain text). Use for **download**.

- `POST /rules/test-inject`, `DELETE /rules/test-inject/{filename}` — test-only (gated by `OFDD_ALLOW_TEST_RULES`); keep for automation.

Add (production, no env gate):

Method	Endpoint	Purpose
POST	`/rules` or `/rules/upload`	Upload a new or replace an existing rule file.
DELETE	`/rules/{filename}`	Delete a rule file from `rules_dir`.

Upload (POST):

- **Body:** Either `multipart/form-data` (file) or JSON `{ "filename": "my_rule.yaml", "content": "..." }`. JSON is easier for "copy/paste YAML" in the UI and matches test-inject.
- **Validation:**
  - `filename`: must end in `.yaml`, no `..`, `/`, `\`.
  - Resolve path under `_rules_dir_resolved()`; ensure path stays inside `rules_dir` (path traversal check).
  - Optional but recommended: parse YAML and check for minimal keys (`name` or `flag`) so invalid rules are rejected.
- **Write:** `path.write_text(content)` (same as test-inject). Create `rules_dir` if missing (optional; currently API returns 503 if not a dir).
- **Response:** 200 + `{ "filename": "...", "path": "..." }` or 201 for create.
- **Auth:** Same as rest of API (Bearer when `OFDD_API_KEY` set).

Delete (DELETE):

- **Path:** `DELETE /rules/{filename}` (filename in path; same safety as GET: `.yaml` only, no path traversal).
- **Behavior:** `path.unlink()`. Return 404 if file doesn't exist.
- **Optional:** After delete, call a small "sync definitions from rules_dir" helper that runs `load_rules_from_dir` + `_sync_fault_definitions_from_rules` so the definitions table (and matrix) update immediately instead of waiting for the next FDD run. If you skip this, the next FDD run will prune the definition anyway.

Optional helper: Expose `POST /rules/sync-definitions` that loads all YAML from `rules_dir` and runs `_sync_fault_definitions_from_rules` (no FDD run). Then upload/delete can call it so the UI updates without waiting for the next run. Low priority if FDD runs every few hours.

124.5 4. Frontend: Upload, Download, Delete

Current (Faults page – FDD rule files section):

- Lists filenames from `GET /rules`.
- Click filename → fetch `GET /rules/{filename}` and show content in a `<pre>`.
- Shows `rules_dir` from the list response.

Add:

1. **Download**
   - Per file: "Download" link that opens `GET /rules/{filename}` (e.g. in a new tab or via `fetch` + blob download). Already supported by existing endpoint.
2. **Upload**
   - "Upload rule" (or "Add rule file"):
     - Option A: File input → read file as text → POST JSON `{ filename: file.name, content }` (normalize filename to `.yaml` if needed).
     - Option B: Textarea "Paste YAML" + filename input → POST same JSON.
   - On success: invalidate `GET /rules` (and definitions query if you add sync), show success toast; optionally refetch rules list and select the new file.
3. **Delete**
   - Per file: "Delete" button (with confirmation: "Remove rule file X? Definition will be removed after next FDD run.").
   - Call `DELETE /rules/{filename}`; on success invalidate list and definitions, clear selection if the deleted file was selected.
4. **Copy from existing**
   - Optional: "Duplicate" that copies current file content into the upload form with a new name (e.g. `sensor_bounds_copy.yaml`) so users can duplicate and edit.

UX: Keep the existing "view content" flow; add Upload above the list and Download/Delete next to each filename. Use existing auth (e.g. `apiFetch` with API key).

124.6 5. Hot reload (no change)

- FDD loop already loads rules from `rules_dir` on **every** run (`load_rules_from_dir(rules_path)` in `run_fdd_loop`). No caching.
- So: user uploads a new YAML → file is on disk in `rules_dir` → next FDD run loads it and runs it; new rule appears in definitions (after sync) and in results. Same for delete: file

gone → next run doesn't load it → prune step removes the definition.
- Optional `POST /rules/sync-definitions` only makes the definitions table (and matrix) update immediately; it doesn't change hot-reload behavior of the FDD engine.

124.7 6. Implementation order

Step	Task	Effort
1	Backend: Prune stale definitions in `_sync_fault_definitions_from_rules`	Small
2	Backend: `POST /rules` (upload) with filename + content, validation	Small
3	Backend: `DELETE /rules/{filename}`	Small
4	Optional: `POST /rules/sync-definitions` and call after upload/delete	Small
5	Frontend: Download link per file	Small
6	Frontend: Upload form (file or paste + filename), call POST /rules	Medium
7	Frontend: Delete button + confirm, call DELETE /rules/{filename}	Small
8	Tests: API tests for upload/delete; e2e or manual for UI	Small

Total: Backend ~half day; frontend ~half day; testing ~half day. Optional sync-definitions adds a bit more.

124.8 7. Security / validation

- **Path traversal:** Reject `..`, `/`, `\` in filename; resolve path and ensure `path.relative_to(base)` (already done in test-inject).
- **Overwrite:** Upload can overwrite existing files (same as test-inject). Optional: return 409 if file exists and require explicit overwrite flag.
- **YAML validation:** At least parse YAML; optionally require `name` or `flag` and reject if missing so broken rules don't get into the matrix until the next run fails.

124.9 8. Summary

- **Fix:** Prune `fault_definitions` in sync so only current rule files have rows → phantom “AHU Short Cycling” goes away after one FDD run.
- **Defaults:** Keep `stack/rules` with only `sensor_bounds.yaml` and `sensor_flatline.yaml`.
- **CRUD:** Add POST (upload) and DELETE for rule files; reuse GET for download; list already exists.
- **Frontend:** Add upload (file or paste), download link, delete with confirm; reuse existing list + view.
- **Hot reload:** Already works; no change. Optional sync-definitions endpoint improves UX by updating definitions immediately after upload/delete.

125 Security

126 Security and Caddy Bootstrap

This document describes how to protect Open-FDD endpoints with **Caddy** (reverse proxy) when you use it, which services are **unencrypted by default** in a typical dev layout, and **hardening** best practices. The project defaults to **non-TLS** for lab stacks; TLS (including self-signed certificates) is optional.

126.1 Current default (VOLTTRON-first)

The **default** path in this monorepo is **`./afdd_stack/scripts/bootstrap.sh`**: **volttron-docker** / **VOLTTRON Central** prep and optional **`--compose-db`** (Postgres/Timescale + Open-FDD init SQL). **Docker Compose** here does **not** start Caddy, the FastAPI container, the React container, or **any** field-bus gateway — **BACnet/Modbus live only in site VOLTTRON**. See **`afdd_stack/legacy/README.md`** and **Getting started**.

The sections below apply when you run **FastAPI + React + Caddy yourself** (local development or a custom deployment). For **VOLTTRON Central** perimeter security, use **upstream VOLTTRON** documentation.

126.2 Reverse proxy: current file vs future hardening

- **Canonical config in the repo:** stack/caddy/Caddyfile — checked-in Caddy config for a **single HTTP entry point** on port **80**: handle /api* (with uri strip_prefix /api), /auth*, /ws*, and /ai* are proxied to the API; handle /* serves the React frontend. This matches the frontend's default **VITE_API_BASE=/api** in Docker Compose: browser calls go to same-origin paths under /api, /auth, and /ws.
- **Status:** When Caddy is running in **your** deployment, use **http://localhost** (or http://<lan-host>) as the main URL; use **http://localhost:8000** (API) and **http://localhost:5173** (frontend direct) when debugging without the proxy. The **default repo compose file** does not start Caddy.
- **Future hardening:** Optional Caddy **basic auth**, **TLS**, rate limiting, and stricter perimeter defaults are documented below as **targets**; they are not all enabled in the committed file.

What is Caddy? A small web server in front of the API and the **React frontend**. The **committed** stack/caddy/Caddyfile is a lightweight starting point; this page also documents **target** patterns (basic auth, X-Caddy-Auth, full API routes) for when you extend or harden the proxy. Optional **Grafana** (compose **grafana** profile) can be exposed via Caddy (e.g. /grafana) once you add a reverse proxy. The **default** bootstrap does not start Caddy; see Quick bootstrap with Caddy (custom deployments).

126.3 Architecture: frontend, API, and Caddy

- **React frontend** (when you run it, e.g. dev server or container on **5173**; Caddy's /* can route there). **App login:** users sign in at **/login**; the API returns a short-lived **JWT access token** (browser **sessionStorage**) and an **HttpOnly** refresh cookie (/auth/*). REST uses **Authorization: Bearer <access_token>**; the WebSocket uses **/ws/events?token=<access_token>** (or the API-key token path in Frontend and API authentication).
- **API** (FastAPI) requires auth when **OFDD_API_KEY** and/or **app-user** config is set (OFDD_APP_USER, OFDD_APP_USER_HASH, **OFDD_JWT_SECRET** — all three together). Valid credentials are **Bearer** matching **OFDD_API_KEY** or a valid **access JWT**. Exempt paths include /, /health, /docs, /redoc, /openapi.json, /app (and /app/*), and **/auth/***. Partial app-user configuration is rejected explicitly: **/auth/login** and related routes return **503** (AUTH_CONFIG_ERROR / AUTH_NOT_CONFIGURED), while the global API middleware returns **500** with AUTH_CONFIG_ERROR for other protected paths—never a silent fallback.

- **Machine clients** (VOLTTRON agents, curl, **Open Claw** on a Windows bench) should rely on **`OFDD_API_KEY`** and **`Authorization: Bearer`**, not the browser cookie flow — see Open-Claw integration.
- **Caddy** can be extended with **basic auth** + **`X-Caddy-Auth`** / **`OFDD_CADDY_INTERNAL_SECRET`** (optional); see Caddyfile for protecting the entire API. **Best practice:** use a reverse proxy as the operator entry point when exposing API + UI to a network.

126.4 Frontend and API authentication

Mode	Typical use	How it works
Dashboard login	Human operators via browser	`POST /auth/login` → access JWT in session + HttpOnly refresh cookie; `apiFetch` attaches `Bearer` access token; WebSocket uses access token in query string.
`OFDD_API_KEY`	Scrapers, scripts, agents, LAN test bench	`Authorization: Bearer <OFDD_API_KEY>` on REST; WebSocket may use `?token=<key>` when no JWT is in use.

Configure dashboard login by setting **`OFDD_APP_USER`**, **`OFDD_APP_USER_HASH`**, and **`OFDD_JWT_SECRET`** in **`afdd_stack/stack/.env`** (see script comments in a legacy fork or generate hashes per FastAPI auth docs). Remove or empty **`OFDD_API_KEY`** / app-user keys for an open local dev API only if you accept the risk.

Legacy note: Older **`./scripts/bootstrap.sh --user`** / **`--mode model`** guardrails applied when Compose started the **model** stack automatically; the **current** `afdd_stack/scripts/bootstrap.sh` does not manage those flags.

Examples (password not on the argv):

```
        # Example: set secrets in afdd_stack/stack/.env (use your org's password hashing for the app user), then start FastAPI/uvicorn and React yourself.
        # Legacy one-liners that combined bootstrap + --user + full Docker rebuild are removed from the default bootstrap script.
```

After **`OFDD_JWT_SECRET`** or the app user changes, sign in again in the browser so tokens match the new secret; use **Sign out** (or `/logout`) if the UI still holds an old session.

`VITE_API_BASE`: When the frontend is built to sit behind a path-based proxy, **`/api`** keeps same-origin requests aligned with a Caddy strip-prefix pattern. Override with a full URL (e.g. `http://api-host:8000`) if the API is on another origin.

Older docs referred to baking **`VITE_OFDD_API_KEY`** into the frontend at build time; the supported operator path is now **login + JWT** (or unauthenticated local dev with auth disabled). A static API key in the bundle is discouraged for production.

HTTPS behind a reverse proxy: The API sees plain HTTP from the proxy. Set **`OFDD_TRUST_FORWARDED_PROTO=true`** on the API container when the edge proxy terminates TLS and sends **`X-Forwarded-Proto: https`**, so refresh cookies get the **`Secure`** flag. Leave **`false`** (default) for HTTP-only lab stacks so cookies stay usable over HTTP.

126.5 Stack hardening (database, Caddy, secrets)

Related tracking: **Stack security hardening** and **dashboard / API authentication**.

Architecture rule: Frontend → API → database. The React app must not talk to Postgres; only the API uses `OFDD_DB_DSN` on the Docker network (`db:5432`).

Area	Repo default / guidance
Postgres publish	`docker-compose.yml` binds the DB port to **`127.0.0.1:5432`** only — tools on the **host** can use `psql` locally; **remote machines cannot** reach the DB via that mapping. For stricter production, **remove** the `ports:` block under `db` so only containers on the compose network can connect (admin via `docker exec` or a throwaway `psql` container).
API / frontend host bind	When using **compose** or **docker run** for API/frontend, **`OFDD_API_HOST_BIND`** and **`OFDD_FRONTEND_HOST_BIND`** control published ports **8000** and **5173**. The **default slim compose** in this repo does not publish those services; bind addresses matter when you add your own compose override or run **`uvicorn/npm run dev`** on the host.
Edge TLS	The committed Caddyfile serves **HTTP on :80** with security headers. For **HTTPS**, use a public DNS name and either Caddy automatic TLS or mounted certs; start with `stack/caddy/Caddyfile.https.example` and set **`OFDD_TRUST_FORWARDED_PROTO=true`** on the API.

Area	Repo default / guidance
	Do not send `Strict-Transport-Security` on plain HTTP; the example adds HSTS only on the HTTPS site block.
Caddy headers	The bundled **`stack/caddy/Caddyfile`** sets `X-Content-Type-Options`, `X-Frame-Options`, and `Referrer-Policy`. Optional: **CSP**, **rate limiting**, Caddy **basic auth** as a second layer — see Caddyfile for protecting the entire API.
Secrets	**`afdd_stack/stack/.env`** is **gitignored**. Store **Argon2 hash**, **JWT secret**, and optional **`OFDD_API_KEY`** (machine-only — not for the browser UI) there. Never commit secrets. For Caddy **`X-Caddy-Auth`** flows, set **`OFDD_CADDY_INTERNAL_SECRET`**. Optional Grafana: **`GF_SECURITY_ADMIN_USER`** / **`GF_SECURITY_ADMIN_PASSWORD`** in `.env` instead of compose defaults.

WebSocket: Same auth model as HTTP (access JWT or API key); no DB access from browser WebSocket code.

126.6 Quick bootstrap with Caddy (custom deployments)

The **default** `afdd_stack/scripts/bootstrap.sh` does **not** start Caddy or the full Docker stack. Use **Getting started** first.

If you run **FastAPI + React + Caddy** yourself (custom compose or manual processes):

1. **Start your stack** so the API listens on **8000**, the frontend on **5173**, and optionally Caddy on **80** using `stack/caddy/Caddyfile` as a template. **Grafana** is optional via the compose **`grafana`** profile from `afdd_stack/stack/`.

2. **Access through Caddy** (**no basic auth** in the committed repo file today):

   - **URL:** `http://localhost` (port **80**)
   - **`/api*`** → API (prefix stripped), **`/auth*`** → API, **`/ws*`** → API, **`/ai*`** → API, **`/*`** → frontend (static build). From another machine on the LAN, use **`http://<server-ip>/`** the same way.

3. **Hardened entry point (future / manual):** To put the **entire** UI and API behind one login and optional TLS, follow Caddyfile for protecting the entire API below and test thoroughly.

Example credentials like `openfdd` / `xyz` apply only after you add **basic_auth** to your Caddyfile—not in the default committed file.

4. **Without Caddy:** You can still use the services directly (no auth if `OFDD_API_KEY` is unset):
   - Frontend: http://localhost:5173
   - API: http://localhost:8000/docs
   - Grafana (if started): http://localhost:3000
     Do not expose these ports to untrusted networks without a reverse proxy and auth.
5. **Browser "Not secure" or "Your connection is not private":** Over plain HTTP (e.g. `http://localhost/` or `http://192.168.x.x/`), browsers show a warning. That is expected without TLS. Use **http://** (not https://) unless you have configured HTTPS in Caddy. Production should use TLS when you deploy with HTTPS.

126.7 Caddyfile for protecting the entire API

The **committed** `stack/caddy/Caddyfile` is intentionally **small** (see Reverse proxy: current file vs future hardening). The block below is an **example / target** for a **future hardened** setup: one entry point, optional **basic auth**, and most API routes (plus WebSocket) proxied with **`X-Caddy-Auth`**. It is **not** drop-in verified for every workflow yet—validate in your environment before relying on it for production. Use the same **secret** in `header_up X-Caddy-Auth` and in the API container env **`OFDD_CADDY_INTERNAL_SECRET`** so the API trusts requests that passed Caddy's basic auth.

```
# Listen on port 80 (or use :8088 and map 8088:8088 in docker-
compose).
:80 {
    # Optional: basic auth (generate hash: docker run --rm caddy:2
caddy hash-password --plaintext 'your_password')
    # basic_auth /* {
    #   openfdd <bcrypt_hash>
    # }

    # API: REST, Swagger, WebSocket. Set
OFDD_CADDY_INTERNAL_SECRET in the API container to match the value
below.
    handle /ws/* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
```

```
handle /docs {
    header_up X-Caddy-Auth "your_internal_secret"
    reverse_proxy api:8000
}
handle /redoc {
    header_up X-Caddy-Auth "your_internal_secret"
    reverse_proxy api:8000
}
handle /openapi.json {
    header_up X-Caddy-Auth "your_internal_secret"
    reverse_proxy api:8000
}
handle /health {
    header_up X-Caddy-Auth "your_internal_secret"
    reverse_proxy api:8000
}
handle /config* {
    header_up X-Caddy-Auth "your_internal_secret"
    reverse_proxy api:8000
}
handle /sites* {
    header_up X-Caddy-Auth "your_internal_secret"
    reverse_proxy api:8000
}
handle /equipment* {
    header_up X-Caddy-Auth "your_internal_secret"
    reverse_proxy api:8000
}
handle /points* {
    header_up X-Caddy-Auth "your_internal_secret"
    reverse_proxy api:8000
}
handle /faults* {
    header_up X-Caddy-Auth "your_internal_secret"
    reverse_proxy api:8000
}
handle /data-model* {
    header_up X-Caddy-Auth "your_internal_secret"
    reverse_proxy api:8000
}
handle /download* {
    header_up X-Caddy-Auth "your_internal_secret"
    reverse_proxy api:8000
}
handle /analytics* {
    header_up X-Caddy-Auth "your_internal_secret"
    reverse_proxy api:8000
}
handle /bacnet* {
    header_up X-Caddy-Auth "your_internal_secret"
    reverse_proxy api:8000
}
```

```
    handle /jobs* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /capabilities* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }
    handle /ai* {
        header_up X-Caddy-Auth "your_internal_secret"
        reverse_proxy api:8000
    }

    # Optional: Grafana when started with --with-grafana
    # handle /grafana/* {
    #   reverse_proxy grafana:3000
    # }

    # Frontend (React)
    handle /* {
        reverse_proxy frontend:5173
    }
}
```

Then restart Caddy and recreate the API so it has `OFDD_CADDY_INTERNAL_SECRET` set: `./scripts/bootstrap.sh --build caddy api` (from repo root). See Troubleshooting if the API returns 401 or the WebSocket fails.

126.8 Default password and how to change it

The **default committed** `stack/caddy/Caddyfile` does **not** include **basic auth** today. The following applies **after** you add a `basic_auth` block (e.g. using the example Caddyfile above):

- Use a **bcrypt hash** for each password (e.g. default example password **`xyz`** only if you paste the matching hash from docs/examples—do not use defaults in production).
- **Change the password** (required before production):

  1. Generate a new hash:

     ```
     docker run --rm caddy:2 caddy hash-password --plaintext 'your_secure_password'
     ```

  2. Open **`stack/caddy/Caddyfile`** and update the `basic_auth` user line(s) with the new hash(es).

3. Restart Caddy: `docker compose -f stack/docker-compose.yml restart caddy` (from repo root).

- **Advanced / multiple users:** Edit the `basic_auth` block in the Caddyfile and add more lines (one per `username hash`). Use `caddy hash-password` for each password. For SSO or advanced auth, consider Caddy's JWT or forward-auth and an IdP instead of basic auth.

126.9 Services: encryption and exposure

Service	Port	Encrypted by default?	Notes
API	8000	No (HTTP)	Put behind Caddy; do not expose directly. Bearer auth when `OFDD_API_KEY` is set.
Frontend	5173	No (HTTP)	React app; use Caddy to serve and protect. Sends Bearer token to API.
Grafana	3000	No (HTTP)	Optional (–with-grafana); use Caddy at /grafana or Grafana's own auth.
TimescaleDB	5432	No (plain PostgreSQL)	Compose maps **`127.0.0.1:5432`** to the container so the host can use `psql` locally; remote clients do not get a LAN-wide DB port.

Service	Port	Encrypted by default?	Notes
			Remove `ports` under `db` for internal-only DB.
Legacy lab gateway	8080	No (HTTP)	**Only if you run** a separate diy-bacnet-style JSON-RPC gateway yourself; **not** part of default Open-FDD.

- **Recommendation:** When hardened, expose only Caddy (compose defaults to **port 80**; use **443** or another port with TLS when ready) to the building/remote network. Do not expose 5432, 8000, 5173, 3000, or ad hoc gateway ports to the internet. Plan for Caddy **basic auth** and **TLS** as security hardening matures; keep the database and **site VOLTTRON** hosts behind the firewall. The React frontend runs in its own container and is served by Caddy (or another reverse proxy), not as static files from the FastAPI process.

126.10 Hardening best practices

1. **Passwords**
   - When you enable Caddy **basic auth**, do not use example defaults like `openfdd` / `xyz` in production. Change via `caddy hash-password` and update `stack/caddy/Caddyfile`.
   - Use strong, unique passwords (or a secrets manager) for Grafana (`GF_SECURITY_ADMIN_PASSWORD`), Postgres (`POSTGRES_PASSWORD`), and Caddy basic auth.
2. **Network**
   - Run TimescaleDB and internal services on a private Docker network; bind DB to localhost or internal IP only if needed.
   - Restrict firewall so only Caddy’s port (and optionally SSH) is reachable from outside.
3. **Reverse proxy**
   - Use Caddy in front of API and Grafana so one place enforces auth and (optionally) TLS.
   - Basic auth over plain HTTP is weak; prefer HTTPS when possible (see below).

4. **Principle of least privilege**
    - Run containers as non-root where possible; avoid sharing host network unless required (e.g. some VOLTTRON/BACnet lab layouts on the edge host).
    - Limit Grafana sign-up and use strong admin credentials.
5. **Updates and hygiene**
    - Keep Caddy, Grafana, TimescaleDB, and the OS updated. Rotate credentials after compromise or periodically.

126.11 Throttling and rate limiting

126.11.1 1. No API rate limiting by default

Open-FDD does **not** rate-limit incoming HTTP requests to the API. There is no built-in cap on how often clients can call the API. If you need to limit how often external clients (e.g. a busy integration or cloud poller) can call the API, add rate limiting at the reverse proxy (e.g. Caddy with a rate-limit module) or with rate-limit middleware in front of the app.

126.11.2 2. Outbound: OT/building network (VOLTTRON)

Open-FDD services in this repo do not originate BACnet/Modbus traffic. Pacing and politeness on the **OT LAN** are configured in **each site's VOLTTRON** deployment (driver scan rates, historian publish rules, etc.—see upstream VOLTTRON documentation). Open-FDD reads **SQL** and optional HTTP to its own API.

Component	Config	Effect
VOLTTRON (site)	Upstream driver / historian settings	Controls how often field devices are read and how often SQL rows are written.
FDD rule loop	`rule_interval_hours` (e.g. 3)	Runs fault detection on a schedule; each run pulls data from the **DB**, not from the wire.
Weather (optional)	`open_meteo_*` / FDD loop	Fetches weather on the configured

Component	Config	Effect
		cadence when enabled.

Legacy: a private fork might still run removed **BACnet scraper** containers; that is **not** documented here as a supported path. See **Site VOLTTRON and the data plane (ZMQ)** and Configuration.

126.11.3 3. Inbound: rate limiting at the reverse proxy (e.g. Caddy)

If you need to **throttle incoming traffic** to the API—for example to protect the API and OT network from aggressive polling, misconfigured integrators, or abuse—enforce rate limiting at the reverse proxy or with middleware. Open-FDD does not implement this itself; use Caddy (with a rate-limit module), nginx, or application middleware.

Using Caddy for rate limiting

Caddy can throttle requests per client (e.g. per IP or per authenticated user) when configured with a rate-limiting capability. The standard Caddy build does not include an HTTP rate-limit directive; you add it by building Caddy with a module (e.g. `xcaddy build --with github.com/mholt/caddy-ratelimit`) or by using a distribution that includes one. Once available, you configure a rate-limit zone in the Caddyfile (e.g. limit by `{remote_host}` or by a header), then apply it to the blocks that proxy to the API (and optionally Grafana). Typical choices are a cap per IP (e.g. 60 requests per minute) or per client identity when using auth. Exceeding the limit returns `429 Too Many Requests`; the client can retry after the window resets.

Example (conceptual; syntax depends on the module you use):

```
:8088 {
  rate_limit {
    zone api { key {remote_host} events 60 duration 1m }
  }
  handle /api/* {
    rate_limit api
    reverse_proxy api:8000
  }
  # ... basic_auth and other handle blocks
}
```

Adjust the zone (events per duration) to match your OT network and integration requirements. For official options and current syntax, see Caddy’s module documentation and the documentation for the rate-limit module you build or install.

126.12 TLS (optional): self-signed certificates

The project **default is non-TLS**. If you want HTTPS in front of Caddy:

1. **Option A – Caddy automatic HTTPS (public hostname)**
   If you have a public hostname and port 80/443, change the Caddyfile address to your domain; Caddy will obtain and renew Let's Encrypt certs automatically.

2. **Option B – Self-signed certificate for Caddy**
   For internal or lab use:

   - Generate a self-signed cert (e.g. with `openssl`).
   - In the Caddyfile, use `tls /path/to/cert.pem /path/to/key.pem` and listen on `:443` (or another port).
   - Mount the cert and key into the Caddy container and point Caddyfile to those paths.
   - Browsers will show a certificate warning; accept or install the CA for your environment.

Example Caddyfile snippet for self-signed (conceptual):

```
https://0.0.0.0:8443 {
  tls /etc/caddy/cert.pem /etc/caddy/key.pem
  basic_auth /* { ... }
  # same handle/reverse_proxy blocks as :8088
}
```

Generate self-signed (host):

```
        openssl req -x509 -newkey rsa:4096 -keyout key.pem -out cert.pem -
days 365 -nodes -subj "/CN=openfdd.local"
```

Mount `cert.pem` and `key.pem` into the Caddy container and reference them in `tls`.

126.13 Troubleshooting

API returns 401 (Unauthorized) after you pass Basic auth in the browser

The API trusts requests that carry the internal header `X-Caddy-Auth` (set by Caddy after Basic auth). The API only checks this when the env var `OFDD_CADDY_INTERNAL_SECRET` is set in the API container and matches the value Caddy sends. If you added or changed this after the stack was already running, the API container may not have the variable. Recreate the API so it picks up the env:

```
cd stack && docker compose up -d --force-recreate api
```

Or from the repo root: `./scripts/bootstrap.sh --build api`. Then hard-refresh or use an incognito window and sign in again with Basic auth.

WebSocket to `/ws/events` fails or closes immediately

When access is through Caddy with Basic auth, the browser may not send credentials on the WebSocket upgrade. The API also accepts the same `X-Caddy-Auth` header on the WebSocket endpoint, so if Caddy is forwarding the request (after Basic auth), the WebSocket should work. Ensure the API has been recreated with `OFDD_CADDY_INTERNAL_SECRET` as above. If the WebSocket still fails (e.g. due to browser or proxy behavior), the UI will keep working; live updates will resume when the connection succeeds.

126.14 Security scanning (Trivy)

During development, use **Trivy** to scan container images and the repo for vulnerabilities and misconfigurations. Run it after building images, when changing Dockerfiles or dependencies, and optionally in CI. See **How-to: Trivy security scanning** for install, when to run, image and filesystem scan commands, and keeping docs up to date.

126.15 Summary

- **Bootstrap:** Start the stack; the committed Caddyfile listens on **`http://localhost`** (port **80**) with **no** basic auth, and routes **`/api`**, **`/auth`**, **`/ws`**, and **`/ai`** to the API. Use **8000** / **5173** for direct debugging.
- **Auth:** Optional dashboard login (JWT + HttpOnly refresh cookie) and/or **`OFDD_API_KEY`** for machine clients; both can be configured together.
- **Passwords:** Change default by running `caddy hash-password` and updating the Caddyfile; use strong passwords for Grafana and Postgres.
- **Unencrypted by default:** API, Grafana, TimescaleDB, and any **optional** lab HTTP gateway you add are plain HTTP/TCP; protect them with network isolation and Caddy (and optional TLS).
- **Hardening:** Strong passwords, expose only Caddy, keep DB and internal services off the public internet, keep software updated.
- **Throttling:** (1) No API rate limiting by default. (2) OT pacing is owned by **VOLTTRON** at the site; Open-FDD reads SQL and

runs FDD/weather on configured intervals. (3) To throttle incoming API traffic, use the reverse proxy (e.g. Caddy with a rate-limiting module) or middleware.

- **TLS:** Optional; default is non-TLS. Add self-signed or Let’s Encrypt via Caddy when you need HTTPS.